GRADE 7 COMMON CORE MATH

Question Collection

Main • Backward Branches • Forward 1 (Grade 8) • Forward 2 (Grade 9)

70 question sets • 1755 questions

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SET 1 • 7.RP.A.1 • RATIOS & PROPORTIONAL RELATIONSHIPS

Compute unit rates associated with ratios of fractions

MAIN	Grade 7 main question	**7.RP.A.1**
BACKWARD 1	Find a unit rate from whole-number quantities	**6.RP.A.2**
BACKWARD 2	Divide a fraction by a fraction	**6.NS.A.1**
FORWARD 1	Compare unit rates shown in different representations	**8.EE.B.5**
FORWARD 2	Convert rates from one unit to another	**HSN-Q.A.1**

SET 1 • 7.RP.A.1

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

Ana walks $\frac{3}{4}$ mile in $\frac{1}{3}$ hour at a steady pace.

How many miles does she walk per hour?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q1/5

A recipe uses $\frac{2}{3}$ cup of sugar for every $\frac{1}{2}$ cup of flour.

How many cups of sugar are used per cup of flour?

Cups of sugar: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q2/5

Leo paints $\frac{5}{6}$ of a wall in $\frac{1}{2}$ hour.

How many walls does he paint per hour?

A. $\frac{3}{5}$ wall

B. $1\frac{2}{3}$ walls

C. $\frac{5}{12}$ wall

D. $2\frac{1}{2}$ walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q3/5

True or false?

A snail crawls $\frac{1}{8}$ meter in $\frac{1}{4}$ minute. Its speed is $\frac{1}{2}$ meter per minute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q4/5

True or false?

$\frac{2}{5}$ pound of cheese costs $3. The price is $1.20 per pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q5/5

SET 1 • 7.RP.A.1

BACKWARD 1

Find a unit rate from whole-number quantities • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A car travels 180 miles in 3 hours at a steady speed.

What is its speed in miles per hour?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q1/5

6 notebooks cost $9.

What is the cost per notebook?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q2/5

A printer prints 120 pages in 8 minutes.

How many pages does it print per minute?

A. 112

B. 960

C. 15

D. 128

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q3/5

True or false?

Jo reads 90 pages in 5 days at the same rate each day. That is 18 pages per day.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q4/5

True or false?

4 pounds of grapes cost $24. The price is $8 per pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 7.RP.A.1

BACKWARD 2

Divide a fraction by a fraction • Grade 6 • The Number System • 6.NS.A.1

Divide.

$\frac{3}{4} \div \frac{1}{2}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q1/5

Divide.

$\frac{5}{6} \div \frac{1}{3}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q2/5

Divide.

$\frac{5}{8} \div \frac{1}{4}$

A. $1\frac{1}{4}$

B. $\frac{5}{32}$

C. $\frac{2}{5}$

D. $2\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{2} \div \frac{1}{4} = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\frac{2}{3} \div \frac{1}{3} = \frac{2}{9}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 7.RP.A.1

FORWARD 1

Compare unit rates shown in different representations • Grade 8 • Expressions & Equations • 8.EE.B.5

Which relationship has the greatest unit rate?

A. y = 4.2x

B. A table with the pairs (2, 8) and (5, 20)

C. A line through (0, 0) and (3, 12.3)

D. y = 3.9x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q1/5

Runner A: a graph of meters (y) against seconds (x) is a line through (0, 0) and (10, 45).

Runner B: d = 4.8t, where d is meters and t is seconds.

Which runner is faster, and by how many meters per second?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q2/5

Plan A charges $0.15 per text message. The cost for Plan B is shown by a line through (0, 0) and (40, 5), where x is texts and y is dollars.

Which plan costs less per text?

Plan: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q3/5

True or false?

The graph of $y = 2.5x$ is steeper than the graph of a proportional relationship that passes through (4, 9).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q4/5

True or false?

A proportional relationship that passes through (6, 15) has a greater unit rate than $y = 2.6x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q5/5

SET 1 • 7.RP.A.1

FORWARD 2

Convert rates from one unit to another • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

A cheetah runs 66 feet per second.

How many miles per hour is this? (1 mile = 5,280 feet)

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q1/5

True or false?

Walking $\frac{1}{2}$ mile in $\frac{1}{10}$ hour is a speed of 440 feet per minute. (1 mile = 5,280 feet)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q2/5

A pump moves 3 gallons every $\frac{1}{4}$ minute.

How many gallons per hour is this?

Gallons per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q3/5

A runner covers 400 meters in 80 seconds at a steady pace.

What is the speed in kilometers per hour?

A. 18

B. 300

C. 0.3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q4**

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

60 miles per hour is the same speed as 1 mile per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q5/5

Decide whether two quantities in a table are in a proportional relationship

MAIN	Grade 7 main question	**7.RP.A.2.a**
BACKWARD 1	Complete a table of equivalent ratios	**6.RP.A.3.a**
BACKWARD 2	Divide multi-digit decimals	**6.NS.B.3**
FORWARD 1	Decide whether a table shows a linear function	**8.F.A.3**
FORWARD 2	Tell equal differences (linear) from equal factors (exponential) ov...	**HSF-LE.A.1.b**

SET 2 • 7.RP.A.2.a

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The table shows a proportional relationship between x and y.

x	2	5	8
y	7	17.5	28

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q1/5

Which table shows a proportional relationship?

A. x: 2, 4, 6 and y: 3, 5, 7

B. x: 1, 2, 3 and y: 5, 6, 7

C. x: 3, 6, 9 and y: 12, 24, 36

D. x: 1, 3, 5 and y: 4, 12, 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q2/5

The table shows the cost of renting a bike.

Is the cost proportional to the number of hours? Explain.

Hours	1	2	3
Cost ($)	4	9	14

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q3/5

The table shows the gallons of water in a tank after some minutes.

Is the number of gallons proportional to the number of minutes? Use the ratios to explain.

Minutes	2	4	6	8
Gallons	5	10	15	21

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q4/5

True or false?

The table shows a proportional relationship between x and y.

x	1	2	4
y	3	6	10

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q5/5

SET 2 • 7.RP.A.2.a

BACKWARD 1

Complete a table of equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

The table shows equivalent ratios.

What is the missing value?

Cups of flour	2	4	6
Cups of milk	3	?	9

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q1/5

The table shows equivalent ratios.

What is the missing value?

Boys	5	10	15
Girls	4	8	?

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q2/5

Which ratio is equivalent to 3 : 5?

A. 9 : 15

B. 6 : 8

C. 3 : 15

D. 5 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q3/5

True or false?

2 : 7 and 6 : 21 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q4/5

True or false?

4 : 6 and 6 : 8 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 7.RP.A.2.a

BACKWARD 2

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

17.5 ÷ 5

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q1/5

Divide.

$8.4 \div 2.4$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q2/5

Divide.

$9.6 \div 3$

A. 3.2

B. 32

C. 0.32

D. 3.02

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$28 \div 3.5 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q4/5

True or false?

$4.5 \div 1.5 = 0.3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 7.RP.A.2.a

FORWARD 1

Decide whether a table shows a linear function • Grade 8 • Functions • 8.F.A.3

True or false?

The table shows a linear function.

x	0	1	2	3
y	4	9	14	19

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q1/5

Which table shows a linear function that is NOT proportional?

A. x: 0, 1, 2 and y: 1, 2, 4

B. x: 0, 1, 2 and y: 0, 4, 8

C. x: 0, 1, 2 and y: 3, 5, 7

D. x: 0, 1, 2 and y: 0, 1, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q2/5

The table shows a linear function.

What is its rate of change?

x	1	2	3	4
y	2	5	8	11

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q3/5

Is the function in the table linear? Explain.

x	1	2	3	4
y	1	4	9	16

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q4/5

True or false?

A table with x-values 0, 1, 2 and y-values 7, 9, 11 shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 7.RP.A.2.a

FORWARD 2

Tell equal differences (linear) from equal factors (exponential) over equal intervals • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b

Does y grow by equal differences or by equal factors over equal intervals of x? Give evidence from the table.

x	0	1	2	3
y	3	6	12	24

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q1**

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q1/5

By how much does y change over each interval of 2 in x? Is the growth linear or exponential?

x	0	2	4	6
y	4	10	16	22

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q2/5

Each table has x-values 0, 1, 2, 3.

Which y-values change by equal differences over equal intervals?

A. y: 2, 4, 8, 16

B. y: 1, 3, 9, 27

C. y: 1, 2, 4, 7

D. y: 7, 4, 1, −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q3/5

True or false?

A table with x-values 0, 1, 2, 3 and y-values 80, 40, 20, 10 shows y decreasing by equal factors over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q4/5

True or false?

A table with x-values 0, 1, 2, 3 and y-values 2, 6, 18, 54 shows equal differences over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q5/5

SET 3 • 7.RP.A.2.a • RATIOS & PROPORTIONAL RELATIONSHIPS

Decide whether a graph shows a proportional relationship

MAIN	Grade 7 main question	**7.RP.A.2.a**
BACKWARD 1	Plot the pairs from a ratio table	**6.RP.A.3.a**
BACKWARD 2	Find the coordinates of points in all four quadrants	**6.NS.C.6.c**
FORWARD 1	Recognize linear functions whose graphs do not pass through the…	**8.F.A.3**
FORWARD 2	Interpret the intercepts of a graph in context	**HSF-IF.B.4**

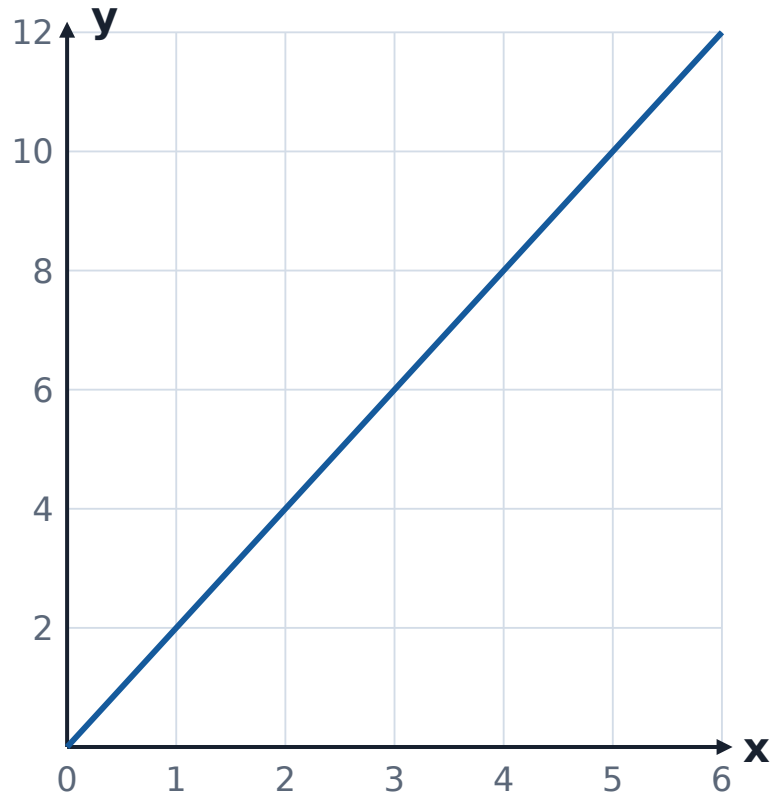
SET 3 • 7.RP.A.2.a

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The graph shows a proportional relationship between x and y.



A. True

B. False

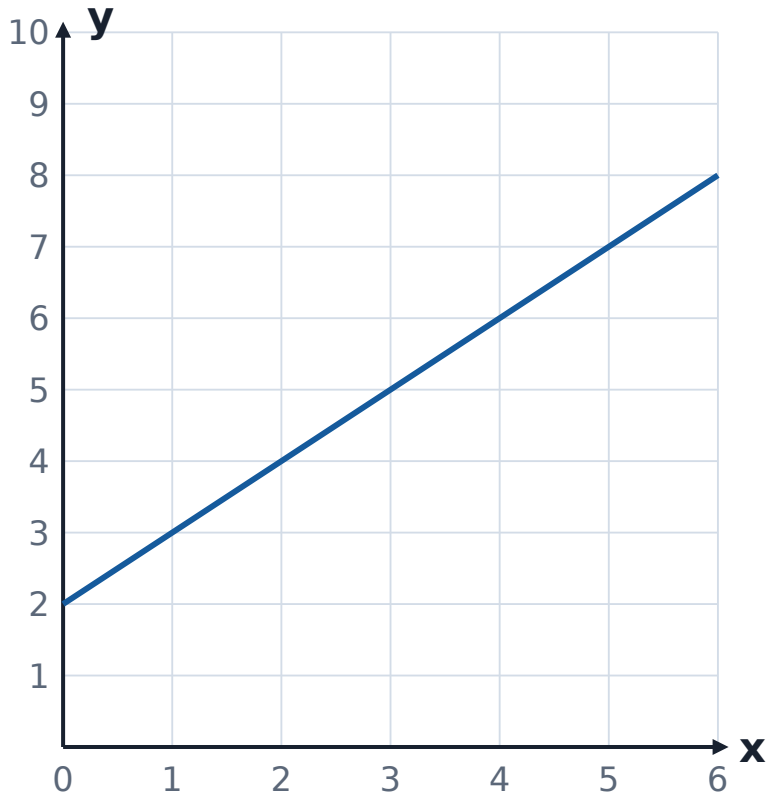
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q1/5

Is the relationship in the graph proportional? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q2/5

Which graph shows a proportional relationship?

A. A straight line that crosses the y-axis at 3

B. Points that do not lie on one line

C. A curve through the origin

D. A straight line through the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

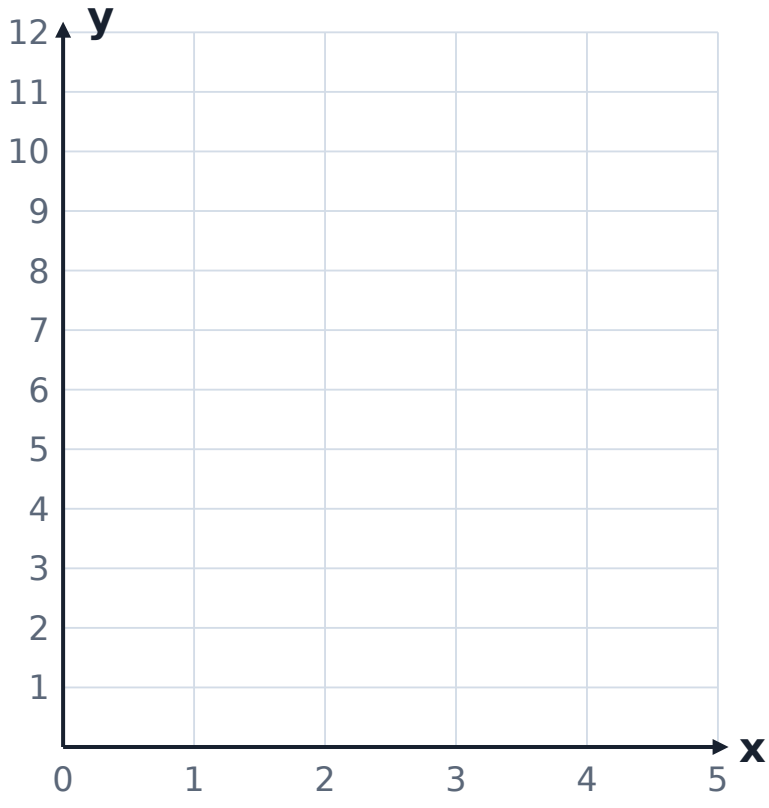
ID S3-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q3/5

Plot the points (1, 2.5), (2, 5), and (4, 10).

Do the points show a proportional relationship? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

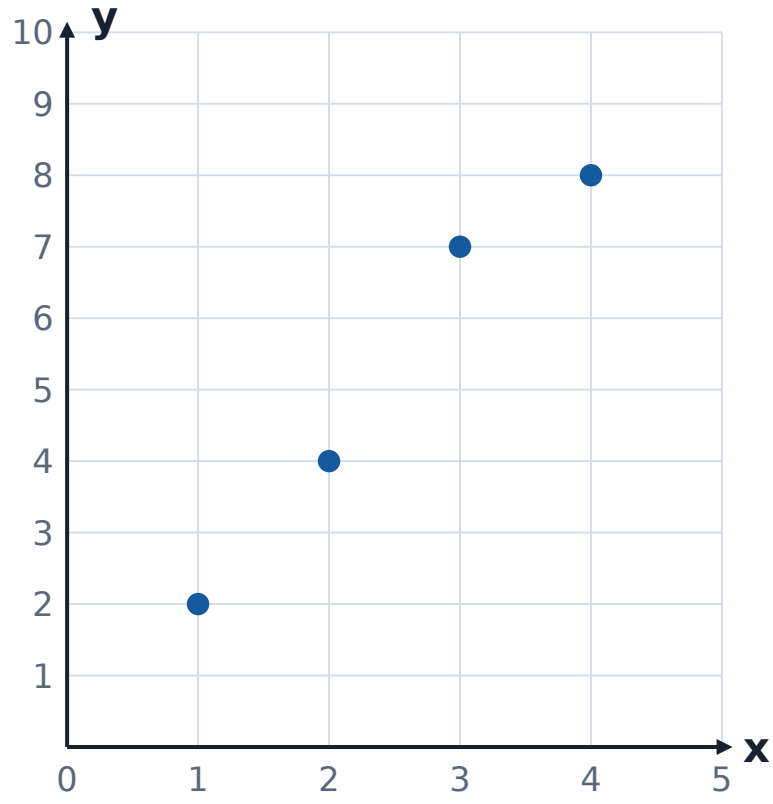
Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q4/5

True or false?

The points in the graph show a proportional relationship.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q5/5

SET 3 • 7.RP.A.2.a

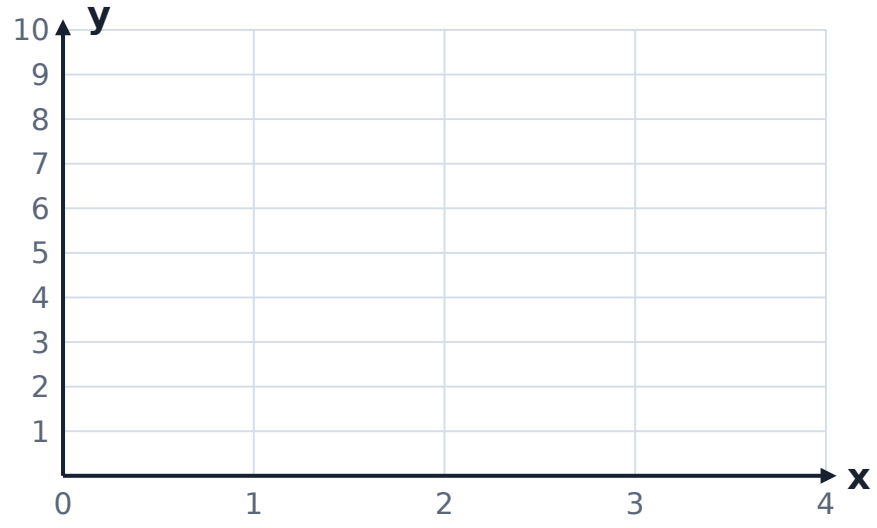
BACKWARD 1

Plot the pairs from a ratio table • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

The table shows equivalent ratios.

Plot each pair (x, y) on the grid.

x	1	2	3
y	3	6	9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q1/5

A ratio table has the pairs (2, 5), (4, 10), and (6, ?).

What is the missing y-value?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q2/5

Which pair belongs in the same ratio table as (3, 4)?

A. (4, 5)

B. (9, 10)

C. (6, 7)

D. (6, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q3/5

True or false?

The pairs (2, 6), (4, 12), and (5, 15) can all come from the same ratio table.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q4/5

True or false?

To plot the pair (3, 7), you start at the origin, move 7 units right, and then move 3 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

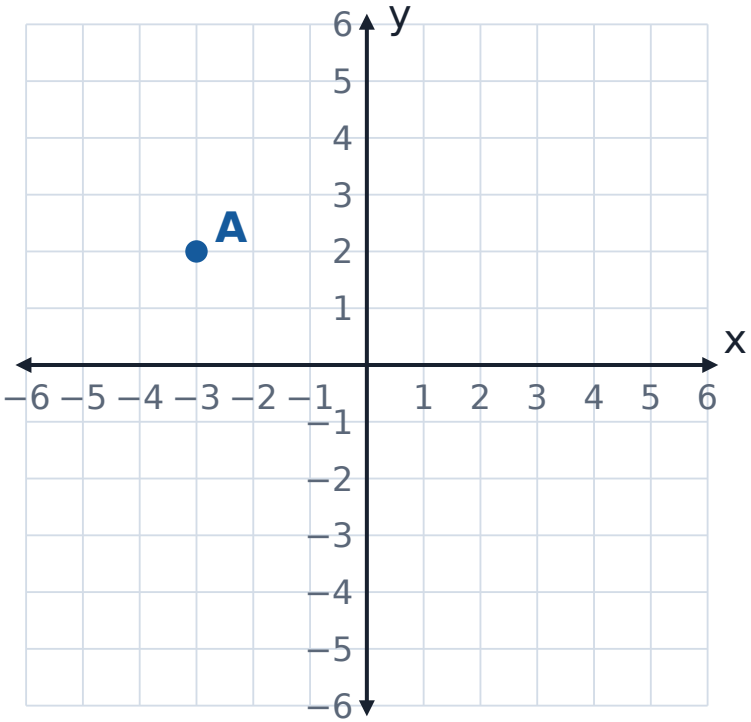
Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 7.RP.A.2.a

BACKWARD 2

Find the coordinates of points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?



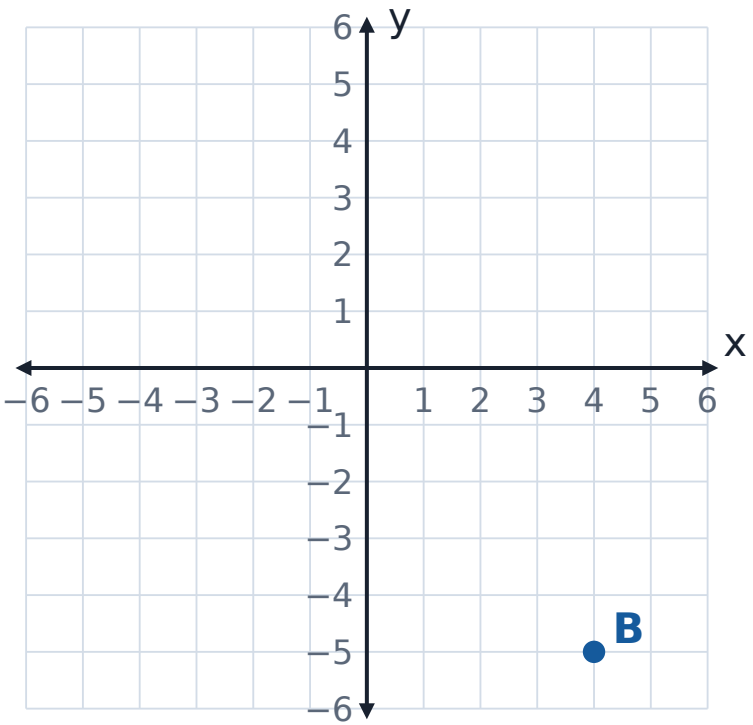
Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q1/5

What are the coordinates of point B?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q2/5

Which point is at the origin?

A. (1, 1)

B. (0, 1)

C. (1, 0)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q3/5

True or false?

The point (3, 0) lies on the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q4/5

True or false?

The point (0, 5) lies on the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q5/5

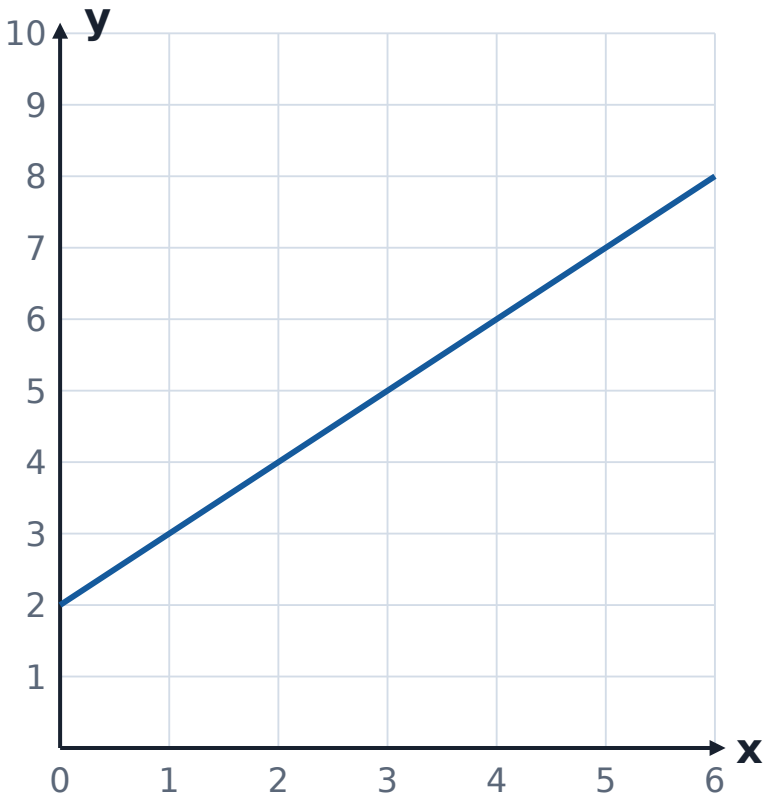
SET 3 • 7.RP.A.2.a

FORWARD 1

Recognize linear functions whose graphs do not pass through the origin • Grade 8 • Functions • 8.F.A.3

The graph shows a line.

Write its equation in the form $y = mx + b$.



Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q1/5

Which equation is linear but NOT proportional?

A. $y = x^2$

B. $y = \frac{3}{x}$

C. $y = 3x$

D. $y = 3x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q2/5

True or false?

The graph of $y = 4x$ is a line through the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q3/5

A line passes through (0, 3) and (2, 7).

Is the relationship proportional? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q4/5

True or false?

Every linear function is a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q5/5

SET 3 • 7.RP.A.2.a

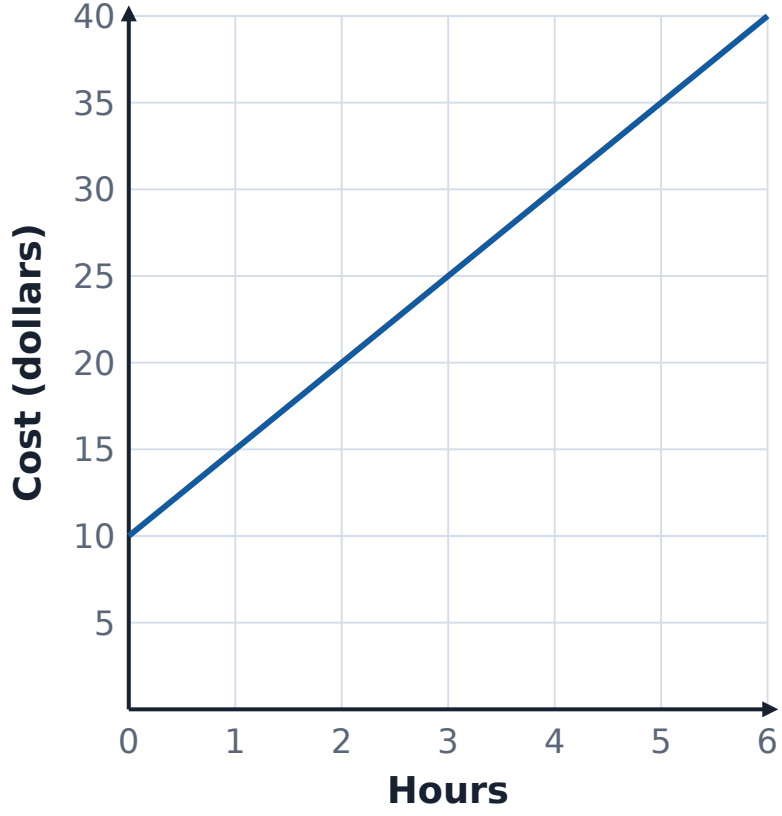
FORWARD 2

Interpret the intercepts of a graph in context • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4

The graph shows the cost of renting a kayak.

What does the point where the line meets the vertical axis mean?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q1/5

The height of a candle after t hours is $h = 24 - 3t$ centimeters.

After how many hours does the candle burn out ($h = 0$)?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q2/5

The water in a tank after m minutes is $w = 50 - 5m$ gallons.

What does the w-intercept, 50, mean?

A. The tank starts with 50 gallons.

B. The tank drains 50 gallons per minute.

C. The tank is empty after 50 minutes.

D. The tank drains 5 gallons in 50 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q3**

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q3/5

True or false?

The cost of a rental is $C = 15 + 2h$ dollars for h hours. The C-intercept, 15, is the cost when $h = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q4/5

True or false?

For $h = 24 - 3t$, the h-intercept is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q5/5

SET 4 • 7.RP.A.2.b • RATIOS & PROPORTIONAL RELATIONSHIPS

Identify the constant of proportionality in tables, graphs, equations, and descriptions

MAIN	Grade 7 main question	**7.RP.A.2.b**
BACKWARD 1	Describe a unit rate	**6.RP.A.2**
BACKWARD 2	Write an equation from a table of values	**6.EE.C.9**
FORWARD 1	Interpret the unit rate as the slope of a graph	**8.EE.B.5**
FORWARD 2	Construct a linear or exponential function from two input-output p…	**HSF-LE.A.2**

SET 4 • 7.RP.A.2.b

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant of proportionality (y per x)?

x	2	4	6
y	5	10	15

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

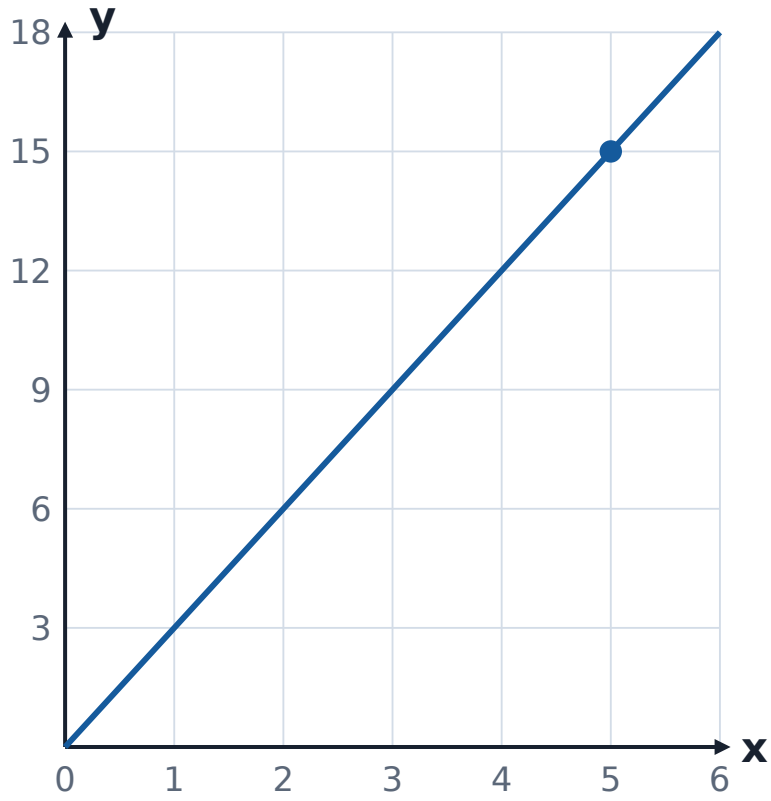
ID S4-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q1/5

The graph shows a proportional relationship.

What is the constant of proportionality?

k = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q2/5

What is the constant of proportionality in $y = \frac{2}{3}x$?

A. 2

B. $\frac{3}{2}$

C. 3

D. $\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q3/5

True or false?

6 pencils cost $2.10. The constant of proportionality, in dollars per pencil, is 0.35.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q4/5

Kim types 216 words in 4 minutes at a steady rate.

What is the constant of proportionality in words per minute?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q5/5

SET 4 • 7.RP.A.2.b

BACKWARD 1

Describe a unit rate • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A pack of 8 juice boxes costs $6.

What is the cost per juice box?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q1/5

A recipe uses 4 cups of flour for 2 loaves.

Which is the unit rate?

A. 6 cups of flour per loaf

B. 8 cups of flour per loaf

C. $\frac{1}{2}$ cup of flour per loaf

D. 2 cups of flour per loaf

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q2/5

Lena swims 600 meters in 12 minutes at a steady pace.

How many meters does she swim per minute?

Meters per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q3/5

True or false?

"$3 per pound" is a unit rate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q4/5

True or false?

5 apples cost $4. The unit rate is $1.25 per apple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 7.RP.A.2.b

BACKWARD 2

Write an equation from a table of values • Grade 6 • Expressions & Equations • 6.EE.C.9

In the table, y is always 4 times x.

Write an equation for y in terms of x.

x	1	2	3
y	4	8	12

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q1/5

Each ticket costs $7.

Which equation gives the cost c of t tickets?

A. $t = 7c$

B. $c = \frac{t}{7}$

C. $c = 7t$

D. $c = t + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q2/5

Write an equation for y in terms of x.

x	2	3	5
y	10	15	25

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q3/5

True or false?

For $y = 6x$, when $x = 3$, $y = 18$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q4/5

True or false?

In d = 50t, d is the independent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 7.RP.A.2.b

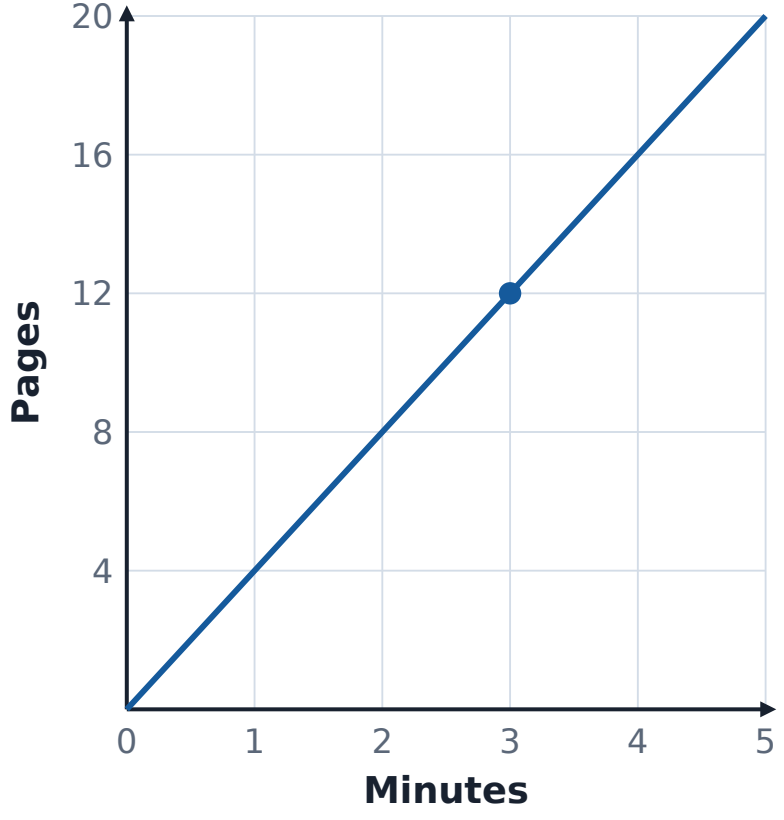
FORWARD 1

Interpret the unit rate as the slope of a graph • Grade 8 • Expressions & Equations • 8.EE.B.5

The graph shows a proportional relationship between minutes and pages read.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q1/5

A line passes through (0, 0) and (8, 6).

What is its slope?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q2/5

The graph of a proportional relationship has a slope of 1.5 liters per minute.

Which point is on the graph?

A. (4, 6)

B. (6, 4)

C. (1.5, 1)

D. (4, 5.5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q3/5

True or false?

The slope of the graph of $y = 0.8x$ is 0.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q4/5

True or false?

A line through the origin and (2, 9) has a slope of $\frac{2}{9}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q5/5

SET 4 • 7.RP.A.2.b

FORWARD 2

Construct a linear or exponential function from two input-output pairs, a sequence, or a description • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

A linear function has f(2) = 7 and f(6) = 19.

Write f(x).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q1/5

An arithmetic sequence is 11, 15, 19, 23, ...

Write a function a(n) for the nth term, starting with n = 1.

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q2/5

A tank drains at a constant rate. It holds 500 gallons after 4 minutes and 380 gallons after 10 minutes.

Which function gives the gallons g(t) after t minutes?

A. $g(t) = -20t + 500$

B. $g(t) = 20t + 420$

C. $g(t) = -20t + 580$

D. $g(t) = -12t + 548$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q3/5

True or false?

The sequence 2, 6, 18, 54, ... is given by $f(n) = 2 \cdot 3^{(n-1)}$ for $n = 1, 2, 3, \ldots$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q4/5

True or false?

A gym's total cost grows at a constant monthly rate. It is $35 after 1 month and $95 after 5 months. The function is C(m) = 15m + 35.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q5/5

Represent a proportional relationship with an equation

MAIN	Grade 7 main question	**7.RP.A.2.c**
BACKWARD 1	Write an equation for a dependent variable	**6.EE.C.9**
BACKWARD 2	Solve unit rate problems	**6.RP.A.3.b**
FORWARD 1	Construct a linear function y = mx + b from a situation	**8.F.B.4**
FORWARD 2	Write an equation in two variables for a situation	**HSA-CED.A.2**

SET 5 • 7.RP.A.2.c

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Peaches cost $2.40 per pound.

Write an equation for the cost c of p pounds.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q1/5

A car travels 29 miles per gallon of gas.

Which equation gives the miles m for g gallons?

A. $m = 29g$

B. $m = g + 29$

C. $g = 29m$

D. $m = \frac{g}{29}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q2/5

The table shows a proportional relationship.

Write an equation for y in terms of x.

x	4	8	12
y	3	6	9

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q3/5

True or false?

Notebooks cost $1.75 each. The equation $c = 1.75n$ gives the cost c of n notebooks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q4/5

True or false?

A printer prints 12 pages every 3 minutes. The equation $p = 12m$ gives the pages p printed in m minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q5/5

SET 5 • 7.RP.A.2.c

BACKWARD 1

Write an equation for a dependent variable • Grade 6 • Expressions & Equations • 6.EE.C.9

A worker earns $18 per hour.

Write an equation for the earnings e after h hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q1/5

Which equation matches the table?

x	1	2	3
y	9	18	27

A. y = x + 8

B. y = 3x

C. x = 9y

D. y = 9x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q2/5

Water fills a tank at 6 liters per minute.

Write an equation for the liters L after t minutes.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q3/5

True or false?

In e = 18h, the value of e depends on the value of h.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q4/5

True or false?

The point (2, 7) is on the graph of y = 5x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 7.RP.A.2.c

BACKWARD 2

Solve unit rate problems • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

5 bags of rice cost $15.

At the same rate, how much do 8 bags cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q1/5

A machine makes 140 parts in 7 hours.

At the same rate, how many parts does it make in 10 hours?

Parts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q2/5

A bus travels at 45 miles per hour.

How long does it take to travel 135 miles?

A. 3 hours

B. 2 hours

C. 4 hours

D. 6 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q3/5

True or false?

At $4 per pound, 6 pounds cost $24.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q4/5

True or false?

A printer prints 12 pages per minute. It takes 6 minutes to print 60 pages.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 7.RP.A.2.c

FORWARD 1

Construct a linear function y = mx + b from a situation • Grade 8 • Functions • 8.F.B.4

A plumber charges $60 for a visit plus $45 per hour.

Write an equation for the cost y after x hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q1/5

The table shows a linear function.

Write its equation in the form $y = mx + b$.

x	0	1	2
y	7	10	13

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q2/5

A line has a slope of 2 and passes through (0, −5).

Which is its equation?

A. $y = -5x + 2$

B. $y = -2x - 5$

C. $y = 2x - 5$

D. $y = 2x + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q3/5

True or false?

A pool has 300 gallons and fills at 20 gallons per minute. The equation $y = 20x + 300$ gives the gallons y after x minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q4/5

True or false?

A candle 15 cm tall burns 2 cm per hour. Its height after x hours is $y = 15x - 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q5/5

SET 5 • 7.RP.A.2.c

FORWARD 2

Write an equation in two variables for a situation • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2

Pens cost $2 each and markers cost $3 each. Kai spends exactly $30.

Write an equation using p for pens and m for markers.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q1/5

Adult tickets cost $12 and child tickets cost $8. A group pays $96.

Write an equation using a for adult tickets and c for child tickets.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q2/5

A farm has 40 animals. Every animal is a chicken or a goat.

Which equation relates the number of chickens c and goats g?

A. $cg = 40$

B. $c + g = 40$

C. $2c + 4g = 40$

D. $c = 40g$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q3/5

True or false?

The equation $5x + 2y = 50$ can describe buying x items at $5 each and y items at $2 each for a total of $50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q4/5

True or false?

$p = 6$ and $m = 4$ is a solution of $2p + 3m = 30$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q5/5

Explain what a point on the graph of a proportional relationship means

MAIN	Grade 7 main question	**7.RP.A.2.d**
BACKWARD 1	Explain a pair in a table of equivalent ratios	**6.RP.A.3.a**
BACKWARD 2	Read values from the graph of an equation	**6.EE.C.9**
FORWARD 1	Interpret the rate of change and initial value of a linear function	**8.F.B.4**
FORWARD 2	Calculate the average rate of change over an interval	**HSF-IF.B.6**

SET 6 • 7.RP.A.2.d

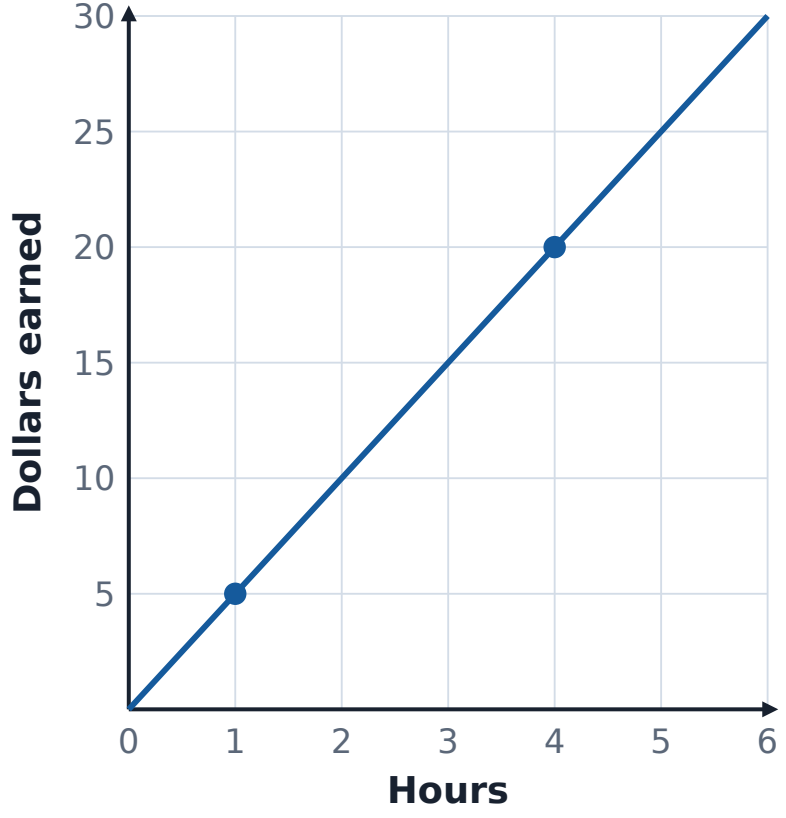
MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the money Raj earns.

What does the point (4, 20) mean in this situation?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

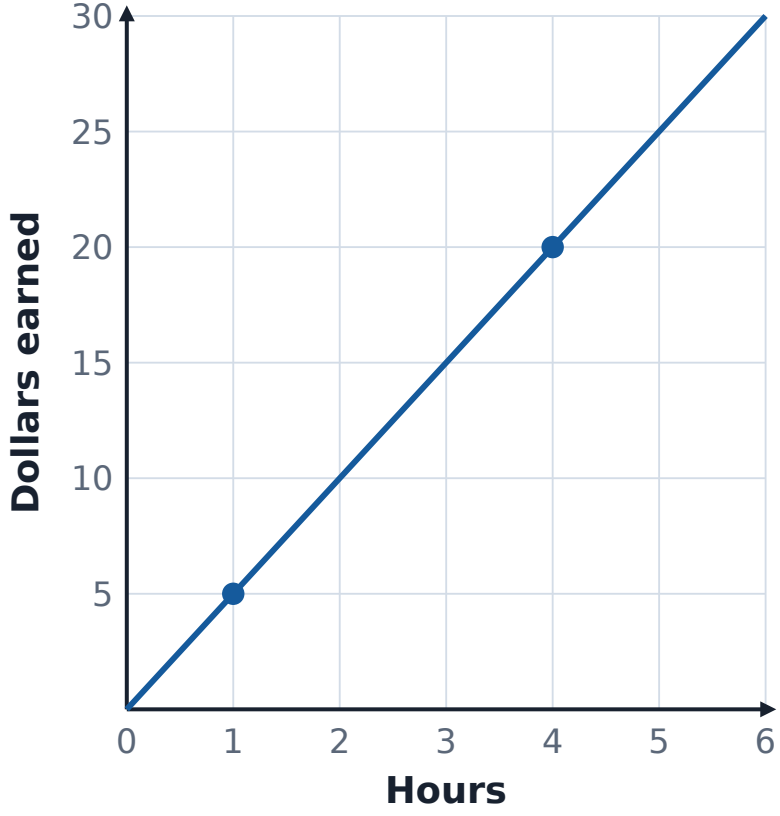
ID S6-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q1/5

The graph shows the money Raj earns.

What does the point (1, 5) tell you?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q2/5

The graph shows the money Raj earns.

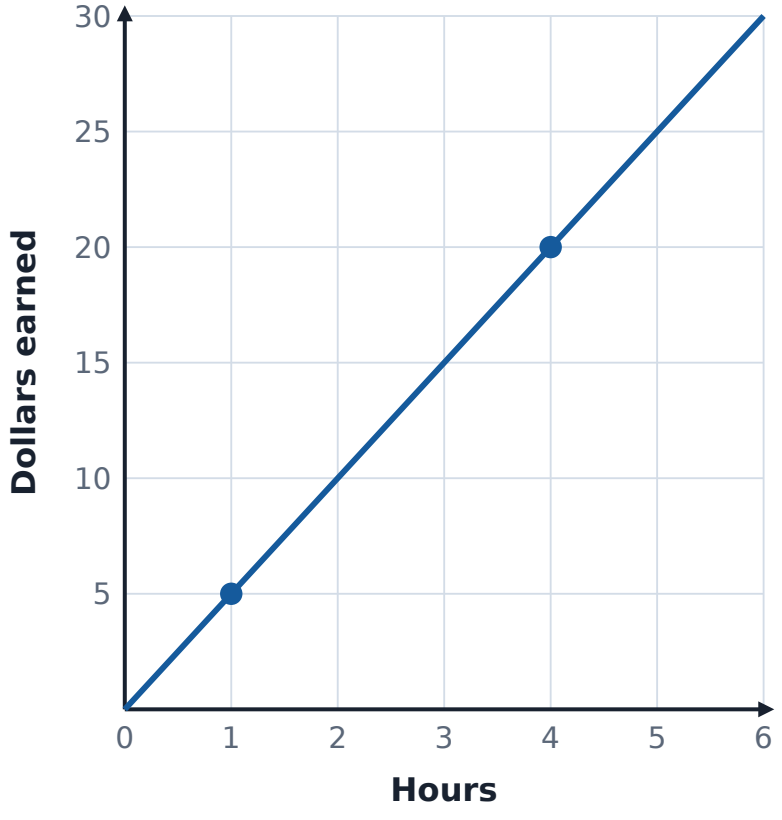
What does the point (0, 0) mean?

A. Raj earns $0 when he works 0 hours.

B. Raj earns $5 per hour.

C. Raj works 0 hours to earn $5.

D. Raj starts with $5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q3/5

True or false?

A proportional graph shows cups of rice (x) and cups of water (y). The point (1, 3.5) means 3.5 cups of water are used for each cup of rice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q4/5

True or false?

A car's distance graph passes through (2, 110), where x is hours and y is miles. This means the car travels 110 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q5/5

SET 6 • 7.RP.A.2.d

BACKWARD 1

Explain a pair in a table of equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

The table shows the cost of apples.

What does the pair (2, 6) mean?

Pounds	1	2	3
Cost ($)	3	6	9

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q1/5

The table shows the time a swimmer takes.

What does the pair (4, 10) mean?

Laps	2	4	6
Minutes	5	10	15

A. Each lap takes 4 minutes.

B. 4 laps take 10 minutes.

C. 10 laps take 4 minutes.

D. Each lap takes 10 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q2/5

True or false?

A table shows hours (x) and miles walked (y). The pair (3, 12) means 12 miles are walked in 3 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q3/5

One ticket costs $8.

What pair (tickets, cost) shows the cost of 5 tickets?

Pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q4/5

True or false?

A table shows boxes (x) and pencils (y). The pair (2, 24) means each box holds 2 pencils.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 7.RP.A.2.d

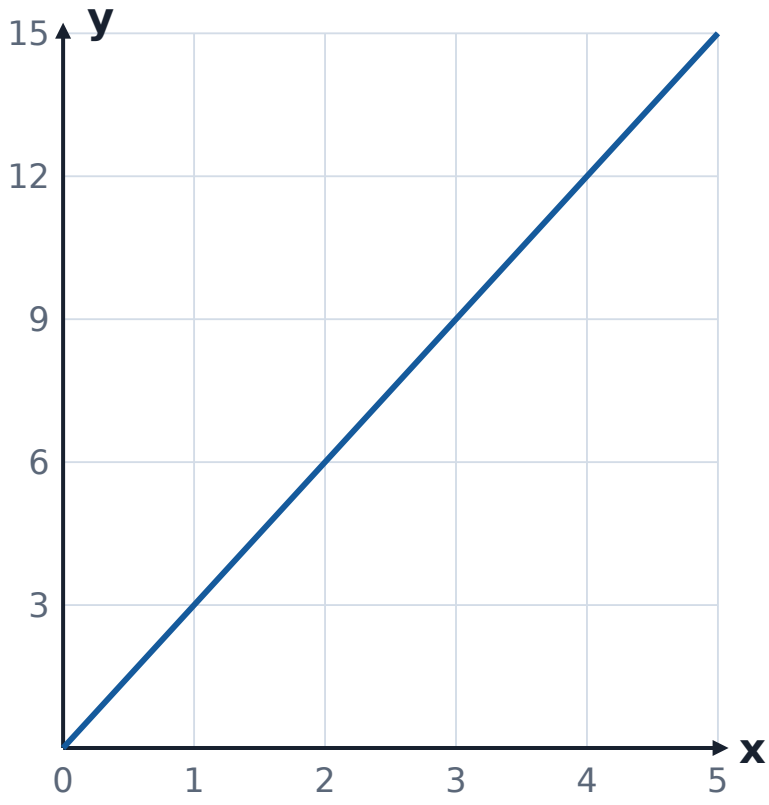
BACKWARD 2

Read values from the graph of an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

The graph shows y = 3x.

What is y when x = 4?

y = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

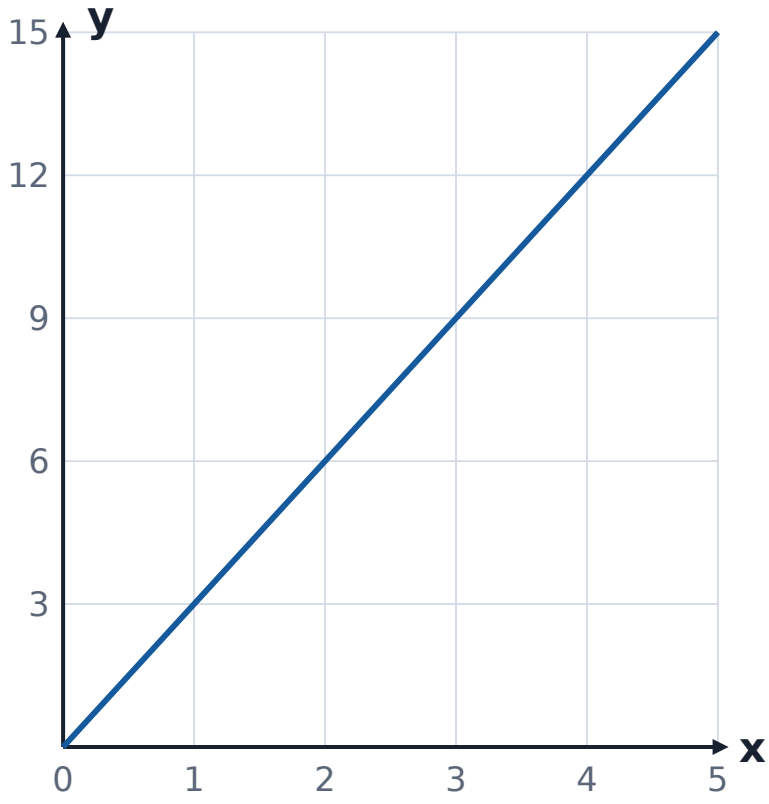
ID S6-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q1/5

The graph shows y = 3x.

What is x when y = 6?

x = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q2/5

Which point is on the graph of y = 2x?

A. (6, 3)

B. (4, 6)

C. (3, 6)

D. (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q3/5

True or false?

The point (5, 15) is on the graph of $y = 3x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q4/5

True or false?

The point (1, 4) is on the graph of $y = 3x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 7.RP.A.2.d

FORWARD 1

Interpret the rate of change and initial value of a linear function • Grade 8 • Functions • 8.F.B.4

A taxi ride costs $y = 2.5x + 4$ dollars for x miles.

What does the 4 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q1/5

A taxi ride costs $y = 2.5x + 4$ dollars for x miles.

What does the 2.5 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q2/5

A tank holds $y = 200 - 8x$ gallons after x minutes.

What is the rate of change?

A. -8 gallons per minute

B. 200 gallons per minute

C. 8 gallons

D. -200 gallons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q3/5

True or false?

A plant's height after x weeks is y = 0.5x + 3 centimeters. The plant was 3 cm tall at the start.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q4/5

True or false?

In $y = 12x + 30$, the initial value is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 7.RP.A.2.d

FORWARD 2

Calculate the average rate of change over an interval • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2$

What is the average rate of change of f from x = 1 to x = 3?

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q1/5

The table shows the height of a balloon.

What is the average rate of change from 2 seconds to 4 seconds?

Time (s)	0	2	4
Height (m)	0	40	60

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q2**

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q2/5

A car is 30 miles from home at 1 p.m. and 150 miles from home at 3 p.m.

What is its average rate of change in distance?

A. 75 miles per hour

B. 60 miles per hour

C. 50 miles per hour

D. 120 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q3**

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q3/5

True or false?

For f(x) = 2x + 1, the average rate of change over any interval is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q4/5

True or false?

For $f(x) = x^2$, the average rate of change from $x = 0$ to $x = 2$ equals the average rate of change from $x = 2$ to $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q5/5

Solve multistep ratio problems

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Use ratio language to describe two quantities	**6.RP.A.1**
BACKWARD 2	Use a table of equivalent ratios to solve a problem	**6.RP.A.3.a**
FORWARD 1	Solve linear equations with rational coefficients	**8.EE.C.7.b**
FORWARD 2	Create one-variable equations to solve ratio problems	**HSA-CED.A.1**

SET 7 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Trail mix uses peanuts, raisins, and almonds in a ratio of $5 : 3 : 2$. Kai makes $4\frac{1}{2}$ cups of trail mix.

How many more cups of peanuts than almonds does he use?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q1/5

A club has boys and girls in a ratio of 4 : 5. There are 36 members.

How many girls are in the club?

Girls: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q2/5

Ali mixes blue and yellow paint in a ratio of 2 : 3. He makes 15 liters of paint.

How many liters are blue?

A. 10 liters

B. 9 liters

C. 7.5 liters

D. 6 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q3/5

True or false?

On a map, 2 cm represents 15 km. Two towns are 7 cm apart on the map. A car drives between them at 60 km per hour. The trip takes less than 1 hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q4/5

True or false?

Juice concentrate and water are mixed in a ratio of 1 : 4. To make 6 liters of juice, you need 6 liters of water.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q5/5

SET 7 • 7.RP.A.3

BACKWARD 1

Use ratio language to describe two quantities • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

A class has 12 girls and 15 boys.

Write the ratio of boys to girls.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q1/5

For every 3 red tiles, there are 5 blue tiles.

What is the ratio of blue tiles to all tiles?

A. 5 : 8

B. 5 : 3

C. 8 : 5

D. 3 : 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q2/5

A team won 9 games and lost 6 games. There were no ties.

Write the ratio of wins to games played.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q3/5

True or false?

There are 2 cats for every 7 dogs. The ratio of dogs to cats is 7 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q4/5

True or false?

A bowl has 4 apples and 9 oranges and no other fruit. The ratio of apples to all the fruit is 4 : 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 7.RP.A.3

BACKWARD 2

Use a table of equivalent ratios to solve a problem • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Lemonade uses 2 cups of mix for every 5 cups of water.

How many cups of water are needed for 8 cups of mix?

Cups of water: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q1/5

A store sells 3 notebooks for $4.

At the same rate, how much do 12 notebooks cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q2/5

Which pair belongs in a table of ratios equivalent to 3 : 4?

A. 9 : 16

B. 12 : 15

C. 6 : 7

D. 15 : 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q3/5

True or false?

A table of ratios equivalent to 5 : 2 can include 20 : 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q4/5

True or false?

A table of ratios equivalent to 2 : 3 can include 8 : 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 7.RP.A.3

FORWARD 1

Solve linear equations with rational coefficients • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for b.

$5b = 3(b + 6)$

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q1/5

Solve for x.

$\frac{2}{3}x + 4 = 10$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q2/5

Solve for x.

$0.4(x - 5) = 2$

A. $x = 0.5$

B. $x = 3$

C. $x = 10$

D. $x = 5.8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q3/5

True or false?

$x = 12$ is the solution of $\frac{3}{4}x - 2 = 7$.

A. True

B. False

True or false?

$x = 4$ is the solution of $2(x + 3) = 5x - 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q5/5

SET 7 • 7.RP.A.3

FORWARD 2

Create one-variable equations to solve ratio problems • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1

The ratio of red marbles to green marbles is 3 : 5. There are 14 more green marbles than red marbles.

Write and solve an equation to find the number of red marbles.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q1/5

A 60-meter rope is cut into two pieces in a ratio of 2 : 3.

Write and solve an equation to find the length of the longer piece.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q2/5

The ages of two cousins are in a ratio of 4 : 7. The sum of their ages is 66.

Which equation can find x, where the ages are 4x and 7x?

A. $4x \cdot 7x = 66$

B. $4x + 7x = 66$

C. $4 + 7x = 66$

D. $7x - 4x = 66$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q3/5

True or false?

The ratio of a to b is 2 : 5, and a + b = 35. Then a = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q4/5

True or false?

The ratio of a to b is 1 : 3, and $b - a = 12$. Then $a = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q5/5

Find a percent increase or a percent decrease

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find a percent of a quantity	**6.RP.A.3.c**
BACKWARD 2	Find the whole from a part and a percent	**6.RP.A.3.c**
FORWARD 1	Compare quantities written in scientific notation — nearest relate…	**8.EE.A.3**
FORWARD 2	Model a quantity that changes by a constant percent rate	**HSF-LE.A.1.c**

SET 8 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A town's population grew from 2,500 to 3,000.

What is the percent increase?

Percent increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q1/5

The price of a jacket dropped from $80 to $60.

What is the percent decrease?

Percent decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q2/5

A plant grew from 40 cm to 46 cm tall.

What is the percent increase?

A. 15%

B. 13%

C. 87%

D. 6%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q3/5

True or false?

A score that falls from 50 to 40 is a 20% decrease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q4/5

True or false?

A price that rises from $60 to $75 is a 15% increase.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q5/5

SET 8 • 7.RP.A.3

BACKWARD 1

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 30% of 70?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q1/5

What is 15% of $80?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q2/5

What is 4% of 250?

A. 40

B. 100

C. 1

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q3/5

True or false?

75% of 40 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q4/5

True or false?

20% of 45 is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 7.RP.A.3

BACKWARD 2

Find the whole from a part and a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

12 is 40% of what number?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q1/5

9 students in a class walk to school. That is 25% of the class.

How many students are in the class?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q2/5

18 is 60% of what number?

A. 48

B. 12

C. 10.8

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q3/5

True or false?

15 is 50% of 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q4/5

True or false?

8 is 20% of 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 7.RP.A.3

FORWARD 1

Compare quantities written in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.3

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

City A has about 3×10^6 people. Town B has about 6×10^4 people.

How many times as many people live in City A as in Town B?

Times as many: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q1/5 • NEAREST RELATED

Write 4,500,000 in scientific notation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which number is the greatest?

A. 9×10^{6}

B. 8×10^{5}

C. 2×10^{7}

D. 1×10^{6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

8×10^8 is 100 times as large as 8×10^6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

3×10^{-2} is greater than 3×10^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 8 • 7.RP.A.3

FORWARD 2

Model a quantity that changes by a constant percent rate • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

A town of 10,000 people grows by 5% each year.

What is the population after 2 years?

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q1/5

Which situation shows growth by a constant percent rate?

A. A plant grows 2 cm each week.

B. An account earns 3% of its current balance each year.

C. A worker earns $15 more each year.

D. A tank fills 4 gallons each minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q2**

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q2/5

A car worth $20,000 loses 10% of its value each year.

What is it worth after 2 years?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q3/5

True or false?

A quantity that is multiplied by 1.08 each year grows by 8% per year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q4/5

True or false?

A quantity that decreases by 10% each year loses the same number of units each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q5/5

Find a price after a markup or a markdown (discount)

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Multiply decimals	**6.NS.B.3**
BACKWARD 2	Find a percent of an amount of money	**6.RP.A.3.c**
FORWARD 1	Solve percent equations with the variable on both sides	**8.EE.C.7.b**
FORWARD 2	Interpret the parts of an expression in context	**HSA-SSE.A.1.a**

SET 9 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A store pays $25 for a backpack and marks the price up by 40%.

What is the selling price?

Selling price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q1/5

A $120 bike is marked down 25%.

What is the sale price?

Sale price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q2/5

A store buys a lamp for $30 and marks the price up by 60%.

What is the selling price?

A. $36

B. $18

C. $90

D. $48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q3/5

True or false?

A pair of shoes costs $64. At 15% off, the sale price is $54.40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q4/5

True or false?

A $50 jacket is marked up 20%. Then the new price is marked down 20%. The final price is $50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q5/5

SET 9 • 7.RP.A.3

BACKWARD 1

Multiply decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

0.15×80

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q1/5

Multiply.

1.08×25

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q2/5

Multiply.

0.06×45

A. 270

B. 2.7

C. 0.27

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$0.2 \times 36.5 = 7.3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$1.5 \times 4.2 = 6.2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 7.RP.A.3

BACKWARD 2

Find a percent of an amount of money • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 10% of $56?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q1/5

What is 25% of $48?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q2/5

What is 8% of $50?

A. $0.40

B. $4

C. $8

D. $40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q3/5

True or false?

20% of $35 is $7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q4/5

True or false?

5% of $60 is $30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 7.RP.A.3

FORWARD 1

Solve percent equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A sells a game for 20% off its price p plus a $6 shipping fee. Store B sells it for the full price p with free shipping.

Solve 0.8p + 6 = p to find the price at which both stores charge the same.

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q1/5

Solve for p.

$0.75p + 12 = 0.9p$

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q2/5

Solve for x.

$1.08(x - 5) = 0.08x + 50$

A. $x = 45$

B. $x = 50.6$

C. $x = 55.4$

D. $x = 55$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q3/5

True or false?

$p = 40$ is the solution of $0.85p + 9 = p + 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q4/5

True or false?

$p = 60$ is the solution of $1.1(p + 10) = p + 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q5/5

SET 9 • 7.RP.A.3

FORWARD 2

Interpret the parts of an expression in context • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A price p is discounted 10%. Then 7% sales tax is added. The final cost is 1.07(0.9p).

What does 0.9p represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q1/5

The final cost of an item is 1.07(0.9p).

What does the factor 1.07 represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q2/5

A store's selling price is p + 0.4p, where p is what the store paid.

What does 0.4p represent?

A. The selling price

B. The amount of a 40% discount

C. The price the store paid

D. The amount of the 40% markup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q3/5

True or false?

The expression 0.85p gives the price after a 15% discount on price p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q4/5

True or false?

The expression 1.2p gives the price after a 20% discount on price p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q5/5

Find simple interest

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Evaluate a formula	**6.EE.A.2.c**
BACKWARD 2	Write a percent as a rate per 100 and as a decimal	**6.RP.A.3.c**
FORWARD 1	Model a growing balance with a linear function	**8.F.B.4**
FORWARD 2	Compare simple interest with compound interest	**HSF-LE.A.1.a**

SET 10 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Ella deposits $800 in an account that earns 3% simple interest per year.

How much interest does she earn in 4 years?

Interest: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q1/5

Leo borrows $2,400 at 5% simple interest per year.

How much interest does he owe after 3 years?

Interest: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q2/5

An account earns 4% simple interest per year.

How many years does it take $500 to earn $100 in interest?

A. 4 years

B. 5 years

C. 2 years

D. 20 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q3/5

True or false?

$1,500 borrowed at 5% simple interest per year for 2 years has $150 in interest.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q4/5

True or false?

$600 earns 2.5% simple interest per year. The balance after 3 years is $660.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q5/5

SET 10 • 7.RP.A.3

BACKWARD 1

Evaluate a formula • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use I = prt.

Find I when p = 200, r = 0.05, and t = 3.

I = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q1/5

Use A = lw.

Find A when l = 12 and w = 7.5.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q2/5

Use d = rt.

Find d when r = 55 and t = 4.

A. 13.75

B. 220

C. 51

D. 59

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q3/5

True or false?

For P = 2l + 2w, when l = 6 and w = 4, P = 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q4/5

True or false?

For C = 5n + 3, when n = 4, C = 32.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q5/5

SET 10 • 7.RP.A.3

BACKWARD 2

Write a percent as a rate per 100 and as a decimal • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write 6% as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q1/5

Write 2.5% as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q2/5

Which decimal is equal to 45%?

A. 4.5

B. 0.45

C. 45

D. 0.045

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q3/5

True or false?

$3\% = \frac{3}{100} = 0.03$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q4/5

True or false?

$8\% = 0.8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 7.RP.A.3

FORWARD 1

Model a growing balance with a linear function • Grade 8 • Functions • 8.F.B.4

An account starts with $500 and earns $20 in simple interest each year.

Write a function for the balance B after t years.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q1/5

A $1,000 loan charges $45 in simple interest each year.

Write a function for the amount A owed after t years.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q2/5

A loan balance after t years is $B = 1{,}200 + 36t$.

What is the rate of change?

A. $36 per year

B. $1,236 per year

C. $1,200 per year

D. 3% per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q3/5

True or false?

For B = 20t + 500, the balance after 3 years is $560.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q4/5

True or false?

For A = 45t + 1,000, the initial value is 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q5/5

SET 10 • 7.RP.A.3

FORWARD 2

Compare simple interest with compound interest • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

$1,000 earns 5% simple interest per year.

What is the balance after 2 years?

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q1/5

$1,000 earns 5% interest compounded once each year.

What is the balance after 2 years?

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q2/5

Which kind of interest makes a balance grow by equal differences over equal time intervals?

A. Both kinds

B. Compound interest

C. Simple interest

D. Neither kind

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q3**

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q3/5

True or false?

With compound interest, a balance grows by equal factors over equal time intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q4/5

True or false?

After 1 year, $500 at 4% compound interest (compounded yearly) earns more than $500 at 4% simple interest.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q5/5

SET 62 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Find a tip and the total cost with the tip

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find 5%, 10%, and 20% of an amount of money	**6.RP.A.3.c**
BACKWARD 2	Add amounts of money	**6.NS.B.3**
FORWARD 1	Solve tip equations with the variable on both sides	**8.EE.C.7.b**
FORWARD 2	Write a function for a total that combines a tip and a fixed fee	**HSF-BF.A.1**

SET 62 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A meal costs $45. Kai leaves a 20% tip.

What is the total cost of the meal with the tip?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q1/5

A haircut costs $28. Mia leaves a 15% tip.

How much is the tip?

Tip: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q2/5

A restaurant bill is $64. A group adds an 18% tip.

What is the total?

A. $66.88

B. $82

C. $11.52

D. $75.52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q3/5

True or false?

A 20% tip on a $36.50 meal is $7.30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q4/5

True or false?

A 15% tip on a $40 meal makes the total $55.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q5/5

SET 62 • 7.RP.A.3

BACKWARD 1

Find 5%, 10%, and 20% of an amount of money • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 10% of $42?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q1/5

What is 5% of $42?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q2/5

What is 20% of $55?

A. $5.50

B. $1.10

C. $11

D. $20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q3/5

True or false?

15% of $20 is $3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q4/5

True or false?

20% of $30 is $3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q5/5

SET 62 • 7.RP.A.3

BACKWARD 2

Add amounts of money • Grade 6 • The Number System • 6.NS.B.3

Add.

$45.80 + $9.16

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q1/5

Add.

$23.45 + $4.69

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q2/5

Add.

$18.75 + $3.75

A. $15.00

B. $22.40

C. $22.50

D. $21.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q3/5

True or false?

$9.99 + $1.50 = $11.49

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q4/5

True or false?

$12.60 + $2.52 = $14.12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • 7.RP.A.3

FORWARD 1

Solve tip equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Two friends split a meal that costs m dollars. Ana pays half of m plus a $5 tip. Ben pays 40% of m plus a $9 tip. They pay the same amount.

Solve 0.5m + 5 = 0.4m + 9.

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q1/5

Solve for m.

$1.2(m + 10) = 1.15m + 14.5$

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q2/5

Solve for x.

$0.15x + 0.85x = 2(x - 20)$

A. $x = -40$

B. $x = 40$

C. $x = 80$

D. $x = 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q3/5

True or false?

$x = 30$ is the solution of $1.2x - 6 = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q4/5

True or false?

$x = 25$ is the solution of $0.18x + 2 = 0.1x + 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q5/5

SET 62 • 7.RP.A.3

FORWARD 2

Write a function for a total that combines a tip and a fixed fee • High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1

A delivery app adds a 15% tip and a $4 fee to the price m of a meal.

Write a function T(m) for the total.

T(m) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q1/5

A delivery total is T(m) = 1.15m + 4, where m is the meal price in dollars.

Find T(30).

T(30) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q2

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q2/5

A 20% tip is added to a meal price m. Then 8% tax is added to that amount.

Which function gives the total?

A. $T(m) = 0.2m + 0.08m$

B. $T(m) = m + 1.2 + 0.08$

C. $T(m) = 1.28m$

D. $T(m) = 1.08(1.2m)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q3/5

True or false?

Adding a fixed $5 fee to a meal price m with a 20% tip gives the function T(m) = 1.2m + 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q4/5

True or false?

For T(m) = 1.15m + 4, doubling the meal price doubles the total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q5/5

SET 63 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Find sales tax and the final cost with tax

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find a percent of an amount of money	**6.RP.A.3.c**
BACKWARD 2	Multiply by a decimal	**6.NS.B.3**
FORWARD 1	Compare tax and flat charges with equations that have the variab…	**8.EE.C.7.b**
FORWARD 2	Interpret a tax expression by seeing its parts as single quantities	**HSA-SSE.A.1.b**

SET 63 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A jacket costs $80. The sales tax is 6%.

How much is the tax?

Tax: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q1/5

A book costs $24.50. The sales tax is 8%.

What is the total cost?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q2/5

A TV costs $350. The sales tax is 7.5%.

What is the final cost?

A. $376.25

B. $357.50

C. $425

D. $26.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q3/5

True or false?

A $12 item with 5% sales tax costs $12.60 in all.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q4/5

True or false?

A shirt costs $35. With 6% sales tax, the total is $37.20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q5/5

SET 63 • 7.RP.A.3

BACKWARD 1

Find a percent of an amount of money • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 6% of $45?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q1/5

What is 8% of $15?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q2/5

What is 5% of $64?

A. $6.40

B. $32

C. $3.20

D. $0.32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q3/5

True or false?

7% of $200 is $14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q4/5

True or false?

4% of $75 is $30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q5/5

SET 63 • 7.RP.A.3

BACKWARD 2

Multiply by a decimal • Grade 6 • The Number System • 6.NS.B.3

Multiply.

1.06×50

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q1/5

Multiply.

0.08×32.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q2/5

Multiply.

1.05×18

A. 19.8

B. 18.9

C. 18.05

D. 1.89

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q3/5

True or false?

$0.07 \times 300 = 21$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q4/5

True or false?

$1.08 \times 25 = 25.08$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • 7.RP.A.3

FORWARD 1

Compare tax and flat charges with equations that have the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A charges the price p plus 8% tax. Store B charges the price p plus a flat $6 with no tax.

Solve 1.08p = p + 6 to find when they charge the same.

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q1/5

Solve for p.

$1.05(p + 20) = p + 24$

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q2/5

Solve for x.

$1.1x - 4 = 0.9x + 6$

A. $x = 5$

B. $x = 20$

C. $x = 10$

D. $x = 50$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q3/5

True or false?

$p = 100$ is the solution of $1.06p = p + 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q4/5

True or false?

$p = 40$ is the solution of $1.07p - 2 = p + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q5/5

SET 63 • 7.RP.A.3

FORWARD 2

Interpret a tax expression by seeing its parts as single quantities • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

n items each cost p dollars. With 6% sales tax, the total is 1.06(np).

What does np represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q1/5

A total with tax is (1 + r)p, where r is the tax rate as a decimal and p is the price.

Find the total when r = 0.075 and p = 40.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q2/5

An order costs p dollars plus s dollars of shipping, and 8% tax is charged on both. The total is 1.08(p + s).

Which part is the amount being taxed?

A. p + s

B. 1.08

C. s

D. p

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F2-Q3**

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q3/5

True or false?

1.06p and p + 0.06p are equivalent expressions for a price p with 6% tax.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q4/5

True or false?

In 1.08(p + 5), the tax is charged only on p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q5/5

SET 64 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Find a commission and pay that includes a commission

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find a percent of a quantity	**6.RP.A.3.c**
BACKWARD 2	Find what percent a part is of a whole	**6.RP.A.3.c**
FORWARD 1	Model pay with a base salary plus a commission	**8.F.B.4**
FORWARD 2	Write and solve an inequality to meet an earnings goal	**HSA-CED.A.3**

SET 64 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A salesperson earns a 6% commission. She sells $2,450 of products.

What is her commission?

Commission: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q1/5

A real estate agent earns a 3% commission on a $280,000 home sale.

What is the commission?

Commission: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q2/5

Jo earns $400 per week plus a 5% commission on her sales. She sells $3,600 in one week.

What is her pay for the week?

A. $4,000

B. $580

C. $180

D. $418

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q3/5

True or false?

A 12% commission on $750 in sales is $90.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q4/5

True or false?

An agent earns a $240 commission on $4,000 in sales. The commission rate is 4%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q5/5

SET 64 • 7.RP.A.3

BACKWARD 1

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 3% of 2,000?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q1/5

What is 12% of 150?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q2/5

What is 5% of 3,600?

A. 720

B. 18

C. 1,800

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q3/5

True or false?

8% of 500 is 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q4/5

True or false?

2% of 900 is 180.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q5/5

SET 64 • 7.RP.A.3

BACKWARD 2

Find what percent a part is of a whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

240 is what percent of 4,000?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q1/5

What percent of 800 is 32?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q2/5

45 out of 300 is what percent?

A. 30%

B. 6.67%

C. 15%

D. 45%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q3/5

True or false?

90 is 12% of 750.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q4/5

True or false?

25 is 5% of 250.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • 7.RP.A.3

FORWARD 1

Model pay with a base salary plus a commission • Grade 8 • Functions • 8.F.B.4

A job pays $350 per week plus 4% of sales s.

Write a function for the weekly pay P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q1/5

Job A pays $500 per week plus 2% of sales. Job B pays $300 per week plus 6% of sales.

For what weekly sales do the two jobs pay the same?

Sales: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q2/5

A line through (0, 200) and (10,000, 900) models weekly pay (y) against sales (x).

What is the commission rate?

A. 7%

B. 9%

C. 70%

D. 0.07%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q3/5

True or false?

For P = 0.05s + 250, the pay with no sales is $250.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q4/5

True or false?

For P = 0.03s + 400, each $100 in sales adds $30 in pay.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q5/5

SET 64 • 7.RP.A.3

FORWARD 2

Write and solve an inequality to meet an earnings goal • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3

Pay is $300 plus 5% of sales s.

Write an inequality for earning at least $800 in a week.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q1/5

Solve $300 + 0.05s \geq 800$, where s is weekly sales in dollars. Explain what the solution means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q2/5

Pay is 250 + 0.08s dollars for sales of s dollars.

Which amount of sales meets the goal
$250 + 0.08s \geq 650$?

A. $4,500

B. $5,500

C. $4,000

D. $3,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q3/5

True or false?

With pay of 200 + 0.1s dollars, sales of $6,000 meet a goal of at least $750.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q4/5

True or false?

With pay of 400 + 0.04s dollars, sales of $5,000 meet a goal of at least $650.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q5/5

SET 65 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Find a percent fee and the total with the fee

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Write a percent as a decimal	**6.RP.A.3.c**
BACKWARD 2	Multiply decimals	**6.NS.B.3**
FORWARD 1	Compare fee plans with equations that have the variable on both s…	**8.EE.C.7.b**
FORWARD 2	Write a function for a total that includes a percent fee and a fixed…	**HSF-BF.A.1**

SET 65 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A ticket website charges a 2.5% fee on a $600 order.

What is the fee?

A. $25

B. $1.50

C. $15

D. $150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q1/5

A bank charges a 3% fee to transfer $1,250.

What is the fee?

Fee: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q2/5

A concert ticket costs $85 plus a 12% service fee.

What is the total cost?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q3/5

True or false?

A 1.5% card fee on a $400 payment is $6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q4/5

True or false?

A 4% fee on $250 is $1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q5/5

SET 65 • 7.RP.A.3

BACKWARD 1

Write a percent as a decimal • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write 1.5% as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q1/5

Write 12% as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q2/5

Which decimal is equal to 3%?

A. 3.0

B. 0.3

C. 0.003

D. 0.03

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q3/5

True or false?

0.5% = 0.005

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q4/5

True or false?

$0.4\% = 0.4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q5/5

SET 65 • 7.RP.A.3

BACKWARD 2

Multiply decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$0.03 \times 1{,}250$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q1/5

Multiply.

1.12×85

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q2/5

Multiply.

0.015×400

A. 6

B. 60

C. 4.15

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q3/5

True or false?

$0.025 \times 80 = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q4/5

True or false?

$0.12 \times 50 = 60$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • 7.RP.A.3

FORWARD 1

Compare fee plans with equations that have the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Site A charges a $3 fee plus 1% of the amount a. Site B charges 2.5% of the amount a.

Solve 3 + 0.01a = 0.025a to find when the fees are equal.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q1/5

Solve for a.

$0.04(a + 50) = 0.03a + 5$

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q2/5

Solve for x.

$1.12x = x + 18$

A. $x = 18$

B. $x = 16.07$

C. $x = 1.5$

D. $x = 150$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q3/5

True or false?

$a = 500$ is the solution of $2 + 0.02a = 0.024a$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q4/5

True or false?

$a = 200$ is the solution of $5 + 0.03a = 0.08a$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q5/5

SET 65 • 7.RP.A.3

FORWARD 2

Write a function for a total that includes a percent fee and a fixed charge • High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1

A ticket site adds a 10% fee and a $2.50 processing charge to a ticket price t.

Write a function T(t) for the total.

T(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q1/5

A ticket total is $T(t) = 1.1t + 2.5$.

What ticket price t gives a total of $68.50?

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q2

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q2/5

A bank fee is 2% of the amount, but never less than $5.

What is the fee on $150?

A. $3

B. $5

C. $7

D. $2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q3/5

True or false?

$T(t) = 1.05t + 1$ and $T(t) = t + 0.05t + 1$ are the same function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q4/5

True or false?

For T(t) = 1.08t + 3, the fee part of the total is 1.08t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q5/5

SET 66 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Solve multistep percent problems that combine a discount with tax or a tip

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find a percent of an amount of money	**6.RP.A.3.c**
BACKWARD 2	Subtract amounts of money	**6.NS.B.3**
FORWARD 1	Solve multistep percent equations with the variable on both sides	**8.EE.C.7.b**
FORWARD 2	Construct an exponential function for repeated percent changes	**HSF-LE.A.2**

SET 66 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A bike costs $120. It is on sale for 25% off. Then 8% sales tax is added to the sale price.

What is the final cost?

Final cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q1/5

A meal costs $60. A $5 coupon is taken off the bill. Then a 20% tip on the $60 price is added.

What is the final amount paid?

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q2/5

A $250 phone is 20% off. Then 6% sales tax is added to the sale price.

What is the final cost?

A. $200

B. $215

C. $212

D. $265

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q3/5

True or false?

A $40 shirt is marked down 10%. Then the sale price is marked down another 10%. The final price is $32.40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q4/5

True or false?

A 30% discount followed by 8% tax on the sale price is the same as a 22% discount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q5/5

SET 66 • 7.RP.A.3

BACKWARD 1

Find a percent of an amount of money • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 25% of $120?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q1/5

What is 80% of $45?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q2/5

What is 6% of $200?

A. $12

B. $120

C. $6

D. $1.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q3/5

True or false?

20% of $250 is $50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q4/5

True or false?

30% of $90 is $30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q5/5

SET 66 • 7.RP.A.3

BACKWARD 2

Subtract amounts of money • Grade 6 • The Number System • 6.NS.B.3

Subtract.

$84.50 − $12.68

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q1/5

Subtract.

$65.00 − $9.75

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q2/5

Subtract.

$100 − $37.45

A. $137.45

B. $62.45

C. $63.55

D. $62.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q3/5

True or false?

$50.00 − $12.50 = $37.50

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q4/5

True or false?

$75.20 − $8.40 = $67.20

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • 7.RP.A.3

FORWARD 1

Solve multistep percent equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A sells an item for 20% off its price p plus $10 shipping. Store B sells it for 10% off with free shipping.

Solve 0.8p + 10 = 0.9p to find when the stores charge the same.

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q1/5

Solve for p.

$1.08(0.75p) = 0.75p + 4.5$

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q2/5

Solve for p.

$1.06(0.9p) = 0.9p + 5.4$

A. p = 90

B. p = 54

C. p = 100

D. p = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q3/5

True or false?

$p = 200$ is the solution of $0.7p + 15 = 0.75p + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q4/5

True or false?

$p = 50$ is the solution of $1.1(0.8p) = p - 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q5/5

SET 66 • 7.RP.A.3

FORWARD 2

Construct an exponential function for repeated percent changes • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

A $200 price is marked down 10% each week.

Write a function P(w) for the price after w weeks.

P(w) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q1/5

The price after w weeks is P(w) = 200(0.9)^w dollars.

Find the price after 2 weeks.

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q2/5

A $500 item increases in price by 4% each year.

Which function gives its price after t years?

A. P(t) = 500 + 1.04t

B. P(t) = 500(1.04)^t

C. P(t) = 500 + 20t

D. P(t) = 500(0.04)^t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-F2-Q3**

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q3/5

True or false?

A 20% discount followed by a 10% discount multiplies the price by 0.72.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q4/5

True or false?

P(t) = 80(0.95)^t models a price that increases 95% each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q5/5

Find the percent error of an estimate or a measurement

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find the absolute value of a number	**6.NS.C.7.c**
BACKWARD 2	Write a fraction as a percent	**6.RP.A.3.c**
FORWARD 1	Compute with numbers in scientific notation — nearest related (n…	**8.EE.A.4**
FORWARD 2	Choose a level of accuracy that fits a measurement	**HSN-Q.A.3**

SET 11 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Ty estimates that 50 people are at a meeting. The actual number is 40.

What is the percent error?

Percent error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q1/5

A scale shows 4.8 kg for a bag whose actual mass is 5 kg.

What is the percent error?

Percent error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q2/5

A student measures a line that is actually 20 cm long as 19 cm.

What is the percent error?

A. 95%

B. 5%

C. 5.3%

D. 1%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q3/5

True or false?

A guess of 90 for an actual value of 100 has a percent error of 10%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q4/5

True or false?

To find percent error, you divide the size of the error by the estimate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q5/5

SET 11 • 7.RP.A.3

BACKWARD 1

Find the absolute value of a number • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$|-12|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q1/5

Find the value.

$|-4.5|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q2/5

Which number has the greatest absolute value?

A. -3

B. 8

C. -9

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q3/5

True or false?

$|-15| = 15$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q4/5

True or false?

$|-2| < |1|$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 7.RP.A.3

BACKWARD 2

Write a fraction as a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write $\frac{3}{20}$ as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q1/5

Write $\frac{7}{25}$ as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q2/5

Which percent is equal to $\frac{9}{12}$?

A. 75%

B. 80%

C. 9%

D. 12%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{8} = 12.5\%$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q4/5

True or false?

$\frac{2}{5} = 25\%$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 7.RP.A.3

FORWARD 1

Compute with numbers in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Multiply. Write the answer in scientific notation.

$(3 \times 10^4) \times (2 \times 10^3)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q1/5 • NEAREST RELATED

Divide. Write the answer in scientific notation.

$(8 \times 10^{9}) \div (4 \times 10^{3})$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q2/5 • NEAREST RELATED

Add.

$(2.5 \times 10^5) + (4 \times 10^4)$

A. 6.5×10^5

B. 2.9×10^5

C. 2.9×10^4

D. 6.5×10^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

$(5 \times 10^3) \times (3 \times 10^2) = 1.5 \times 10^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

$(6 \times 10^8) \div (2 \times 10^2) = 3 \times 10^4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 11 • 7.RP.A.3

FORWARD 2

Choose a level of accuracy that fits a measurement • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

A ruler is marked in millimeters. A length is read as 12.4 cm.

To what place value is this measurement given?

Place: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q1/5

A scale measures mass to the nearest 0.1 kg.

Which reading is reported to an accuracy that fits this scale?

A. 3 kg

B. 3.4 kg

C. 3.4172 kg

D. 3.41 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q2/5

The actual length of a track is 400 m. A measurement of it is off by at most 2 m.

What is the greatest possible percent error?

Percent error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q3/5

True or false?

A length measured with a meter stick marked only in centimeters should not be reported as 1.2347 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q4/5

True or false?

Writing more decimal places always makes a measurement more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q5/5

SET 12 • 7.NS.A.1.a • THE NUMBER SYSTEM

Describe situations in which opposite quantities combine to make 0

MAIN	Grade 7 main question	**7.NS.A.1.a**
BACKWARD 1	Use positive and negative numbers to represent quantities	**6.NS.C.5**
BACKWARD 2	Recognize opposite numbers	**6.NS.C.6.a**
FORWARD 1	Use opposite exponents with the properties of integer exponents…	**8.EE.A.1**
FORWARD 2	Justify each step in solving an equation and find steps that chang…	**HSA-REI.A.1**

SET 12 • 7.NS.A.1.a

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.a

Ari has $15 in his wallet. He spends all $15.

Write an addition expression for this situation and find its value.

Expression and value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q1/5

Which situation combines to make 0?

A. Gaining 5 points, then gaining 5 more points

B. A temperature drop of 6°, then another drop of 6°

C. Climbing up 8 floors, then down 8 floors

D. Walking 3 miles east, then 2 miles west

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q2/5

A diver is 12 feet below the surface of the water (−12 feet).

How many feet must she rise to be at the surface (0 feet)?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q3/5

True or false?

A gain of 9 yards followed by a loss of 9 yards is a total change of 0 yards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q4/5

True or false?

A deposit of $40 followed by a withdrawal of $30 leaves an account balance unchanged.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q5/5

SET 12 • 7.NS.A.1.a

BACKWARD 1

Use positive and negative numbers to represent quantities • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 20 feet below sea level.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q1/5

Write an integer for a gain of 7 yards.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q2/5

Which integer represents a withdrawal of $35?

A. 35

B. −35

C. −3.5

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q3/5

True or false?

A temperature of 4 degrees below zero can be written as $-4°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q4/5

True or false?

The integer -10 represents a deposit of \$10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 7.NS.A.1.a

BACKWARD 2

Recognize opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 8?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q1/5

What is the opposite of −13?

Opposite: _____

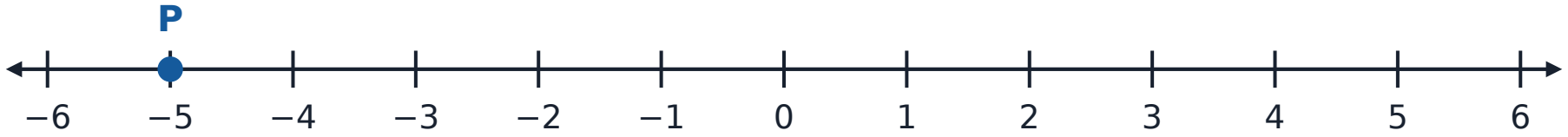
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q2/5

Point P is at −5.

Which number is the same distance from 0 on the other side of 0?



A. 5 **B.** −10 **C.** 10 **D.** 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q3/5

True or false?

The opposite of the opposite of 6 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q4/5

True or false?

0 and 1 are opposites.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 7.NS.A.1.a

FORWARD 1

Use opposite exponents with the properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Find the value.

$5^{3} \times 5^{-3}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q1/5 • NEAREST RELATED

Write as a single power. Then find its value.

$2^{4} \times 2^{-6}$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which expression is equal to 1?

A. $7^2 \times 7^2$

B. $7^{-2} \times 7^{-2}$

C. $7^{-2} + 7^2$

D. $7^{-2} \times 7^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

$10^{-2} = \frac{1}{100}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

$3^{-2} = -9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 12 • 7.NS.A.1.a

FORWARD 2

Justify each step in solving an equation and find steps that change the solution • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Solve $\frac{1}{2}(x - 6) = 4 + x$. Write each step and the reason for it.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q1/5

Lia solved $3(x - 4) = 2x + 5$ like this:

Step 1: $3x - 4 = 2x + 5$

Step 2: $x = 9$

Which step is wrong? Give the correct solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q2/5

Which step can change the solutions of an equation?

A. Dividing both sides of $-4x = 12$ by -4

B. Multiplying both sides of $\frac{x}{3} = 5$ by 3

C. Multiplying both sides of $2x = 6$ by 0

D. Adding -7 to both sides of $x + 7 = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q3/5

True or false?

Adding 2x to both sides of $5 - 2x = x + 11$ gives $5 = 3x + 11$, which has the same solution as the original equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q4/5

True or false?

Squaring both sides of $x = 3$ gives $x^2 = 9$, which has the same solutions as $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q5/5

SET 13 • 7.NS.A.1.b • THE NUMBER SYSTEM

Add rational numbers on a number line

MAIN	Grade 7 main question	**7.NS.A.1.b**
BACKWARD 1	Find and position integers on a number line	**6.NS.C.6.c**
BACKWARD 2	Add decimals	**5.NBT.B.7**
FORWARD 1	Locate irrational numbers on a number line	**8.NS.A.2**
FORWARD 2	Decide whether a sum is rational or irrational	**HSN-RN.B.3**

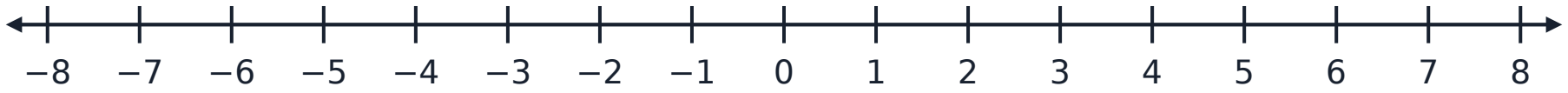
SET 13 • 7.NS.A.1.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.b

Use the number line to add.

$-3 + 7$



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q1/5

Use the number line to add.

$-\frac{3}{4} + 1\frac{1}{2}$



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q2/5

Start at 2 on a number line and move 5 units to the left.

Which sum does this show?

A. $-5 + (-2)$

B. $-2 + 5$

C. $2 + 5$

D. $2 + (-5)$

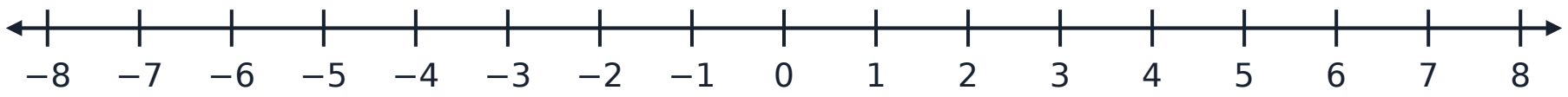
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q3/5

Show $-2 + 6$ on the number line. Start at -2 and draw an arrow 6 units long.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q4/5

True or false?

$$-\frac{3}{4} + \frac{5}{6} = \frac{1}{12}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

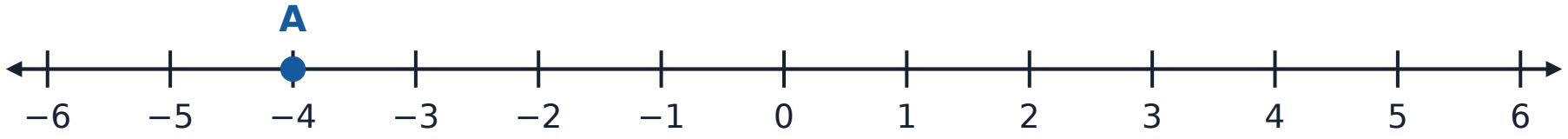
Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q5/5

SET 13 • 7.NS.A.1.b

BACKWARD 1

Find and position integers on a number line • Grade 6 • The Number System • 6.NS.C.6.c

What number is at point A?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q1/5

Plot −3 and 5 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q2/5

Which number is farthest to the left on a number line?

A. -2

B. 0

C. 3

D. -7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q3/5

True or false?

-1 is to the right of -5 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q4/5

True or false?

-6 is between -3 and 0 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 7.NS.A.1.b

BACKWARD 2

Add decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Add.

3.75 + 2.5

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q1/5

Add.

0.8 + 4.65

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q2/5

Add.

12.4 + 3.06

A. 46.4

B. 15.10

C. 12.71

D. 15.46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q3/5

True or false?

$2.7 + 1.3 = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q4/5

True or false?

$5.6 + 0.04 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 7.NS.A.1.b

FORWARD 1

Locate irrational numbers on a number line • Grade 8 • The Number System • 8.NS.A.2

Between which two whole numbers is $\sqrt{20}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q1/5

Plot an approximate location of $\sqrt{2}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q2/5

Which is the best approximation of $\sqrt{50}$ to the nearest tenth?

A. 25

B. 7.1

C. 7.5

D. 7.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q3/5

True or false?

$\sqrt{10}$ is between 3 and 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q4/5

True or false?

π is less than 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q5/5

SET 13 • 7.NS.A.1.b

FORWARD 2

Decide whether a sum is rational or irrational • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3

Is 3 + √2 rational or irrational?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q1/5

Is $\frac{1}{2} + \frac{3}{4}$ rational or irrational? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q2/5

Which sum is irrational?

A. $\sqrt{9} + 4$

B. $0.5 + \frac{1}{2}$

C. $-2 + 2$

D. $\sqrt{3} + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q3/5

True or false?

The sum of two rational numbers is always rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q4/5

True or false?

The sum of a rational number and an irrational number can be rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q5/5

SET 14 • 7.NS.A.1.b • THE NUMBER SYSTEM

Use additive inverses: a number plus its opposite is 0

MAIN	Grade 7 main question	**7.NS.A.1.b**
BACKWARD 1	Opposites are the same distance from 0	**6.NS.C.6.a**
BACKWARD 2	Interpret absolute value as distance from 0	**6.NS.C.7.c**
FORWARD 1	Tell whether a linear equation has one, no, or infinitely many solut…	**8.EE.C.7.a**
FORWARD 2	Solve linear equations with letter coefficients or fraction coefficien…	**HSA-REI.B.3**

SET 14 • 7.NS.A.1.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.b

What number added to −9.5 gives 0?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q1/5

Fill in the blank.

$\frac{3}{4} + ___ = 0$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q2/5

Which sum is equal to 0?

A. $-2.6 + (-2.6)$

B. $-2.6 + 2.6$

C. $2.6 + 0$

D. $-2.6 + 6.2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q3/5

True or false?

For any rational number p, $-p + p = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q4/5

True or false?

The additive inverse of -7 is -7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q5/5

SET 14 • 7.NS.A.1.b

BACKWARD 1

Opposites are the same distance from 0 • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 2.5?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q1/5

Which pair of numbers are opposites?

A. -11 and 11

B. -11 and -11

C. 11 and $\frac{1}{11}$

D. 1 and -11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q2/5

How far is −9 from 0 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q3/5

True or false?

$-\frac{2}{3}$ and $\frac{2}{3}$ are the same distance from 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q4/5

True or false?

The opposite of 0 is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 7.NS.A.1.b

BACKWARD 2

Interpret absolute value as distance from 0 • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$|-18|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q1/5

A diver is at −30 feet.

How far is she from the surface (0 feet)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q2/5

Which number has an absolute value of 6?

A. $\frac{1}{6}$

B. -6

C. $-\frac{1}{6}$

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q3/5

True or false?

$|-3.5| = |3.5|$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q4/5

True or false?

$|-20| = -20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 7.NS.A.1.b

FORWARD 1

Tell whether a linear equation has one, no, or infinitely many solutions • Grade 8 • Expressions & Equations • 8.EE.C.7.a

How many solutions does this equation have?

$3x + 5 = 3x - 2$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q1/5

How many solutions does this equation have?

$2(x + 4) = 2x + 8$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q2/5

Which equation has exactly one solution?

A. $4x + 1 = 2x + 7$

B. $2(2x) = 4x$

C. $4x + 1 = 4x + 7$

D. $4x + 1 = 4x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q3/5

True or false?

$x + 6 = x - 6$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q4/5

True or false?

$5x - 5x = 0$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q5/5

SET 14 • 7.NS.A.1.b

FORWARD 2

Solve linear equations with letter coefficients or fraction coefficients • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve for x. Assume $a \neq 0$.

$ax + 5 = 2$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q1/5

Solve for x. Assume $a \neq b$.

$ax + c = bx$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q2/5

Solve for x.

$\frac{2}{3}(x - 4) = \frac{1}{4}x + 1$

A. $x = \frac{44}{5}$

B. $x = 12$

C. $x = \frac{20}{3}$

D. $x = \frac{11}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q3/5

True or false?

If $k \neq 0$, the solution of $k(x - 2) = 3k$ is $x = 5$.

A. True

B. False

True or false?

The solution of $mx + 4 = 4$ is $x = 1$ for every nonzero value of m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q5/5

SET 15 • 7.NS.A.1.b • THE NUMBER SYSTEM

Interpret sums of rational numbers in real-world situations

MAIN	Grade 7 main question	**7.NS.A.1.b**
BACKWARD 1	Use signed numbers in a real-world situation	**6.NS.C.5**
BACKWARD 2	Compare signed numbers in a real-world situation	**6.NS.C.7.b**
FORWARD 1	Model a changing quantity with a linear function	**8.F.B.4**
FORWARD 2	Evaluate a function and interpret the value in context	**HSF-IF.A.2**

SET 15 • 7.NS.A.1.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.b

The temperature was −5°F at dawn. It rose 12°F by noon.

What was the temperature at noon?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q1/5

A stock's price changed by $-1\frac{1}{4}$ dollars on Monday and by $+2\frac{3}{4}$ dollars on Tuesday.

What was the total change over the two days?

Total change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q2/5

Jo's account balance is −$18. She deposits $25.

Which sum gives her new balance?

A. $-18 + (-25) = -43$

B. $18 + 25 = 43$

C. $-18 + 25 = 7$

D. $18 + (-25) = -7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q3/5

True or false?

A pond's water level changed by $-\frac{2}{3}$ foot one day and by $+\frac{1}{2}$ foot the next day. The total change was $-\frac{1}{6}$ foot.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q4/5

True or false?

A hiker 350 feet below sea level climbs 400 feet. She ends 50 feet below sea level.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q5/5

SET 15 • 7.NS.A.1.b

BACKWARD 1

Use signed numbers in a real-world situation • Grade 6 • The Number System • 6.NS.C.5

A stock's price changed by $-\$3.25$.

Was this a gain or a loss, and of how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q1/5

Which number describes 15 meters above sea level?

A. $\frac{1}{15}$

B. 15

C. 0

D. -15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q2/5

Write a number for a temperature of 8 degrees below zero.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q3/5

True or false?

In a bank account, a balance of −$50 can mean $50 is owed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q4/5

True or false?

A temperature of $-2°$ is warmer than $0°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • 7.NS.A.1.b

BACKWARD 2

Compare signed numbers in a real-world situation • Grade 6 • The Number System • 6.NS.C.7.b

Which temperature is colder, −8°C or −3°C?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q1/5

Which statement is true?

A. $-7 < -12$

B. $-12 > -7$

C. $-12 < -7$

D. $-7 = -12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q2/5

Write an inequality to compare elevations of −15 feet and −9 feet.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q3/5

True or false?

A balance of −$20 is less than a balance of −$5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q4/5

True or false?

$-4°F$ is colder than $-10°F$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q5/5

SET 15 • 7.NS.A.1.b

FORWARD 1

Model a changing quantity with a linear function • Grade 8 • Functions • 8.F.B.4

A diver starts at −10 feet and descends 4 feet per minute.

Write a function for her position d after t minutes.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q1/5

The temperature is −6°F and rises 3°F per hour.

Write a function for the temperature T after h hours.

T = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q2/5

An account balance starts at −$40 and goes up $15 per week.

What is the balance after 4 weeks?

A. $20

B. −$100

C. −$20

D. $60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q3/5

True or false?

For $T = 3h - 6$, the temperature reaches 0°F after 2 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q4/5

True or false?

For $d = -4t - 10$, the starting position is -4 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 7.NS.A.1.b

FORWARD 2

Evaluate a function and interpret the value in context • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2

$h(t) = -16t^2 + 64t$ gives the height, in feet, of a ball after t seconds.

Find h(1).

h(1) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q1/5

$T(h) = -2 + 1.5h$ gives the temperature in °C after h hours.

Find $T(6)$. What does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q2/5

$f(x) = 3x - 8$

What is $f(-2)$?

A. 14

B. -14

C. -2

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q3**

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q3/5

True or false?

If $g(x) = -x + 5$, then $g(-3) = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q4/5

True or false?

If $f(x) = x^2 - 4$, then $f(-3) = -13$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q5/5

SET 16 • 7.NS.A.1.c • THE NUMBER SYSTEM

Subtract rational numbers by adding the additive inverse

MAIN	Grade 7 main question	**7.NS.A.1.c**
BACKWARD 1	Subtract decimals	**5.NBT.B.7**
BACKWARD 2	Subtract fractions with unlike denominators	**5.NF.A.1**
FORWARD 1	Find a rate of change from two points with negative coordinates	**8.F.B.4**
FORWARD 2	Subtract polynomials	**HSA-APR.A.1**

SET 16 • 7.NS.A.1.c

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.c

Rewrite as an addition expression. Then find the value.

$4 - 9$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q1/5

Subtract.

$-7 - (-12)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q2/5

Which expression is equal to $-3 - 8$?

A. $3 + (-8)$

B. $3 + 8$

C. $-3 + 8$

D. $-3 + (-8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q3/5

True or false?

$-2.5 - 4 = -6.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q4/5

True or false?

$\frac{1}{2} - (-\frac{1}{4}) = \frac{1}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q5/5

SET 16 • 7.NS.A.1.c

BACKWARD 1

Subtract decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Subtract.

$8.2 - 3.75$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q1/5

Subtract.

$10 - 6.38$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q2/5

Subtract.

$5.4 - 2.9$

A. 8.3

B. 2.5

C. 3.5

D. 2.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q3/5

True or false?

$7.06 - 2.5 = 4.56$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q4/5

True or false?

$9.3 - 0.7 = 9.6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • 7.NS.A.1.c

BACKWARD 2

Subtract fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$\frac{5}{6} - \frac{1}{4}$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q1/5

Subtract.

$3\frac{1}{2} - 1\frac{2}{3}$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q2/5

Subtract.

$\frac{3}{4} - \frac{2}{5}$

A. $\frac{7}{20}$

B. $\frac{1}{20}$

C. $\frac{1}{2}$

D. $\frac{5}{9}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q3/5

True or false?

$\frac{2}{3} - \frac{1}{6} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q4/5

True or false?

$\frac{4}{5} - \frac{1}{3} = \frac{3}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 7.NS.A.1.c

FORWARD 1

Find a rate of change from two points with negative coordinates • Grade 8 • Functions • 8.F.B.4

Find the rate of change of the line through $(-2, 5)$ and $(3, -5)$.

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q1/5

Find the slope of the line through $(-4, -1)$ and $(2, 8)$.

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q2/5

What is the slope of the line through $(1, -3)$ and $(-1, 5)$?

A. 1

B. 4

C. -1

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q3/5

True or false?

The slope of the line through $(-3, -2)$ and $(0, 4)$ is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q4/5

True or false?

The slope of the line through $(2, -1)$ and $(5, -7)$ is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q5/5

SET 16 • 7.NS.A.1.c

FORWARD 2

Subtract polynomials • High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Subtract.

$(5x^2 - 3x + 2) - (2x^2 - 7x - 4)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q1/5

Subtract.

$(4x - 9) - (-x + 3)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q2/5

Subtract.

$(x^2 + 2x) - (3x^2 - 5x)$

A. $-2x^2 + 7x$

B. $-2x^2 - 3x$

C. $-2x^2 + 3x$

D. $4x^2 + 7x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q3**

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q3/5

True or false?

$$(6x + 1) - (2x - 5) = 4x + 6$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q4/5

True or false?

$$(3x^2 - x) - (x^2 + x) = 2x^2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q5/5

SET 17 • 7.NS.A.1.c • THE NUMBER SYSTEM

Find the distance between two rational numbers as the absolute value of their difference

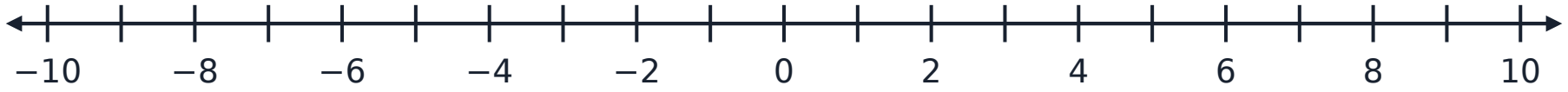
MAIN	Grade 7 main question	**7.NS.A.1.c**
BACKWARD 1	Find the distance between points with the same x- or y-coordinate	**6.NS.C.8**
BACKWARD 2	Position fractions and decimals on a number line	**6.NS.C.6.c**
FORWARD 1	Use the Pythagorean theorem to find the distance between two po…	**8.G.B.8**
FORWARD 2	Use coordinates to find the perimeter of a polygon	**HSG-GPE.B.7**

SET 17 • 7.NS.A.1.c

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.c

Find the distance between -6 and 9 on the number line.



Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q1/5

Find the distance between -4.5 and -1.2 on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q2/5

Which expression gives the distance between -3 and 8 on a number line?

A. $-3 - 8$

B. $|-3 - 8|$

C. $-3 + 8$

D. $|-3 + 8|$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q3/5

True or false?

$|2 - (-5)| = 7$, so 2 and -5 are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q4/5

True or false?

The distance between -10 and -2 on a number line is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q5/5

SET 17 • 7.NS.A.1.c

BACKWARD 1

Find the distance between points with the same x- or y-coordinate • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-3, 4)$ and $(5, 4)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q1/5

Find the distance between $(2, -6)$ and $(2, 1)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q2/5

Which two points are 5 units apart?

A. (1, −2) and (1, 3)

B. (1, −2) and (6, 3)

C. (−2, 1) and (2, 1)

D. (1, −2) and (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q3/5

True or false?

$(-4, -1)$ and $(-4, 6)$ are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q4/5

True or false?

$(-5, 2)$ and $(3, 2)$ are 2 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

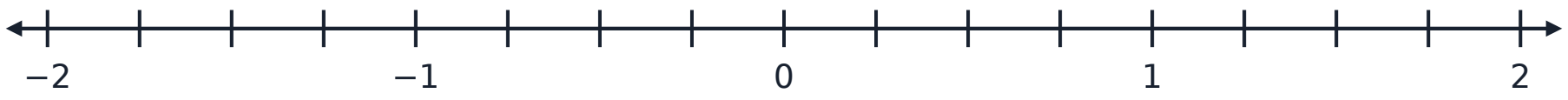
Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 7.NS.A.1.c

BACKWARD 2

Position fractions and decimals on a number line • Grade 6 • The Number System • 6.NS.C.6.c

Plot -1.5 and $\frac{3}{4}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q1/5

Name a number between -2 and -1.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q2/5

Which number is between −0.5 and 0?

A. −0.75

B. −1

C. 0.25

D. −0.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q3/5

True or false?

$-\frac{1}{2}$ is to the left of 0 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q4/5

True or false?

-2.3 is to the right of -2 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 7.NS.A.1.c

FORWARD 1

Use the Pythagorean theorem to find the distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q1/5

Find the distance between $(-2, -3)$ and $(4, 5)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q2/5

What is the distance between (0, 0) and (5, 12)?

A. 13 units

B. 7 units

C. 60 units

D. 17 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q3/5

True or false?

The distance between $(-1, 1)$ and $(2, 5)$ is 5 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q4/5

True or false?

The distance between (0, 0) and (3, 3) is 6 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q5/5

SET 17 • 7.NS.A.1.c

FORWARD 2

Use coordinates to find the perimeter of a polygon • High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

A triangle has vertices (0, 0), (6, 0), and (6, 8).

Find its perimeter.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q1/5

A rectangle has vertices $(-3, -2)$, $(5, -2)$, $(5, 4)$, and $(-3, 4)$.

Find its perimeter.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q2/5

A triangle has vertices $(-2, 1)$, $(2, 1)$, and $(2, 4)$.

What is its perimeter?

A. 10 units

B. 6 units

C. 12 units

D. 7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q3**

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q3/5

True or false?

A square with vertices (1, 1), (4, 1), (4, 4), and (1, 4) has a perimeter of 12 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q4/5

True or false?

A triangle with vertices (0, 0), (8, 0), and (0, 6) has a perimeter of 14 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q5/5

SET 18 • 7.NS.A.1.d • THE NUMBER SYSTEM

Use properties of operations to add and subtract rational numbers

MAIN	Grade 7 main question	**7.NS.A.1.d**
BACKWARD 1	Apply properties to write equivalent expressions	**6.EE.A.3**
BACKWARD 2	Add multi-digit decimals	**6.NS.B.3**
BACKWARD 3	Reorder and regroup addends to add within 100	**2.NBT.B.5**
FORWARD 1	Solve equations by collecting like terms	**8.EE.C.7.b**
FORWARD 2	Use the structure of an expression to rewrite it	**HSA-SSE.A.2**

SET 18 • 7.NS.A.1.d

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.d

Find the sum by grouping numbers to make the work easier. Show your work.

$-8 + 13.5 + 8$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q1/5

Rewrite each subtraction as adding the opposite. Then use properties to find the value. Show your work.

$5.6 - 8 - 3.6 + 8$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q2/5

Which property is shown?

$-5 + 2.7 = 2.7 + (-5)$

A. Associative property of addition

B. Additive inverse property

C. Commutative property of addition

D. Additive identity property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q3/5

True or false?

$$-6 + (4 + 6) = (-6 + 6) + 4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q4/5

True or false?

$7 - 3 = 3 - 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q5/5

SET 18 • 7.NS.A.1.d

BACKWARD 1

Apply properties to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Use the distributive property to rewrite 3(x + 5).

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q1/5

Combine like terms.

$4y + 2 + 6y$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q2/5

Which expression is equivalent to 2(4 + n)?

A. 4 + 2n

B. 6 + n

C. 8 + 2n

D. 8 + n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q3/5

True or false?

$5 + (a + 7) = (5 + 7) + a$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q4/5

True or false?

$$6(x + 2) = 6x + 2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 7.NS.A.1.d

BACKWARD 2

Add multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

14.62 + 7.9

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q1/5

Add.

0.75 + 3.6 + 1.25

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q2/5

Add.

9.08 + 2.5

A. 34.08

B. 11.58

C. 9.33

D. 11.13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q3/5

True or false?

$2.25 + 3.75 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q4/5

True or false?

$4.5 + 0.55 = 5.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 7.NS.A.1.d

BACKWARD 3

Reorder and regroup addends to add within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add by first adding the two numbers that make a ten.

28 + 17 + 2

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q1/5

Add by first adding the two numbers that make a ten.

36 + 25 + 4

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q2/5

Which grouping makes 46 + 19 + 4 easier to add?

A. (46 + 19) − 4

B. (46 + 4) + 19

C. 46 + 19 − 4

D. (46 − 4) + 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q3/5

True or false?

$27 + 18 + 3 = 27 + 3 + 18$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q4/5

True or false?

$(35 + 15) + 9 = 35 + (15 + 8)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q5/5

SET 18 • 7.NS.A.1.d

FORWARD 1

Solve equations by collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$-2.5x + 4 + 1.5x = -3$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q1/5

Solve for x.

$\frac{1}{3}x - 5 + \frac{2}{3}x = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q2/5

Solve for x.

$3x - 7 + x = 2x + 5$

A. $x = 12$

B. $x = 2$

C. $x = -1$

D. $x = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q3/5

True or false?

$x = -4$ is the solution of $5x + 3 - 2x = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q4/5

True or false?

$x = 2$ is the solution of $-x + 6 + 3x = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q5/5

SET 18 • 7.NS.A.1.d

FORWARD 2

Use the structure of an expression to rewrite it • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Rewrite $x^2 - 9$ as a product of two factors.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q1/5

Factor out the greatest common factor.

$4x^2 + 12x$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q2/5

Which expression is equivalent to $x^2 - 25$?

A. $(x - 5)(x + 5)$

B. $(x + 5)^2$

C. $(x - 5)^2$

D. $(x - 25)(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q3/5

True or false?

$x^2 + 6x + 9 = (x + 3)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q4/5

True or false?

$x^2 - 16 = (x - 4)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q5/5

SET 19 • 7.NS.A.2.a • THE NUMBER SYSTEM

Multiply rational numbers using the rules for signs

MAIN	Grade 7 main question	**7.NS.A.2.a**
BACKWARD 1	Multiply fractions	**5.NF.B.4.a**
BACKWARD 2	Multiply decimals	**5.NBT.B.7**
FORWARD 1	Apply the properties of integer exponents with negative bases	**8.EE.A.1**
FORWARD 2	Multiply polynomials	**HSA-APR.A.1**

SET 19 • 7.NS.A.2.a

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.a

Multiply.

-6×7

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q1/5

Multiply.

$-2.5 \times (-4)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q2/5

Multiply.

$$-\frac{2}{3} \times \frac{9}{10}$$

A. $-\frac{3}{5}$

B. $\frac{3}{5}$

C. $-\frac{27}{20}$

D. $-\frac{11}{13}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q3/5

True or false?

$(-1)(-1) = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q4/5

True or false?

The product of three negative numbers is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q5/5

SET 19 • 7.NS.A.2.a

BACKWARD 1

Multiply fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$\frac{3}{5} \times \frac{10}{9}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q1/5

Multiply.

$\frac{4}{7} \times 14$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q2/5

Multiply.

$\frac{5}{6} \times \frac{3}{4}$

A. $\frac{5}{8}$

B. $\frac{3}{8}$

C. $\frac{15}{10}$

D. $\frac{8}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q4/5

True or false?

$\frac{3}{4} \times 8 = 24$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q5/5

SET 19 • 7.NS.A.2.a

BACKWARD 2

Multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

2.4×3.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q1/5

Multiply.

0.6×0.7

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q2/5

Multiply.

1.25×8

A. 100

B. 1

C. 9.25

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q3/5

True or false?

$0.3 \times 0.3 = 0.09$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q4/5

True or false?

$4.2 \times 10 = 4.20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 7.NS.A.2.a

FORWARD 1

Apply the properties of integer exponents with negative bases • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power of -2. Then find its value.

$(-2)^5 \div (-2)^3$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q1/5

Write as a single power of -3. Then find its value.

$(-3)^4 \times (-3)^{-1}$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q2/5

Which expression is equal to $\frac{1}{16}$?

A. -4^{-2}

B. $(-4)^{-1} \times 4$

C. $(-4)^{2}$

D. $(-4)^{-2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q3/5

True or false?

$((-5)^2)^3 = (-5)^6 = 15{,}625$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q4/5

True or false?

$(-7)^0 = -1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q5/5

SET 19 • 7.NS.A.2.a

FORWARD 2

Multiply polynomials • High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply.

$-3x(2x - 5)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q1/5

Multiply.

$(x - 4)(x + 6)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q2/5

Multiply.

$(-2x)(-7x)$

A. $14x$

B. $14x^2$

C. $-14x^2$

D. $-9x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F2-Q3**

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q3/5

True or false?

$$(x - 3)(x - 3) = x^2 - 6x + 9$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q4/5

True or false?

$$-4(x - 2) = -4x - 8$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q5/5

SET 20 • 7.NS.A.2.a • THE NUMBER SYSTEM

Interpret products of rational numbers in real-world situations

MAIN	Grade 7 main question	**7.NS.A.2.a**
BACKWARD 1	Multiply fractions and mixed numbers in word problems	**5.NF.B.6**
BACKWARD 2	Explain the meaning of signed numbers and 0	**6.NS.C.5**
FORWARD 1	Solve equations that compare two quantities changing at negative…	**8.EE.C.7.b**
FORWARD 2	Create a one-variable equation for a situation with a negative rate	**HSA-CED.A.1**

SET 20 • 7.NS.A.2.a

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.a

A stock's price drops $1.75 each day for 6 days.

What is the total change in price?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q1/5

A diver descends 2.5 feet each second.

What is the change in her position after 8 seconds?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q2/5

A tank loses 3 gallons each minute.

Which product gives the change in the amount of water after 12 minutes?

A. $12 \times 3 = 36$

B. $12 \times (-3) = -36$

C. $12 + (-3) = 9$

D. $-12 \times (-3) = 36$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q3/5

True or false?

The temperature falls 4°F each hour for 5 hours. The total change is −20°F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q4/5

True or false?

An account balance changes by −$15 each week. The change over 3 weeks is $45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q5/5

SET 20 • 7.NS.A.2.a

BACKWARD 1

Multiply fractions and mixed numbers in word problems • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A recipe uses $\frac{3}{4}$ cup of oats for each batch.

How many cups of oats are needed for 6 batches?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q1/5

A path is $2\frac{1}{2}$ miles long. Sam walks $\frac{2}{5}$ of it.

How far does he walk?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q2/5

A board is $\frac{7}{8}$ yard long. Ty cuts a piece that is $\frac{1}{2}$ as long as the board.

How long is the piece?

A. $1\frac{3}{8}$ yards

B. $\frac{7}{4}$ yards

C. $\frac{8}{14}$ yard

D. $\frac{7}{16}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q3/5

True or false?

$\frac{2}{3}$ of a 12-foot rope is 8 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q4/5

True or false?

4 bags each hold $\frac{2}{3}$ pound of nuts. Together they hold 3 pounds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • 7.NS.A.2.a

BACKWARD 2

Explain the meaning of signed numbers and 0 • Grade 6 • The Number System • 6.NS.C.5

In a game, a score of 0 means no points won or lost.

What does a score of −12 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q1/5

Sea level is 0 meters.

Which elevation is 30 meters below sea level?

A. −30 m

B. 30 m

C. 0 m

D. −3 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q2/5

What does a change of −$8 in a bank account mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q3/5

True or false?

A temperature change of −5° means it got colder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q4/5

True or false?

On a horizontal number line, positive numbers are to the left of 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 7.NS.A.2.a

FORWARD 1

Solve equations that compare two quantities changing at negative rates • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Diver A starts at −10 m and rises 0.5 m each second. Diver B starts at −22 m and rises 1.25 m each second.

Solve −10 + 0.5t = −22 + 1.25t to find when they are at the same depth.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q1/5

Tank A has 50 liters and drains 2.5 liters each minute. Tank B has 20 liters and drains 1 liter each minute.

Solve $50 - 2.5t = 20 - t$ to find when they hold the same amount.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q2/5

Solve for x.

$-3(x + 4) = 2(x - 1) - 5$

A. $x = 1$

B. $x = 3$

C. $x = -1$

D. $x = -3.4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q3/5

True or false?

$t = 8$ is the solution of $300 - 12.5t = 180 + 2.5t$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q4/5

True or false?

$x = -2$ is the solution of $-1.5x + 4 = 2(x + 5)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q5/5

SET 20 • 7.NS.A.2.a

FORWARD 2

Create a one-variable equation for a situation with a negative rate • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1

A phone battery is at 90% of its full charge. It drops by 6 percentage points of its full capacity each hour.

Write and solve an equation to find when it reaches 30%.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q1/5

A submarine at −50 m dives 20 m each minute.

Write and solve an equation to find when it reaches −170 m.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q2/5

A store's profit was −$400 and goes up $125 each month.

Which equation finds the month m when the profit is $600?

A. $-400 + 125m = 600$

B. $125 - 400m = 600$

C. $-400 - 125m = 600$

D. $400 + 125m = 600$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q3/5

True or false?

A 72°F room cools 4°F each hour. The equation $72 - 4t = 40$ finds when the room is 40°F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q4/5

True or false?

The solution of $72 - 4t = 40$ is $t = 28$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q5/5

SET 21 • 7.NS.A.2.b • THE NUMBER SYSTEM

Divide integers and explain the sign of a quotient

MAIN	Grade 7 main question	**7.NS.A.2.b**
BACKWARD 1	Divide multi-digit whole numbers	**6.NS.B.2**
BACKWARD 2	Divide fractions	**6.NS.A.1**
FORWARD 1	Classify numbers as rational or irrational	**8.NS.A.1**
FORWARD 2	Decide whether a product is rational or irrational	**HSN-RN.B.3**

SET 21 • 7.NS.A.2.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.b

Divide.

$-48 \div 6$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q1/5

Divide.

$-63 \div (-9)$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q2/5

Which fraction is equal to $-\frac{3}{4}$?

A. $\frac{-3}{4}$

B. $\frac{4}{-3}$

C. $\frac{-3}{-4}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q3/5

True or false?

$$\frac{-10}{2} = \frac{10}{-2} = -5$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q4/5

True or false?

$-7 \div 0 = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q5/5

SET 21 • 7.NS.A.2.b

BACKWARD 1

Divide multi-digit whole numbers • Grade 6 • The Number System • 6.NS.B.2

Divide.

1,872 ÷ 24

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q1/5

Divide.

3,915 ÷ 45

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q2/5

Divide.

2,436 ÷ 12

A. 203

B. 2,030

C. 23

D. 230

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q3/5

True or false?

$5{,}040 \div 35 = 144$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q4/5

True or false?

$1{,}950 \div 15 = 13$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 7.NS.A.2.b

BACKWARD 2

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$\frac{4}{5} \div \frac{2}{15}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q1/5

Divide.

$3 \div \frac{3}{8}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q2/5

Divide.

$\frac{7}{10} \div \frac{1}{5}$

A. $\frac{2}{7}$

B. $\frac{7}{50}$

C. $1\frac{2}{5}$

D. $3\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q3/5

True or false?

$\frac{3}{8} \div \frac{3}{4} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q4/5

True or false?

$$\frac{5}{6} \div \frac{5}{12} = \frac{1}{2}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 7.NS.A.2.b

FORWARD 1

Classify numbers as rational or irrational • Grade 8 • The Number System • 8.NS.A.1

Is $\frac{-5}{8}$ rational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q1/5

Which number is irrational?

A. $\sqrt{7}$

B. 0.25

C. $\frac{-22}{7}$

D. $-\sqrt{16}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q2/5

Write -2.4 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q3/5

True or false?

Every integer is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q4/5

True or false?

$\sqrt{25}$ is irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 7.NS.A.2.b

FORWARD 2

Decide whether a product is rational or irrational • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3

Is $-3 \times \sqrt{5}$ rational or irrational?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q1/5

Is $\frac{-2}{3} \times \frac{9}{4}$ rational or irrational? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q2/5

Which product is rational?

A. $-1 \times \pi$

B. $5 \times \sqrt{2}$

C. $2 \times \sqrt{3}$

D. $\sqrt{2} \times \sqrt{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q3/5

True or false?

The product of a nonzero rational number and an irrational number is irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q4/5

True or false?

The quotient of two rational numbers (with a nonzero divisor) can be irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q5/5

SET 22 • 7.NS.A.2.b • THE NUMBER SYSTEM

Interpret quotients of rational numbers in real-world situations

MAIN	Grade 7 main question	**7.NS.A.2.b**
BACKWARD 1	Divide fractions in word problems	**6.NS.A.1**
BACKWARD 2	Divide unit fractions and whole numbers in word problems	**5.NF.B.7.c**
FORWARD 1	Find a rate of change from a description	**8.F.B.4**
FORWARD 2	Calculate and interpret a negative average rate of change	**HSF-IF.B.6**

SET 22 • 7.NS.A.2.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.b

A snail's position changed by $-\frac{3}{4}$ meter in $\frac{1}{2}$ minute at a steady rate.

What was the change in position per minute?

Change per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q1/5

A diver's position changed by -45 feet in 9 seconds at a steady rate.

What was the change in position per second?

Change per second: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q2/5

The temperature changed by $-4\frac{1}{2}$°F over $1\frac{1}{2}$ hours at a steady rate.

What was the change per hour?

A. $-\frac{1}{3}$°F

B. 3°F

C. -3°F

D. $-6\frac{3}{4}$°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q3/5

True or false?

A balance changes by −$120 over 8 weeks at a steady rate. The change is −$15 per week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q4/5

True or false?

A water level changed by $-2\frac{2}{5}$ inches over 4 days at a steady rate. The change was $\frac{3}{5}$ inch per day.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q5/5

SET 22 • 7.NS.A.2.b

BACKWARD 1

Divide fractions in word problems • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{3}{4}$-cup servings are in 6 cups of rice?

Servings: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q1/5

A $\frac{2}{3}$-mile trail is split into 4 equal parts.

How long is each part?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q2/5

A strip $\frac{5}{6}$ meter long is cut into pieces $\frac{1}{12}$ meter long.

How many pieces are there?

A. $\frac{5}{72}$

B. 12

C. 6

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q3/5

True or false?

There are 6 half-pound portions in 3 pounds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q4/5

True or false?

3 people share $\frac{3}{4}$ pound of nuts equally. Each gets $\frac{1}{2}$ pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 7.NS.A.2.b

BACKWARD 2

Divide unit fractions and whole numbers in word problems • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c

How many $\frac{1}{4}$-pound burgers can be made from 5 pounds of meat?

Burgers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q1/5

3 friends share $\frac{1}{2}$ pound of grapes equally.

How much does each friend get?

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q2/5

2 people share $\frac{1}{3}$ of a pizza equally.

How much of the whole pizza does each person get?

A. $\frac{2}{3}$ of the pizza

B. $\frac{1}{6}$ of the pizza

C. 6 pizzas

D. $\frac{3}{2}$ of the pizza

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q3/5

True or false?

4 pounds of rice is packed into bags that each hold $\frac{1}{5}$ pound. There are 20 bags.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q4/5

True or false?

2 people share $\frac{1}{8}$ pound of nuts equally. Each person gets $\frac{1}{4}$ pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 7.NS.A.2.b

FORWARD 1

Find a rate of change from a description • Grade 8 • Functions • 8.F.B.4

A tank held 240 gallons at 2:00 and 180 gallons at 2:15. It drained at a constant rate.

What is the rate of change in gallons per minute?

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q1/5

A plane descends from 30,000 feet to 21,000 feet in 15 minutes at a constant rate.

What is the rate of change in feet per minute?

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q2/5

A line passes through (0, 50) and (10, 20).

What is its rate of change?

A. $-\frac{1}{3}$

B. 3

C. -30

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q3/5

True or false?

A balance that goes from $80 to $20 in 12 weeks at a constant rate changes by −$5 per week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q4/5

True or false?

A temperature that goes from 10° to −6° in 4 hours at a constant rate changes by −1° per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q5/5

SET 22 • 7.NS.A.2.b

FORWARD 2

Calculate and interpret a negative average rate of change • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

The table shows temperatures.

What is the average rate of change from 0 hours to 6 hours?

Time (h)	0	3	6
Temperature (°F)	12	3	−6

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q1/5

$f(x) = -x^2 + 10$

What is the average rate of change from $x = 1$ to $x = 3$?

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q2/5

A company's profit was $5,000 in year 2 and $2,000 in year 5.

What is the average rate of change?

A. −$600 per year

B. $1,000 per year

C. −$3,000 per year

D. −$1,000 per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q3/5

True or false?

A negative average rate of change means the quantity decreased over the interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q4/5

True or false?

For $f(x) = 5 - 2x$, the average rate of change from $x = 0$ to $x = 4$ is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q5/5

SET 23 • 7.NS.A.2.c • THE NUMBER SYSTEM

Use properties of operations to multiply and divide rational numbers

MAIN	Grade 7 main question	**7.NS.A.2.c**
BACKWARD 1	Apply the distributive property	**6.EE.A.3**
BACKWARD 2	Use the commutative and associative properties of multiplication	**3.OA.B.5**
FORWARD 1	Solve equations with fraction coefficients on both sides	**8.EE.C.7.b**
FORWARD 2	Rewrite expressions with rational exponents	**HSN-RN.A.2**

SET 23 • 7.NS.A.2.c

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.c

Use properties to multiply. Show how you grouped or ordered the factors.

$-4 \times 7.3 \times 25$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q1/5

Rewrite the division as multiplication by a reciprocal. Then use properties to find the value. Show your work.

$$-18 \div \frac{3}{4} \times \frac{1}{6}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q2/5

Which property is shown?

$-2 \times (5 \times 3.1) = (-2 \times 5) \times 3.1$

A. Multiplicative identity property

B. Associative property of multiplication

C. Distributive property

D. Commutative property of multiplication

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q3/5

True or false?

$$-6 \times (10 + \tfrac{1}{2}) = -60 + (-3)$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q4/5

True or false?

$$-24 \div 6 \div 2 = -24 \div (6 \div 2)$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q5/5

SET 23 • 7.NS.A.2.c

BACKWARD 1

Apply the distributive property • Grade 6 • Expressions & Equations • 6.EE.A.3

Rewrite using the distributive property.

$7(2y + 3)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q1/5

Use the distributive property to find 8×47 as $8 \times (40 + 7)$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q2/5

Which expression is equivalent to 5(x + 9)?

A. 5x + 14

B. x + 45

C. 5x + 45

D. 5x + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q3/5

True or false?

$4(3 + 2.5) = 12 + 10$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q4/5

True or false?

$3(a + 4) = 3a + 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 7.NS.A.2.c

BACKWARD 2

Use the commutative and associative properties of multiplication • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

Fill in the blank.

$6 \times 4 = 4 \times ___$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q1/5

Find 2 × 7 × 5 by first multiplying 2 × 5.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q2/5

Which expression is equal to (3 × 2) × 4?

A. 3 × 2 + 4

B. 3 + (2 × 4)

C. 3 × (2 × 4)

D. (3 + 2) × 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q3/5

True or false?

$8 \times 3 = 3 \times 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q4/5

True or false?

$(2 \times 3) \times 5 = 2 \times 3 + 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 7.NS.A.2.c

FORWARD 1

Solve equations with fraction coefficients on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$\frac{3}{4}x - 2 = \frac{1}{2}x + 3$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q1/5

Solve for x.

$$-\frac{2}{3}(x + 6) = \frac{1}{3}x - 1$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q2/5

Solve for x.

$\frac{x}{4} + \frac{x}{6} = 5$

A. x = 12

B. x = 6

C. x = 50

D. x = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q3/5

True or false?

$x = -10$ is the solution of $-\frac{1}{2}x + 1 = \frac{1}{5}x + 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q4/5

True or false?

$x = 3$ is the solution of $\frac{5}{6}x - 1 = \frac{1}{2}x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q5/5

SET 23 • 7.NS.A.2.c

FORWARD 2

Rewrite expressions with rational exponents • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2

Find the value.

$27^{(1/3)}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q1/5

For a real number $x \geq 0$, write $\sqrt{x}$ using a rational exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q2/5

Which number is equal to 16^(1/2)?

A. 4

B. 32

C. 2

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q3**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q3/5

True or false?

$(x^3)^{(1/3)} = x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q4/5

True or false?

$25^{(1/2)} = 12.5$

A. True

B. False

SET 24 • 7.NS.A.2.d • THE NUMBER SYSTEM

Use long division to write a rational number as a terminating or repeating decimal

MAIN	Grade 7 main question	**7.NS.A.2.d**
BACKWARD 1	Write fractions with denominators of 10 or 100 as decimals	**4.NF.C.6**
BACKWARD 2	Divide multi-digit whole numbers	**6.NS.B.2**
FORWARD 1	Write a repeating decimal as a fraction	**8.NS.A.1**
FORWARD 2	Report a decimal result to a sensible level of accuracy	**HSN-Q.A.3**

SET 24 • 7.NS.A.2.d

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.d

Use long division to write $\frac{3}{8}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q1/5

Use long division to write $\frac{5}{11}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q2/5

Which decimal is equal to $-\frac{7}{12}$?

A. $-0.58333...$ (the 3 repeats)

B. -0.712

C. $-1.714...$

D. -0.5833 (the decimal ends)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q3/5

True or false?

$\frac{2}{9} = 0.222\ldots$, a repeating decimal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q4/5

True or false?

$\frac{7}{20}$ is a repeating decimal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q5/5

SET 24 • 7.NS.A.2.d

BACKWARD 1

Write fractions with denominators of 10 or 100 as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{7}{10}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q1/5

Write $\frac{43}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q2/5

Which decimal is equal to $\frac{6}{100}$?

A. 0.06

B. 0.006

C. 0.6

D. 6.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q3/5

True or false?

$\frac{9}{10} = 0.9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q4/5

True or false?

$\frac{25}{100} = 2.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 7.NS.A.2.d

BACKWARD 2

Divide multi-digit whole numbers • Grade 6 • The Number System • 6.NS.B.2

Divide.

7,425 ÷ 25

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q1/5

Divide.

4,368 ÷ 52

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q2/5

Divide.

9,072 ÷ 36

A. 2,520

B. 252

C. 25.2

D. 242

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

$6{,}912 \div 48 = 144$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q4/5

True or false?

$2{,}925 \div 75 = 49$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 7.NS.A.2.d

FORWARD 1

Write a repeating decimal as a fraction • Grade 8 • The Number System • 8.NS.A.1

Write 0.777... (the 7 repeats) as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q1/5

Write 0.1818... (the digits 18 repeat) as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q2/5

Which fraction is equal to 0.333... (the 3 repeats)?

A. $\frac{1}{3}$

B. $\frac{3}{100}$

C. $\frac{3}{10}$

D. $\frac{33}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q3/5

True or false?

0.999... (the 9 repeats) is equal to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q4/5

True or false?

0.4545... (the digits 45 repeat) is equal to $\frac{45}{100}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q5/5

SET 24 • 7.NS.A.2.d

FORWARD 2

Report a decimal result to a sensible level of accuracy • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

A 10-foot board is cut into 3 equal pieces. The tape measure reads to the nearest hundredth of a foot.

Report the length of each piece to that accuracy.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q1/5

$\frac{2}{7}$ = 0.285714... Round it to the nearest thousandth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q2/5

A scale reads to the nearest 0.1 kg. 5 kg of flour is split into 3 equal bags.

Which report of each bag's mass fits the scale?

A. 1.666666 kg

B. 1.7 kg

C. 2 kg

D. 1.67 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q3/5

True or false?

Reporting $\frac{1}{3}$ hour as 0.33 hour is accurate to the nearest hundredth of an hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q4/5

True or false?

A length reported as 4.0 cm is less precise than a length reported as 4 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q5/5

SET 25 • 7.NS.A.3 • THE NUMBER SYSTEM

Solve real-world problems with the four operations on rational numbers

MAIN	Grade 7 main question	**7.NS.A.3**
BACKWARD 1	Multiply and divide decimals in word problems	**6.NS.B.3**
BACKWARD 2	Solve one-step equations	**6.EE.B.7**
FORWARD 1	Solve real-world problems with scientific notation	**8.EE.A.4**
FORWARD 2	Use units to solve multistep problems	**HSN-Q.A.1**

SET 25 • 7.NS.A.3

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.3

A freezer is at 4°C. It cools 1.5°C every hour.

What is the temperature after 6 hours?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q1/5

Maya owes $42.60. She pays it off in 6 equal payments.

How much is each payment?

Payment: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q2/5

A diver is at -18 m. She rises $\frac{1}{3}$ of the distance to the surface.

What is her new position?

A. -6 m

B. 12 m

C. -12 m

D. -24 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q3/5

True or false?

A hiker is at −86 m. She climbs 120 m. She ends 34 m above sea level.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q4/5

True or false?

Three stock prices change by −$2.50, +$4, and −$3.25. The total change is +$1.75.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q5/5

SET 25 • 7.NS.A.3

BACKWARD 1

Multiply and divide decimals in word problems • Grade 6 • The Number System • 6.NS.B.3

Gas costs $3.85 per gallon.

How much do 12 gallons cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q1/5

A 7.5-meter rope is cut into 3 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q2/5

A 4.8-liter jug fills cups that hold 0.3 liter each.

How many cups can it fill?

A. 14.4

B. 16

C. 160

D. 1.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q3/5

True or false?

A 2.5-pound bag of nuts costs $3.20 per pound. The bag costs $8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q4/5

True or false?

$9.60 shared equally by 4 people is $2.60 each.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • 7.NS.A.3

BACKWARD 2

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$x + 4.5 = 12$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q1/5

Solve for y.

$6y = 39$

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q2/5

Solve for n.

$\frac{2}{3}n = 10$

A. $n = 30$

B. $n = \frac{20}{3}$

C. $n = 15$

D. $n = 9\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q3/5

True or false?

$m = 8$ is the solution of $m - 2.75 = 5.25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q4/5

True or false?

$k = 3$ is the solution of $4k = 1.2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 7.NS.A.3

FORWARD 1

Solve real-world problems with scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

A hard drive holds 2×10^{12} bytes. A file is 4×10^{8} bytes.

How many files of this size fit on the drive?

Files: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q1/5

Earth is about 1.5×10^8 km from the Sun. Mars is about 2.3×10^8 km from the Sun.

About how much farther from the Sun is Mars? Write the answer in scientific notation.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q2/5

A cell is 2×10^{-5} m wide.

How wide are 3,000 of these cells placed side by side?

A. 6×10^{-8} m

B. 5×10^{-2} m

C. 6×10^{-2} m

D. 6×10^{2} m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q3/5

True or false?

$(6 \times 10^4) \times 5 = 3 \times 10^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q4/5

True or false?

$(9 \times 10^6) \div (3 \times 10^2) = 3 \times 10^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q5/5

SET 25 • 7.NS.A.3

FORWARD 2

Use units to solve multistep problems • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

A car gets 32 miles per gallon. Gas costs $3.60 per gallon.

Use units to find the gas cost per mile. Then find the cost of a 400-mile trip.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q1/5

A faucet leaks 12 milliliters per minute.

How many liters leak in 1 day? Show the unit conversions.

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q2/5

A printer uses 0.8 mL of ink per page. A 30 mL ink cartridge costs $24.

What is the cost of ink per page?

A. $0.64

B. $1.25

C. $0.32

D. $0.80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q3/5

True or false?

A speed of 12 meters per second is 43.2 kilometers per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q4/5

True or false?

Walking 3 miles per hour for 45 minutes covers 2.5 miles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q5/5

SET 26 • 7.EE.A.1 • EXPRESSIONS & EQUATIONS

Add and subtract linear expressions with rational coefficients

MAIN	Grade 7 main question	**7.EE.A.1**
BACKWARD 1	Combine like terms	**6.EE.A.3**
BACKWARD 2	Identify equivalent expressions	**6.EE.A.4**
FORWARD 1	Solve equations by expanding and collecting like terms	**8.EE.C.7.b**
FORWARD 2	Interpret the terms, factors, and coefficients of an expression in a…	**HSA-SSE.A.1.a**

SET 26 • 7.EE.A.1

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$(3x + 5) + (2x - 8)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q1/5

Simplify.

$(4.5y - 2) - (1.5y + 6)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q2/5

Simplify.

$\frac{1}{2}a + 3 - \frac{3}{4}a + 1$

A. $-\frac{1}{4}a + 4$

B. $-\frac{1}{4}a + 2$

C. $\frac{1}{4}a + 4$

D. $\frac{5}{4}a + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q3/5

True or false?

$$(6m - 1) - (2m - 4) = 4m + 3$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q4/5

True or false?

$$-3k + 7 + 5k - 2 = 8k + 5$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q5/5

SET 26 • 7.EE.A.1

BACKWARD 1

Combine like terms • Grade 6 • Expressions & Equations • 6.EE.A.3

Combine like terms.

$5n + 3n + 4$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q1/5

Combine like terms.

$2y + 7 + y$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q2/5

Which expression is equivalent to x + x + x + 6?

A. 9x

B. 3x + 2

C. 3x + 6

D. $x^3 + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q3/5

True or false?

$4b + 2b = 6b$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q4/5

True or false?

$3a + 2 = 5a$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 7.EE.A.1

BACKWARD 2

Identify equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.4

Are 2(x + 3) and 2x + 6 equivalent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q1/5

Which expression is equivalent to y + y + y?

A. 3y

B. y + 3

C. y^3

D. 3 + y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q2/5

Write an expression with no parentheses that is equivalent to $4(n + 1)$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q3/5

True or false?

5m − 2m and 3m are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q4/5

True or false?

a + 4 and 4a are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 7.EE.A.1

FORWARD 1

Solve equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$2(3x - 1) - (x + 4) = 9$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q1/5

Solve for x.

$\frac{1}{2}(4x + 6) = 3x - 5$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q2/5

Solve for x.

$0.5(x + 8) - 2x = 1$

A. $x = 6$

B. $x = 3$

C. $x = 2$

D. $x = -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q3/5

True or false?

$x = 5$ is the solution of $3(x - 2) = 2x - 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q4/5

True or false?

$x = 1$ is the solution of $4x - (x + 2) = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q5/5

SET 26 • 7.EE.A.1

FORWARD 2

Interpret the terms, factors, and coefficients of an expression in context • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A phone plan costs 0.1m + 25 dollars for m minutes in a month.

What do the terms 0.1m and 25 represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q1/5

The perimeter of a rectangle is 2(l + 4) centimeters, where l is its length.

What does the factor (l + 4) represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q2/5

A ride costs $2.5 + 1.75d$ dollars for d miles.

What does 1.75 represent?

A. The starting fee

B. The cost per mile

C. The total cost of a 1-mile ride

D. The number of miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F2-Q3**

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q3/5

True or false?

A repair costs 40 + 15h dollars for h hours of work. The term 15h is the labor cost for h hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q4/5

True or false?

A savings account holds $500 - 20w$ dollars after w weeks. The coefficient -20 means \$20 is added each week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q5/5

SET 27 • 7.EE.A.1 • EXPRESSIONS & EQUATIONS

Factor and expand linear expressions with rational coefficients

MAIN	Grade 7 main question	**7.EE.A.1**
BACKWARD 1	Use the greatest common factor and the distributive property	**6.NS.B.4**
BACKWARD 2	Identify the parts of an expression	**6.EE.A.2.b**
FORWARD 1	Expand to decide how many solutions an equation has	**8.EE.C.7.a**
FORWARD 2	Factor quadratic expressions	**HSA-SSE.A.2**

SET 27 • 7.EE.A.1

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

$-4(2x - 3)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q1/5

Factor out the greatest common factor.

$6y + 15$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q2/5

Which expression is equivalent to $\frac{1}{2}(8a - 6)$?

A. $8a - 3$

B. $4a - 3$

C. $4a - 6$

D. $16a - 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q3/5

True or false?

$$12m - 18 = 6(2m - 3)$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q4/5

True or false?

$$-5(n + 2) = -5n + 10$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q5/5

SET 27 • 7.EE.A.1

BACKWARD 1

Use the greatest common factor and the distributive property • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q1/5

Use the greatest common factor to write 30 + 42 as a product.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q2/5

Which expression is equal to 18 + 27?

A. 9(2 + 3)

B. 9(2 + 27)

C. 9(18 + 3)

D. 3(6 + 27)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q3/5

True or false?

The greatest common factor of 16 and 40 is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q4/5

True or false?

$20 + 35 = 5(4 + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • 7.EE.A.1

BACKWARD 2

Identify the parts of an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.b

How many terms are in $3x + 7y - 2$?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q1/5

What is the coefficient of w in 9 + 5w?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q2/5

Which part of 4(n + 6) is a sum?

A. 4n

B. 4

C. n + 6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q3/5

True or false?

In 2(a + b), the factor 2 multiplies the sum a + b.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q4/5

True or false?

In $8k - 3$, the term $8k$ is a constant.

A. True

B. False

SET 27 • 7.EE.A.1

FORWARD 1

Expand to decide how many solutions an equation has • Grade 8 • Expressions & Equations • 8.EE.C.7.a

How many solutions does this equation have?

$4(x - 2) = 4x - 8$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q1/5

How many solutions does this equation have?

$-3(x + 1) = -3x + 3$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q2/5

Which equation has no solution?

A. $2(x + 5) = x + 10$

B. $2(x + 5) = 3x$

C. $2(x + 5) = 2x + 10$

D. $2(x + 5) = 2x + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q3/5

True or false?

$6(x - 1) = 2(3x - 3)$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q4/5

True or false?

$5(x + 1) = 5x + 1$ has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q5/5

SET 27 • 7.EE.A.1

FORWARD 2

Factor quadratic expressions • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 7x + 12$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q1/5

Factor completely.

$3x^2 - 12$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q2/5

Factor.

$x^2 - 2x - 15$

A. $(x - 5)(x - 3)$

B. $(x - 15)(x + 1)$

C. $(x + 5)(x - 3)$

D. $(x - 5)(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q3**

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q3/5

True or false?

$2x^2 + 10x = 2x(x + 5)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q4/5

True or false?

$x^2 + 5x + 6 = (x + 1)(x + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q5/5

SET 28 • 7.EE.A.2 • EXPRESSIONS & EQUATIONS

Rewrite an expression to show how the quantities in a problem are related

MAIN	Grade 7 main question	**7.EE.A.2**
BACKWARD 1	Write expressions from words	**6.EE.A.2.a**
BACKWARD 2	Write and interpret numerical expressions	**5.OA.A.2**
FORWARD 1	Write equivalent expressions with the properties of integer expon…	**8.EE.A.1**
FORWARD 2	Rewrite percent-growth expressions to reveal the growth factor	**HSA-SSE.B.3**

SET 28 • 7.EE.A.2

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.A.2

A shirt costs p dollars. The price is raised by 30%.

Write the new price as a single term.

New price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q1/5

A rectangle's length is 4 cm more than its width w.

Write its perimeter in two equivalent ways.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q2/5

A price p is lowered by 15%.

Which expression shows the new price?

A. 1.15p

B. 0.85p

C. 0.15p

D. p − 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q3/5

True or false?

$n + 0.08n = 1.08n$ shows that increasing n by 8% is the same as multiplying n by 1.08.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q4/5

True or false?

The price after a 20% discount on a price of d dollars is d − 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q5/5

SET 28 • 7.EE.A.2

BACKWARD 1

Write expressions from words • Grade 6 • Expressions & Equations • 6.EE.A.2.a

Write an expression for "5 less than y."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q1/5

Write an expression for "the product of 8 and n."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q2/5

Which expression means "the sum of x and 3, times 4"?

A. $4(x + 3)$

B. $x + 3 \times 4$

C. $4x + 3$

D. $4 + x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q3/5

True or false?

"Twice a number m" can be written as 2m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q4/5

True or false?

"7 fewer than k" can be written as $7 - k$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q5/5

SET 28 • 7.EE.A.2

BACKWARD 2

Write and interpret numerical expressions • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for "add 6 and 9, then multiply by 2."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q1/5

How does $5 \times (24 + 13)$ compare with $24 + 13$?

A. It is 5 more.

B. It is 5 less.

C. It is 5 times as large.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q2/5

Write an expression for "subtract 4 from 20, then divide by 2."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q3/5

True or false?

$(7 + 8) \times 3$ is three times as large as $7 + 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q4/5

True or false?

$10 \times (5 - 2)$ is 10 more than $5 - 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • 7.EE.A.2

FORWARD 1

Write equivalent expressions with the properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$4^{5} \times 4^{-2}$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q1/5

Write as a single power.

$(2^3)^4$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q2/5

Which number is equal to $5^6 \div 5^8$?

A. 25

B. 5^{14}

C. $\frac{1}{25}$

D. -25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q3/5

True or false?

$7^0 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q4/5

True or false?

$3^2 \times 3^3 = 9^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q5/5

SET 28 • 7.EE.A.2

FORWARD 2

Rewrite percent-growth expressions to reveal the growth factor • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3

A town's population after t years is P(1.02)^t.

Rewrite the expression to show the growth factor for each 10 years, rounded to the nearest hundredth.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q1/5

A price p is raised 10%, and then the new price is raised 10% again: p(1.1)(1.1).

Write this as a single factor times p, and give the total percent increase.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q2/5

A value after t years is 50(1.03)^t.

What is the yearly percent increase?

A. 50%

B. 3%

C. 0.3%

D. 103%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q3/5

True or false?

The expression 1,000(0.97)^t shows a value that decreases by 3% each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q4/5

True or false?

A 10% increase followed by a 10% decrease leaves a price unchanged.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q5/5

SET 29 • 7.EE.B.3 • EXPRESSIONS & EQUATIONS

Solve multistep problems with positive and negative rational numbers in any form

MAIN	Grade 7 main question	**7.EE.B.3**
BACKWARD 1	Interpret a fraction as division	**5.NF.B.3**
BACKWARD 2	Find a percent of a quantity	**6.RP.A.3.c**
FORWARD 1	Solve real-world problems with two linear equations	**8.EE.C.8.c**
FORWARD 2	Represent constraints with equations or inequalities	**HSA-CED.A.3**

SET 29 • 7.EE.B.3

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.3

Jon earns $16.50 per hour. He gets a raise of $\frac{1}{5}$ of his wage. Then he works 30 hours.

How much does he earn for the 30 hours?

Earnings: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q1/5

A board is $8\frac{1}{4}$ feet long. Ana cuts off 2.5 feet. She cuts the rest into 3 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q2/5

A 60-gallon tank is $\frac{3}{4}$ full. Then 12.5 gallons are used.

How many gallons are left?

A. 57.5

B. 47.5

C. 32.5

D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q3/5

True or false?

A $40 sweater is 25% off. Mia pays with a $50 bill. Her change is $20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q4/5

True or false?

Leo runs $2\frac{1}{2}$ miles on Monday and 3.4 miles on Tuesday. He runs 6.4 miles in all.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q5/5

SET 29 • 7.EE.B.3

BACKWARD 1

Interpret a fraction as division • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $\frac{3}{4}$ as a division expression. Then find its decimal value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q1/5

5 friends share 2 pizzas equally.

How much pizza does each friend get?

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q2/5

Which number is equal to $7 \div 8$?

A. 7.8

B. $\frac{7}{8}$

C. $\frac{8}{7}$

D. 0.78

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q3/5

True or false?

$\frac{9}{2} = 9 \div 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q4/5

True or false?

$3 \div 5 = \frac{5}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • 7.EE.B.3

BACKWARD 2

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 35% of 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q1/5

What is 12.5% of 64?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q2/5

What is 150% of 20?

A. 130

B. 30

C. 3

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q3/5

True or false?

60% of 25 is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q4/5

True or false?

5% of 300 is 1.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q5/5

SET 29 • 7.EE.B.3

FORWARD 1

Solve real-world problems with two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.c

Adult tickets cost $9 and child tickets cost $5. Someone buys 20 tickets for $140.

How many adult tickets were bought?

Adult tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q1/5

Plan A costs $20 plus $0.10 per minute. Plan B costs $0.35 per minute.

For how many minutes do the plans cost the same?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q2/5

Two numbers have a sum of 30 and a difference of 8.

What is the larger number?

A. 15

B. 11

C. 19

D. 22

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q3/5

True or false?

$x = 3$ and $y = 5$ is a solution of both $x + y = 8$ and $2x - y = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q4/5

True or false?

$x = 4$ and $y = 2$ is a solution of both $x + y = 6$ and $x - y = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q5/5

SET 29 • 7.EE.B.3

FORWARD 2

Represent constraints with equations or inequalities • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3

A baker has 24 cups of flour. Each muffin uses $\frac{3}{4}$ cup and each loaf uses 2.5 cups.

Write an inequality for m muffins and l loaves.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q1/5

A baker has 24 cups of flour. Each muffin uses $\frac{3}{4}$ cup and each loaf uses 2.5 cups.

Can she make 12 muffins and 7 loaves? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q2/5

A trip budget is at most $150. Gas costs $3.50 per gallon (g) and meals cost $12 each (f).

Which inequality shows the budget?

A. $3.5g + 12f \leq 150$

B. $12g + 3.5f \leq 150$

C. $3.5 + 12 \leq 150gf$

D. $3.5g + 12f \geq 150$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q3/5

True or false?

$m = 8$ and $l = 4$ makes $0.75m + 2.5l \leq 24$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q4/5

True or false?

$m = 10$ and $l = 9$ makes $0.75m + 2.5l \leq 24$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q5/5

SET 30 • 7.EE.B.3 • EXPRESSIONS & EQUATIONS

Use estimation to judge whether an answer is reasonable

MAIN	Grade 7 main question	**7.EE.B.3**
BACKWARD 1	Round decimals	**5.NBT.A.4**
BACKWARD 2	Use rounding to check the answer to a two-step word problem	**4.OA.A.3**
FORWARD 1	Estimate very large or very small quantities with powers of 10	**8.EE.A.3**
FORWARD 2	Define appropriate quantities and units for a model	**HSN-Q.A.2**

SET 30 • 7.EE.B.3

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.3

Estimate 19.8×4.1 by rounding each number to the nearest whole number.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q1/5

Kim says 48.7 + 31.2 = 799.

Use estimation to explain whether her answer is reasonable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q2/5

A meal costs $38.75. Ana adds a 20% tip.

Which is the best estimate of the total?

A. About $40

B. About $58

C. About $78

D. About $47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q3/5

True or false?

$6\frac{7}{8} \times 3\frac{1}{10}$ is about 7×3, so a product of about 21 is reasonable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q4/5

True or false?

A $59.99 jacket is 30% off. A sale price of about $18 is reasonable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q5/5

SET 30 • 7.EE.B.3

BACKWARD 1

Round decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 7.86 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q1/5

Round 12.348 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q2/5

Round 4.165 to the nearest hundredth.

A. 4.17

B. 4.1

C. 4.2

D. 4.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q3/5

True or false?

3.97 rounded to the nearest tenth is 4.0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q4/5

True or false?

15.52 rounded to the nearest whole number is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • 7.EE.B.3

BACKWARD 2

Use rounding to check the answer to a two-step word problem • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

A store sells 4 packs of cards for $52 each and charges $11 for shipping.

Round 52 to the nearest ten, and keep 4 and 11 unchanged, to estimate the total cost.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q1/5

A store sells 4 packs of cards for $52 each and charges $11 for shipping. Ty says the total is $2,019.

Round 52 to the nearest ten to decide whether his answer is reasonable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q2/5

A school buys 6 boxes of 48 pencils and gives away 95 pencils.

Round 48 and 95 to the nearest ten, and keep 6 unchanged. What is the best estimate of the pencils left?

A. 300

B. 200

C. 400

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q3/5

True or false?

A farm picks 312 apples on Monday and 289 apples on Tuesday, then sells 198 apples. Rounding each number to the nearest hundred gives an estimate of 400 apples left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q4/5

True or false?

A bus carries 38 riders on each of 7 trips and 21 riders on one more trip. Rounding 38 and 21 to the nearest ten, and keeping 7 unchanged, gives an estimate of 220 riders.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • 7.EE.B.3

FORWARD 1

Estimate very large or very small quantities with powers of 10 • Grade 8 • Expressions & Equations • 8.EE.A.3

Estimate 38,900,000 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q1/5

Estimate 0.00062 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q2/5

About how many times as large as 3×10^5 is 9×10^8?

A. 300

B. 3

C. 30

D. 3,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q3/5

True or false?

3×10^8 is a reasonable estimate for 296,000,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q4/5

True or false?

6×10^{-3} is about 6,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q5/5

SET 30 • 7.EE.B.3

FORWARD 2

Define appropriate quantities and units for a model • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2

You want to estimate how many times a heart beats in one day.

Which two quantities do you need, and what are their units?

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q1/5

You want to model the cost of gas for a car trip to another city.

Name three quantities you would define, with their units.

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q2/5

Which quantity is most useful for estimating the cost to paint the walls of a room?

A. The volume of the room

B. The height of the door

C. The number of windows

D. The area of the walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q3/5

True or false?

Gallons of water used per day is a reasonable quantity for modeling a family's water use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q4/5

True or false?

The color of a car is a useful quantity for modeling how much fuel it uses on a trip.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q5/5

SET 31 • 7.EE.B.4.a • EXPRESSIONS & EQUATIONS

Solve word problems that lead to equations of the form px + q = r

MAIN	Grade 7 main question	**7.EE.B.4.a**
BACKWARD 1	Solve one-step equations	**6.EE.B.7**
BACKWARD 2	Write an expression with a variable for a situation	**6.EE.B.6**
FORWARD 1	Solve equations with the variable on both sides	**8.EE.C.7.b**
FORWARD 2	Rearrange a formula to solve for a chosen quantity	**HSA-CED.A.4**

SET 31 • 7.EE.B.4.a

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A gym charges a $25 sign-up fee plus $18 per month. Ty has paid $151 in all.

Write and solve an equation to find the number of months m.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q1/5

Solve for x.

$4x + 9 = -11$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q2/5

Ava has $60. She buys movie tickets for $7.50 each and has $15 left.

Which equation finds the number of tickets t?

A. $60t - 7.5 = 15$

B. $7.5 - 60t = 15$

C. $7.5t + 60 = 15$

D. $60 - 7.5t = 15$

True or false?

$x = 6$ is the solution of $\frac{1}{2}x + 3 = 6$.

A. True

B. False

True or false?

A plumber charges $45 plus $30 per hour. A bill of $165 means the plumber worked 5 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q5/5

SET 31 • 7.EE.B.4.a

BACKWARD 1

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$x + 13 = 41$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q1/5

Solve for y.

$9y = 72$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q2/5

Solve for n.

$n - 6.4 = 3.6$

A. $n = 10$

B. $n = 2.8$

C. $n = -2.8$

D. $n = 23.04$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q3/5

True or false?

$k = 7$ is the solution of $5k = 35$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q4/5

True or false?

$p = 9$ is the solution of $p + 4 = 15$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • 7.EE.B.4.a

BACKWARD 2

Write an expression with a variable for a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

A ticket costs $12.

Write an expression for the cost of t tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q1/5

Ana has 5 more stickers than Ben. Ben has b stickers.

Write an expression for the number of stickers Ana has.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q2/5

A pizza has s slices. 4 friends share it equally.

Which expression gives the number of slices each friend gets?

A. $s - 4$

B. $4s$

C. $s + 4$

D. $\frac{s}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q3/5

True or false?

A pen costs p dollars. 3 pens cost 3p dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q4/5

True or false?

Mo is m years old. His age in 6 years is 6m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • 7.EE.B.4.a

FORWARD 1

Solve equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A charges $40 plus $5 per day. Store B charges $25 plus $8 per day.

After how many days do the stores charge the same amount?

Days: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q1/5

Solve for x.

$7x - 4 = 3x + 20$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q2/5

Solve for x.

$-2x + 9 = 4x - 15$

A. $x = -4$

B. $x = 1$

C. $x = 4$

D. $x = 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q3/5

True or false?

$x = -1$ is the solution of $5x + 3 = 2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q4/5

True or false?

$x = 2$ is the solution of $6x - 1 = 4x + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q5/5

SET 31 • 7.EE.B.4.a

FORWARD 2

Rearrange a formula to solve for a chosen quantity • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4

Solve $P = 2l + 2w$ for l.

$l =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q1/5

Solve $y = mx + b$ for x. Assume $m \neq 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q2/5

Solve $A = \frac{1}{2}bh$ for h, where the base $b > 0$.

A. $h = 2Ab$

B. $h = \frac{A}{2b}$

C. $h = \frac{b}{2A}$

D. $h = \frac{2A}{b}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q3/5

True or false?

For a speed $r > 0$, solving $d = rt$ for t gives $t = \frac{d}{r}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q4/5

True or false?

Solving $C = 5n + 20$ for n gives $n = 5C - 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q5/5

SET 32 • 7.EE.B.4.a • EXPRESSIONS & EQUATIONS

Solve word problems that lead to equations of the form p(x + q) = r

MAIN	Grade 7 main question	**7.EE.B.4.a**
BACKWARD 1	Evaluate expressions with parentheses	**5.OA.A.1**
BACKWARD 2	Use the distributive property to expand an expression	**6.EE.A.3**
FORWARD 1	Solve equations with the distributive property on both sides	**8.EE.C.7.b**
FORWARD 2	Solve quadratic equations by square roots or factoring	**HSA-REI.B.4.b**

SET 32 • 7.EE.B.4.a

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A rectangle is 7 cm wide and (x + 3) cm long. Its area is 63 cm².

Write and solve an equation to find x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q1/5

Solve for x.

$-3(x - 4) = 27$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q2/5

5 friends each buy a $4 snack and a drink that costs d dollars. They spend $32.50 in all.

Which equation finds d?

A. $d + 5(4) = 32.5$

B. $5(d + 4) = 32.5$

C. $5d + 4 = 32.5$

D. $4(d + 5) = 32.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q3/5

True or false?

$x = 2.5$ is the solution of $2(x + 1.5) = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q4/5

True or false?

The solution of $\frac{1}{3}(x - 6) = 4$ is $x = 6$.

A. True

B. False

SET 32 • 7.EE.B.4.a

BACKWARD 1

Evaluate expressions with parentheses • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$3 \times (8 + 4)$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q1/5

Evaluate.

$(15 - 6) \times 2$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q2/5

Evaluate.

$4 \times (9 - 2) + 1$

A. 25

B. 29

C. 36

D. 33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q3/5

True or false?

$2 \times (5 + 5) = 20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q4/5

True or false?

$(12 + 3) \times 2 = 18$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • 7.EE.B.4.a

BACKWARD 2

Use the distributive property to expand an expression • Grade 6 • Expressions & Equations • 6.EE.A.3

Rewrite using the distributive property.

$9(m + 2)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q1/5

Rewrite using the distributive property.

$4(3k + 5)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q2/5

Which expression is equivalent to 6(x + 1.5)?

A. 6x + 1.5

B. 6x + 9

C. 6x + 7.5

D. 7x + 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q3/5

True or false?

$$5(2 + y) = 10 + 5y$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q4/5

True or false?

$$8(p + 3) = 8p + 3$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 7.EE.B.4.a

FORWARD 1

Solve equations with the distributive property on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$3(x + 2) = 2(x + 7)$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q1/5

Solve for x.

$-2(x - 5) = 4(x + 1)$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q2/5

Solve for x.

$\frac{1}{2}(6x - 4) = 2(x + 3)$

A. $x = 2$

B. $x = 8$

C. $x = -8$

D. $x = 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q3/5

True or false?

$x = 4$ is the solution of $5(x - 1) = 3(x + 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q4/5

True or false?

$x = 3$ is the solution of $2(x + 6) = 4(x - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q5/5

SET 32 • 7.EE.B.4.a

FORWARD 2

Solve quadratic equations by square roots or factoring • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve for x. Give all solutions.

$(x + 3)^2 = 49$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q1/5

Solve for x. Give all solutions.

$x^2 = 81$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q2/5

Solve for x.

$(x - 2)(x + 5) = 0$

A. $x = -5$ only

B. $x = 2$ only

C. $x = -2$ or $x = 5$

D. $x = 2$ or $x = -5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q3/5

True or false?

The solutions of $2(x - 1)^2 = 18$ are $x = 4$ and $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q4/5

True or false?

$x^2 = 16$ has only one solution, $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q5/5

SET 33 • 7.EE.B.4.a • EXPRESSIONS & EQUATIONS

Compare an algebraic solution with an arithmetic solution

MAIN	Grade 7 main question	**7.EE.B.4.a**
BACKWARD 1	Solve two-step word problems	**3.OA.D.8**
BACKWARD 2	Decide whether a value makes an equation true	**6.EE.B.5**
FORWARD 1	Solve systems of two linear equations algebraically and by inspec…	**8.EE.C.8.b**
FORWARD 2	Justify a complete solution and find an invalid step	**HSA-REI.A.1**

SET 33 • 7.EE.B.4.a

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A gym charges $15 to join plus $8 for each class. Max paid $79 in all.

(a) Solve with arithmetic. (b) Write and solve an equation. (c) Compare the steps of the two solutions.

Classes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q1/5

Four friends each buy a ticket and a $3 popcorn. They pay $46 in all.

(a) Solve with arithmetic. (b) Write and solve an equation. (c) Compare the steps of the two solutions.

Ticket price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q2/5

An algebraic solution of $5x - 8 = 27$ adds 8 to both sides and then divides both sides by 5.

Which arithmetic steps match it?

A. Multiply 27 by 5, then add 8.

B. Add 8 to 27, then divide by 5.

C. Divide 27 by 5, then add 8.

D. Subtract 8 from 27, then divide by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q3/5

An arithmetic solution of $3(x + 4) = 33$ is $33 \div 3 - 4$.

For each arithmetic operation, name the algebra step that matches it. Then give x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q4/5

True or false?

An equation multiplies x by 2 and then adds 6: $2x + 6 = 20$. The arithmetic $(20 - 6) \div 2$ undoes those operations in reverse order, just as the algebra steps do.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q5/5

SET 33 • 7.EE.B.4.a

BACKWARD 1

Solve two-step word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

Jo has 45 stickers. She gives away 9. Then she shares the rest equally among 4 friends.

How many stickers does each friend get?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q1/5

There are 6 boxes with 8 pens in each box. 5 pens are lost.

How many pens are left?

Pens: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q2/5

A book has 90 pages. Kim reads 20 pages on Monday and 35 pages on Tuesday.

How many pages are left?

A. 15

B. 55

C. 145

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q3/5

True or false?

3 packs of 7 cards and 4 more cards make 25 cards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q4/5

True or false?

40 apples are shared equally in 5 bags. Then 2 apples are taken out of each bag. Each bag now has 8 apples.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • 7.EE.B.4.a

BACKWARD 2

Decide whether a value makes an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 4 a solution of 3x + 2 = 14? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q1/5

Which value of n makes n + 9 = 15 true?

A. 9

B. 24

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q2/5

Which of the values 2, 5, and 8 makes 7y = 35 true?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q3/5

True or false?

m = 3 makes 12 ÷ m = 4 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q4/5

True or false?

$a = 5$ makes $2a - 1 = 11$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • 7.EE.B.4.a

FORWARD 1

Solve systems of two linear equations algebraically and by inspection • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Solve the system.

$y = 2x$

$x + y = 18$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q1/5

Solve the system.

$y = x + 3$

$y = -2x + 9$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q2/5

Look at the system. How many solutions does it have?

$3x + 2y = 5$

$3x + 2y = 6$

A. Infinitely many solutions

B. Two solutions

C. No solution

D. One solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q3/5

True or false?

The system $y = 4x$ and $y = 4x + 1$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q4/5

True or false?

$x = 4$ and $y = 6$ is the solution of the system $x + y = 10$ and $x - y = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q5/5

SET 33 • 7.EE.B.4.a

FORWARD 2

Justify a complete solution and find an invalid step • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Solve $2(x + 3) = \frac{1}{2}(4x - 6) + x$. Write each step and the reason for it.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q1/5

Mo solved $\frac{x}{2} + 3 = x - 1$ like this:

Step 1: $x + 3 = 2x - 1$

Step 2: $x = 4$

Which step is invalid? Correct it and give the solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q2/5

Which reason justifies rewriting $4x - 7 = 2x + 5$ as $2x - 7 = 5$?

A. Commutative property of multiplication

B. Distributive property

C. Subtraction property of equality (subtract 2x from both sides)

D. Division property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q3/5

True or false?

Dividing both sides of $3(x - 2) = 3x - 6$ by 3 gives $x - 2 = x - 2$, so every number is a solution of the original equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q4/5

True or false?

Dividing both sides of $x^2 = 5x$ by x keeps every solution of the equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q5/5

SET 34 • 7.EE.B.4.b • EXPRESSIONS & EQUATIONS

Solve word problems that lead to inequalities of the form px + q > r or px + q < r

MAIN	Grade 7 main question	**7.EE.B.4.b**
BACKWARD 1	Write an inequality for a situation	**6.EE.B.8**
BACKWARD 2	Decide whether a value makes an inequality true	**6.EE.B.5**
FORWARD 1	Solve multistep equations with the variable on both sides — near…	**8.EE.C.7.b**
FORWARD 2	Solve linear inequalities with the variable on both sides	**HSA-REI.B.3**

SET 34 • 7.EE.B.4.b

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.b

Sam has $40. He buys a $12 shirt and some pairs of socks that cost $3.50 each.

Write and solve an inequality for the number of pairs of socks p he can buy.

Inequality and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q1/5

A plumber charges $55 plus $40 per hour. Ana can spend no more than $255.

Write and solve an inequality for the number of hours h.

Inequality and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q2/5

A club has $120 and earns $15 for each car it washes. It needs more than $300.

Which inequality and solution fit the problem?

A. $120c + 15 > 300$; $c > 2.375$

B. $15c > 300 + 120$; $c > 28$

C. $15c + 120 > 300$; $c > 12$

D. $15c + 120 < 300$; $c < 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q3/5

True or false?

A tank holds 500 liters and drains 25 liters per minute. The water stays above 200 liters while $500 - 25t > 200$, which is while $t < 12$ minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q4/5

True or false?

Kim has $20 and saves $6 each week. To have at least $80, she solves $6w + 20 \geq 80$, so she needs at most 10 weeks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q5/5

SET 34 • 7.EE.B.4.b

BACKWARD 1

Write an inequality for a situation • Grade 6 • Expressions & Equations • 6.EE.B.8

Riders must be at least 48 inches tall.

Write an inequality for a rider's height h.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q1/5

Write an inequality for "a temperature t below −2°F."

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q2/5

Which inequality means "x is more than 7"?

A. $x < 7$

B. $x > 7$

C. $x \leq 7$

D. $x = 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q3/5

True or false?

$x < 3$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q4/5

True or false?

5 is a solution of $x < 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • 7.EE.B.4.b

BACKWARD 2

Decide whether a value makes an inequality true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is 9 a solution of $2x > 16$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q1/5

Which value makes y + 4 < 10 true?

A. 8

B. 10

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q2/5

Which of the values 3, 6, and 9 make $n - 2 \geq 4$ true?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q3/5

True or false?

$x = 0$ makes $3x < 1$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q4/5

True or false?

k = 2 makes 5k > 10 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • 7.EE.B.4.b

FORWARD 1

Solve multistep equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Solve for x.

$0.5x + 7 = 2x - 2$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q1/5 • NEAREST RELATED

Solve for x.

$\frac{2}{3}(x + 3) = x - 1$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q2/5 • NEAREST RELATED

Solve for x.

$6 - 3x = x + 22$

A. $x = -4$

B. $x = 7$

C. $x = -7$

D. $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

$x = -2$ is the solution of $4(x + 5) = x + 14$.

A. True

B. False

True or false?

$x = 1$ is the solution of $9x - 3 = 2x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 34 • 7.EE.B.4.b

FORWARD 2

Solve linear inequalities with the variable on both sides • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve.

$5x - 3 > 2x + 9$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q1/5

Solve.

$-4(x + 1) \leq 2x + 8$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q2/5

Solve.

$7 - 2x < x - 8$

A. $x < -5$

B. $x > -5$

C. $x > 5$

D. $x < 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q3/5

True or false?

$x = 0$ is a solution of $3(x - 2) < x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q4/5

True or false?

The solution of $6x + 1 \geq 4x + 11$ is $x \leq 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q5/5

SET 35 • 7.EE.B.4.b • EXPRESSIONS & EQUATIONS

Graph the solution set of an inequality and interpret it in context

MAIN	Grade 7 main question	**7.EE.B.4.b**
BACKWARD 1	Graph an inequality on a number line	**6.EE.B.8**
BACKWARD 2	Read an inequality as positions on a number line	**6.NS.C.7.a**
FORWARD 1	Find the solution of a system of equations from its graph	**8.EE.C.8.a**
FORWARD 2	Graph the solutions of a linear inequality in two variables	**HSA-REI.D.12**

SET 35 • 7.EE.B.4.b

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.b

A taxi charges $3 plus $2 per mile. Ty can spend at most $15.

Solve $2m + 3 \leq 15$ for the miles m. Then graph the possible distances on the number line. (Distances cannot be negative.)



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q1/5

A freezer is at 2°C and cools 1.5°C each hour.

Solve $2 - 1.5t < -4$ to find when it is colder than −4°C. Then graph the solution on the number line.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q2/5

Tickets cost $6 each. Jo has $40 and must keep at least $10, so $6t \leq 30$. The graph shows the solution as whole numbers.

Which description fits the graph?



A. Jo can buy any number of tickets greater than 5.

B. Jo can buy 2.5 tickets.

C. Jo can buy exactly 5 tickets.

D. Jo can buy 0, 1, 2, 3, 4, or 5 tickets.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q3/5

A club has $30 and earns $5 for each ticket it sells. It needs more than $230.

Solve 5t + 30 > 230. Then graph the possible numbers of tickets (whole numbers) on the number line.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q4/5

True or false?

A plant is 5 cm tall and grows 2 cm per week. The graph shows the weeks w when the plant is taller than 11 cm, the solution of $2w + 5 > 11$.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

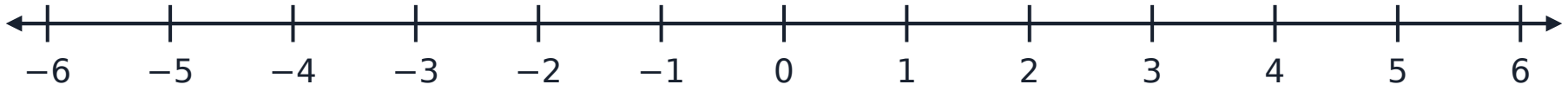
Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q5/5

SET 35 • 7.EE.B.4.b

BACKWARD 1

Graph an inequality on a number line • Grade 6 • Expressions & Equations • 6.EE.B.8

Graph $x < 4$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q1/5

Graph $x \geq -1$ on the number line.

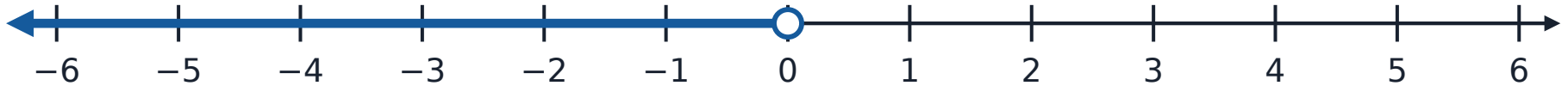


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q2/5

Which inequality does the graph show?



A. $x < 0$

B. $x \geq 0$

C. $x \leq 0$

D. $x > 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q3/5

True or false?

A closed circle on the graph of an inequality shows that the endpoint is a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q4/5

True or false?

The graph of $x > 5$ has a closed circle at 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • 7.EE.B.4.b

BACKWARD 2

Read an inequality as positions on a number line • Grade 6 • The Number System • 6.NS.C.7.a

-4 is to the left of 1 on a number line.

Write an inequality that shows this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q1/5

Which statement says that -2 is to the right of -6 on a number line?

A. $-2 < -6$

B. $-2 = -6$

C. $-6 > -2$

D. $-2 > -6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q2/5

What does $-8 < -3$ tell you about the positions of -8 and -3 on a number line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q3/5

True or false?

$-1 > -9$ because -1 is to the right of -9 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q4/5

True or false?

$-5 > 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q5/5

SET 35 • 7.EE.B.4.b

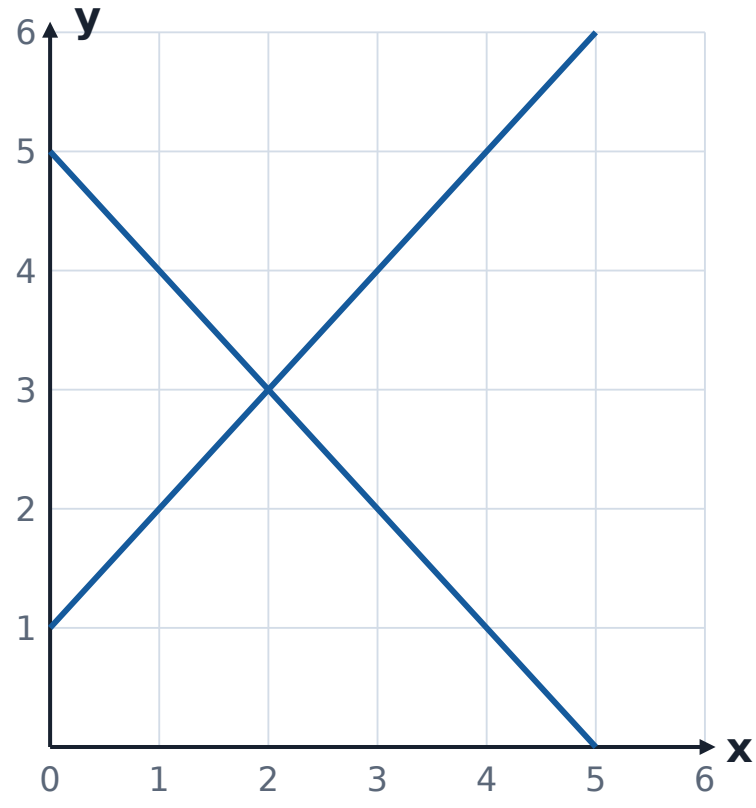
FORWARD 1

Find the solution of a system of equations from its graph • Grade 8 • Expressions & Equations • 8.EE.C.8.a

The graph shows $y = x + 1$ and $y = -x + 5$.

At what point do the lines intersect?

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q1/5

Two lines are graphed on the same grid.

What does their point of intersection represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q2/5

The graphs of two linear equations are parallel lines.

How many solutions does the system have?

A. One solution

B. Two solutions

C. Infinitely many solutions

D. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q3/5

True or false?

(1, 4) is the solution of the system $y = 2x + 2$ and $y = 3x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q4/5

True or false?

If two equations have the same graph, the system has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

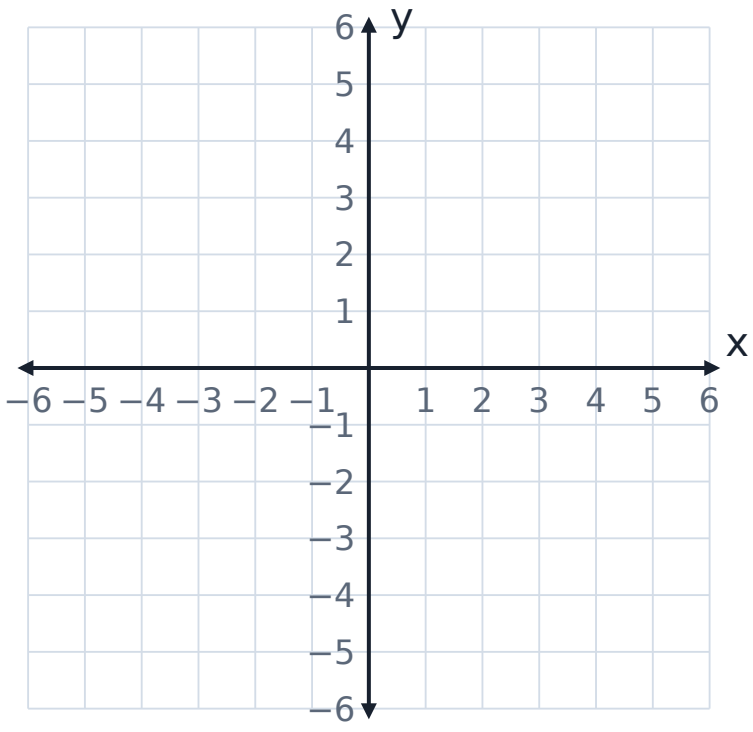
Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q5/5

SET 35 • 7.EE.B.4.b

FORWARD 2

Graph the solutions of a linear inequality in two variables • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

Graph $y > x + 1$ on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q1/5

Is (2, 5) a solution of $y \geq 2x$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q2/5

Which point is in the solution set of $y < -x + 3$?

A. (0, 0)

B. (0, 5)

C. (3, 1)

D. (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F2-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q3/5

True or false?

The boundary line of the graph of $y \leq 3x - 2$ is drawn solid.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q4/5

True or false?

The solution set of $y > 2$ is the region below the line $y = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q5/5

SET 36 • 7.G.A.1 • GEOMETRY

Find actual lengths and areas from a scale drawing

MAIN	Grade 7 main question	**7.G.A.1**
BACKWARD 1	Convert measurement units	**6.RP.A.3.d**
BACKWARD 2	Find the area of a rectangle	**4.MD.A.3**
FORWARD 1	Describe similar figures with dilations	**8.G.A.4**
FORWARD 2	Decide whether two triangles are similar	**HSG-SRT.A.2**

SET 36 • 7.G.A.1

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.1

A scale drawing of a garden uses the scale 1 cm : 2 m. The garden is 6 cm long on the drawing.

What is the actual length of the garden?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q1/5

A floor plan uses the scale 1 inch : 4 feet. A room is 3 inches by 2.5 inches on the plan.

What is the actual area of the room?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q2/5

A map uses the scale 1 cm : 25 km. Two towns are 4.4 cm apart on the map.

What is the actual distance between the towns?

A. 100 km

B. 110 km

C. 5.68 km

D. 29.4 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q3/5

True or false?

A drawing uses the scale 1 cm : 3 m. A 5 cm by 2 cm rectangle on the drawing has an actual area of 90 m².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q4/5

True or false?

A drawing uses the scale 1 cm : 10 cm. The actual area is 10 times the area on the drawing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q5/5

SET 36 • 7.G.A.1

BACKWARD 1

Convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

How many inches are in 4.5 feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q1/5

How many meters are in 350 centimeters?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q2/5

How many yards are in 27 feet?

A. 24

B. 9

C. 3

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q3/5

True or false?

2.5 kilometers = 2,500 meters

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q4/5

True or false?

6 feet = 60 inches

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q5/5

SET 36 • 7.G.A.1

BACKWARD 2

Find the area of a rectangle • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 12 m long and 9 m wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q1/5

A rectangle has an area of 84 square feet. It is 12 feet long.

How wide is it?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q2/5

A rectangle is 15 cm long and 6 cm wide.

What is its area?

A. 96 square centimeters

B. 90 square centimeters

C. 21 square centimeters

D. 42 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q3/5

True or false?

A square with 7-inch sides has an area of 49 square inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q4/5

True or false?

A rectangle 10 m long and 4 m wide has an area of 28 square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • 7.G.A.1

FORWARD 1

Describe similar figures with dilations • Grade 8 • Geometry • 8.G.A.4

A triangle has sides 3, 4, and 5 units. It is dilated by a scale factor of 2.5.

What are the new side lengths?

Side lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q1/5

A 4-by-6 rectangle is similar to a rectangle that is 10 by w. The 4 side matches the 10 side.

Find w.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q2/5

A figure is dilated by a scale factor of $\frac{1}{3}$. One side is 18 cm long.

How long is the matching side of the image?

A. $18\frac{1}{3}$ cm

B. 3 cm

C. 6 cm

D. 54 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q3/5

True or false?

Dilations keep angle measures the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q4/5

True or false?

A 2-by-3 rectangle and a 4-by-5 rectangle are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q5/5

SET 36 • 7.G.A.1

FORWARD 2

Decide whether two triangles are similar • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has angles of 40°, 60°, and 80°. Triangle DEF has angles of 40°, 60°, and 80°.

Are the triangles similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q1/5

One triangle has sides 6, 8, and 10. Another has sides 9, 12, and 15.

What is the scale factor from the first triangle to the second?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q2/5

Which two triangles must be similar?

A. Two right triangles

B. Two triangles that each have a 5 cm side

C. Two isosceles triangles

D. Two triangles that each have a 50° angle and a 70° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q3/5

True or false?

Two triangles are similar with a scale factor of 3. Each pair of corresponding sides has a ratio of 3 : 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q4/5

True or false?

All rectangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q5/5

SET 37 • 7.G.A.1 • GEOMETRY

Reproduce a scale drawing at a different scale

MAIN	Grade 7 main question	**7.G.A.1**
BACKWARD 1	Draw a polygon on the coordinate plane	**6.G.A.3**
BACKWARD 2	Compare the size of a product to the size of one factor	**5.NF.B.5.a**
FORWARD 1	Describe the effect of a dilation on coordinates	**8.G.A.3**
FORWARD 2	Verify the properties of dilations	**HSG-SRT.A.1**

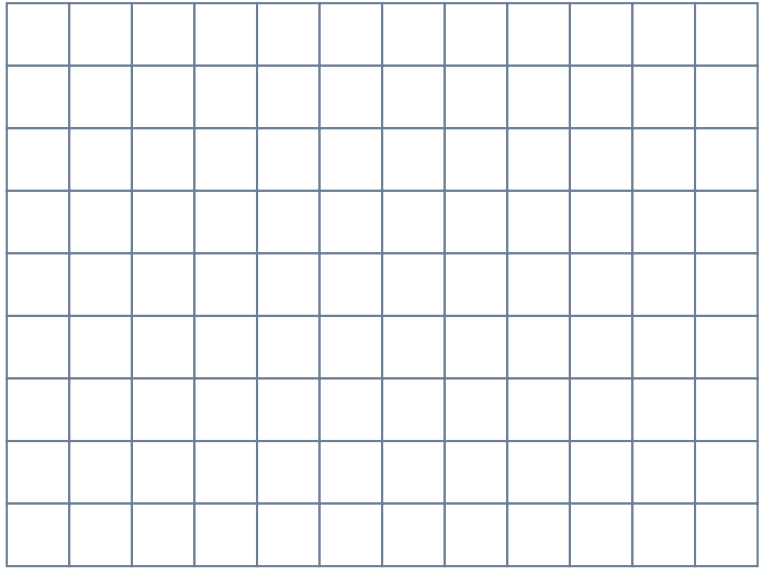
SET 37 • 7.G.A.1

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.1

A scale drawing of a garden is a rectangle 6 units long and 4 units wide. Its scale is 1 unit : 2 m.

Redraw the garden on the grid at a scale of 1 unit : 1 m.



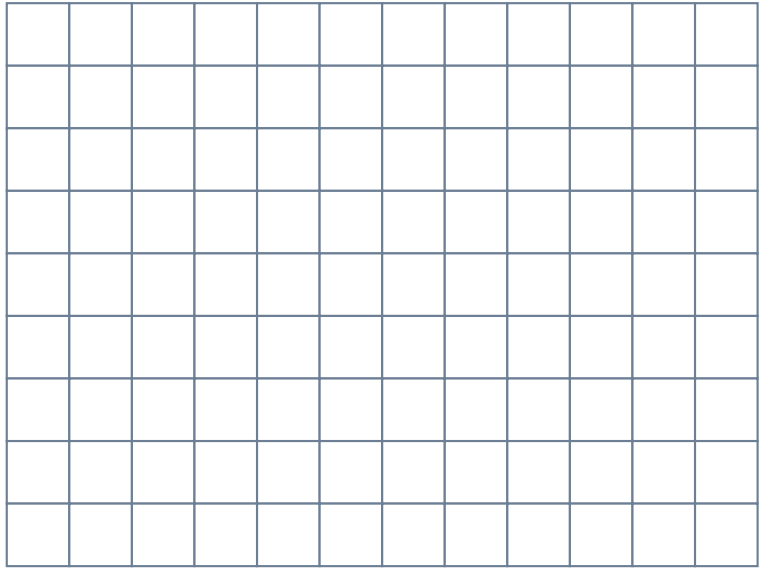
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q1/5

A rectangle on a drawing is 4 units by 2 units.

Draw the rectangle at a scale twice as large on the grid.



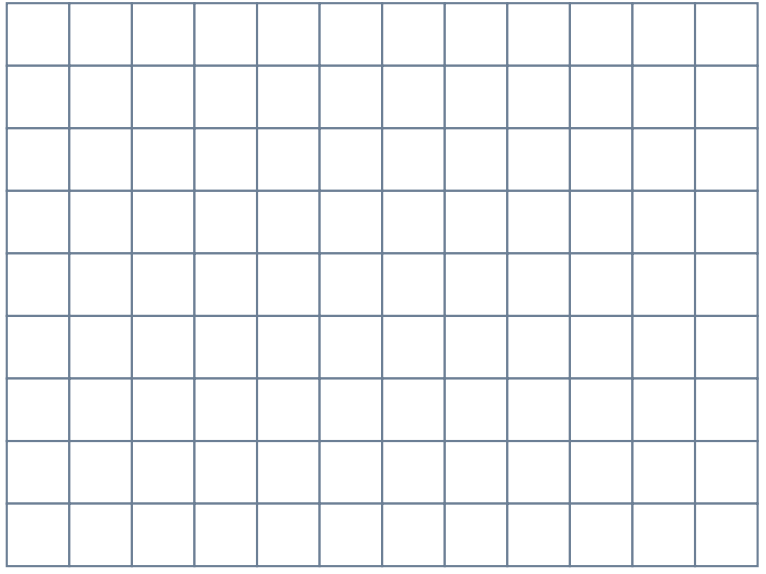
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q2/5

A floor plan shows a room as a rectangle 9 units by 6 units. Its scale is 1 unit : 2 feet.

Redraw the room on the grid at a scale of 1 unit : 3 feet.



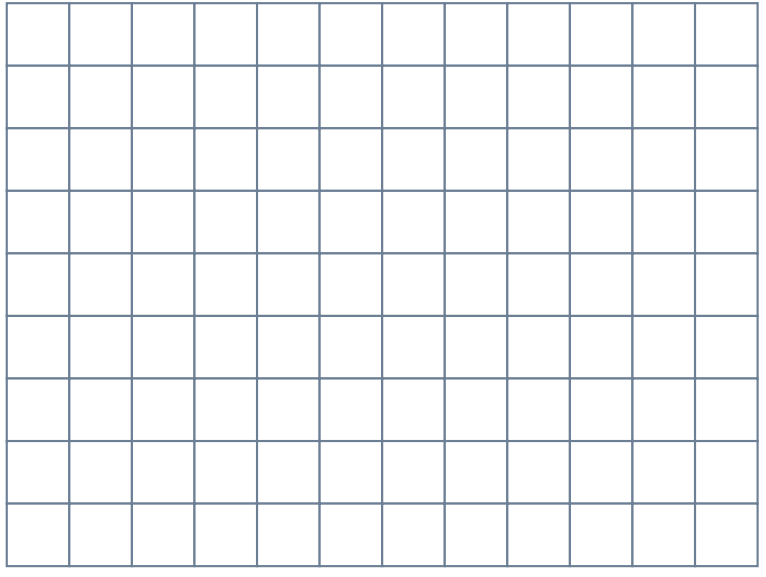
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q3/5

A scale drawing of a ramp is a right triangle whose two sides that form the right angle are 3 units and 4 units. Its scale is 1 unit : 5 m.

Redraw it on the grid at a scale of 1 unit : 2.5 m.



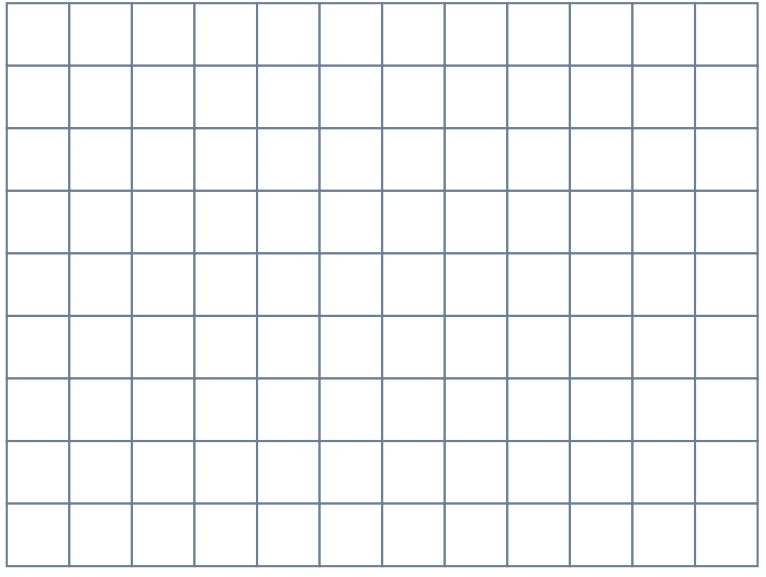
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q4/5

A scale drawing of a patio is a rectangle 5 units by 3 units. Its scale is 1 unit : 6 m.

Redraw the patio on the grid at a scale of 1 unit : 3 m.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q5/5

SET 37 • 7.G.A.1

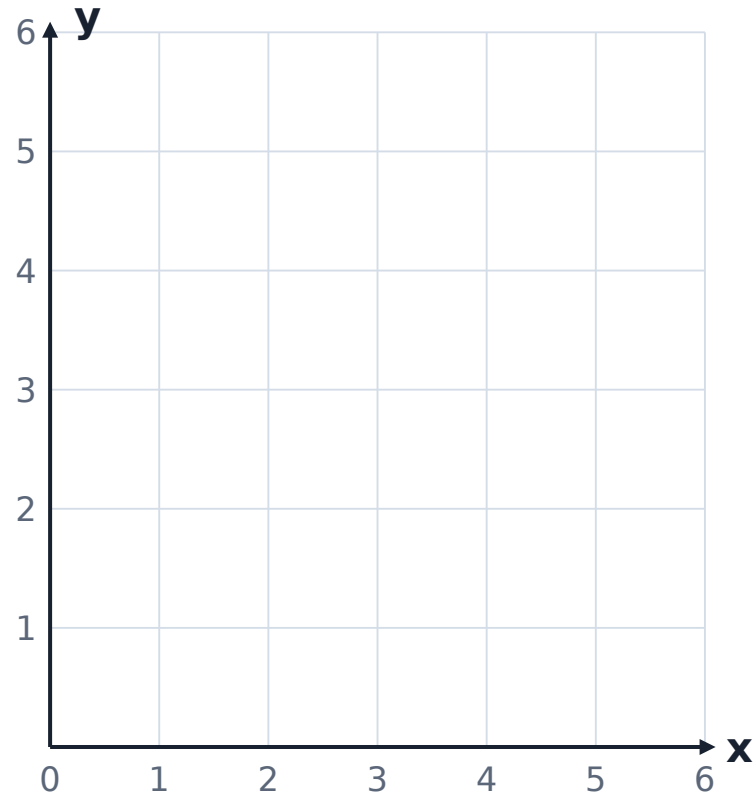
BACKWARD 1

Draw a polygon on the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Draw the polygon with vertices (1, 1), (5, 1), (5, 4), and (1, 4).

How long is the side from (1, 1) to (5, 1)?

Length: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

A rectangle has vertices (2, 1), (2, 6), (7, 6), and (7, 1).

How long is the side from (2, 1) to (7, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

Which point makes a square with (1, 1), (4, 1), and (4, 4)?

A. (4, 7)

B. (3, 1)

C. (1, 3)

D. (1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

True or false?

The points (0, 2) and (0, 7) are 5 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

True or false?

The side from (3, 2) to (3, 8) is 5 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 7.G.A.1

BACKWARD 2

Compare the size of a product to the size of one factor • Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a

Without multiplying, is 3×12 greater than or less than 12?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q1/5

Without multiplying, is $\frac{1}{2} \times 18$ greater than or less than 18?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q2/5

Which product is greater than 15?

A. 2×15

B. $\frac{3}{4} \times 15$

C. $\frac{1}{2} \times 15$

D. $\frac{1}{3} \times 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q3/5

True or false?

$\frac{2}{3} \times 9$ is less than 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q4/5

True or false?

4×8 is less than 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 7.G.A.1

FORWARD 1

Describe the effect of a dilation on coordinates • Grade 8 • Geometry • 8.G.A.3

The point $(4, -2)$ is dilated by a scale factor of 3 with the origin as the center.

What are the coordinates of the image?

Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q1/5

A triangle has vertices (2, 2), (6, 2), and (2, 4). It is dilated by a scale factor of $\frac{1}{2}$ with the origin as the center.

What are the vertices of the image?

Vertices: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q2/5

A dilation centered at the origin maps (5, 10) to (2, 4).

What is the scale factor?

A. $\frac{2}{5}$

B. $\frac{5}{2}$

C. -3

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q3/5

True or false?

A dilation by a scale factor of 2 with the origin as the center maps $(-3, 1)$ to $(-6, 2)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q4/5

True or false?

A dilation by a scale factor of 4 with the origin as the center maps (2, 3) to (6, 7).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 7.G.A.1

FORWARD 2

Verify the properties of dilations • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

A segment 5 cm long is dilated by a scale factor of 1.8.

How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q1/5

A line passes through the center of a dilation.

What is the image of the line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q2/5

A line that does not pass through the center of a dilation is dilated.

What is its image?

A. The same line

B. A curve

C. A perpendicular line

D. A parallel line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q3/5

True or false?

A dilation with a scale factor between 0 and 1 makes segments shorter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q4/5

True or false?

A dilation always changes the slope of a line that does not pass through its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q5/5

SET 38 • 7.G.A.2 • GEOMETRY

Draw triangles from given conditions

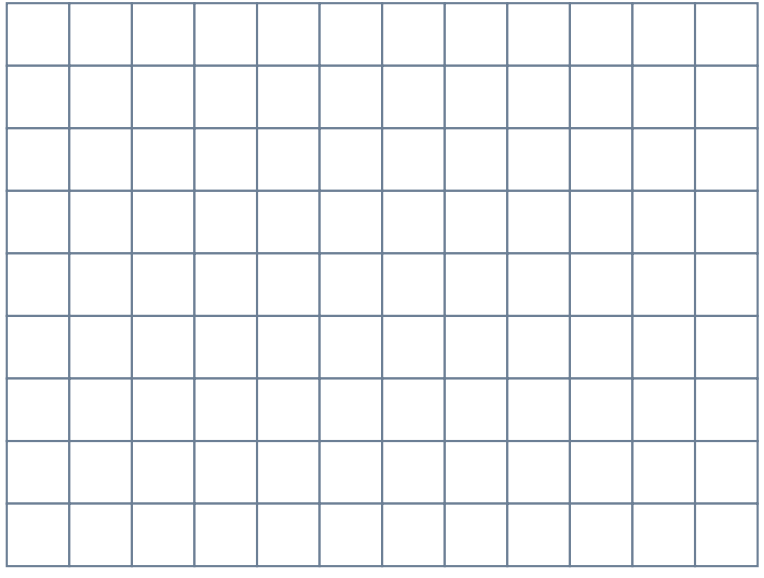
MAIN	Grade 7 main question	**7.G.A.2**
BACKWARD 1	Measure and draw angles with a protractor	**4.MD.C.6**
BACKWARD 2	Classify triangles by their angles	**4.G.A.2**
FORWARD 1	Use the Pythagorean theorem to find an unknown side length	**8.G.B.7**
FORWARD 2	Make formal constructions with a compass and straightedge	**HSG-CO.D.12**

SET 38 • 7.G.A.2

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.2

On the grid, draw a triangle with a right angle. The two sides that form the right angle should be 4 units and 6 units long.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q1/5

Use a ruler and a protractor. Draw a triangle with a 5 cm side that has a 50° angle at one end and a 60° angle at the other end.

Measure the third angle.

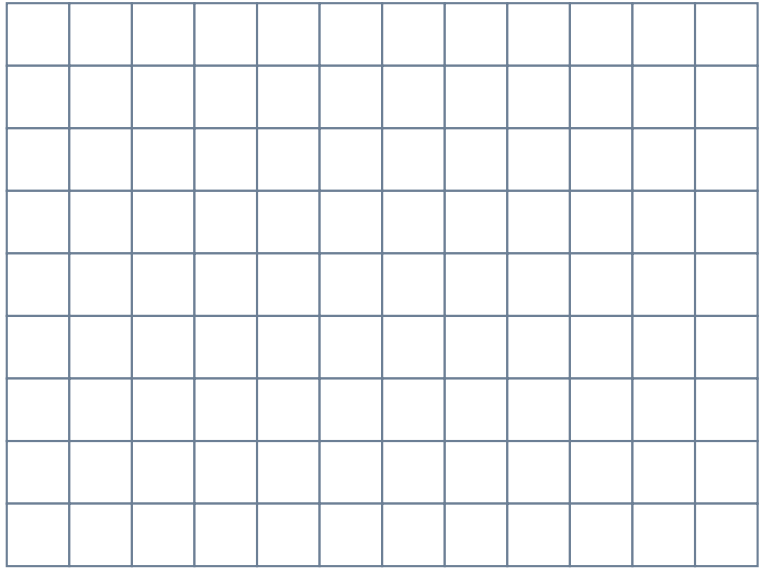
Third angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q2/5

On the grid, draw an obtuse triangle that has one side 5 units long along a grid line.

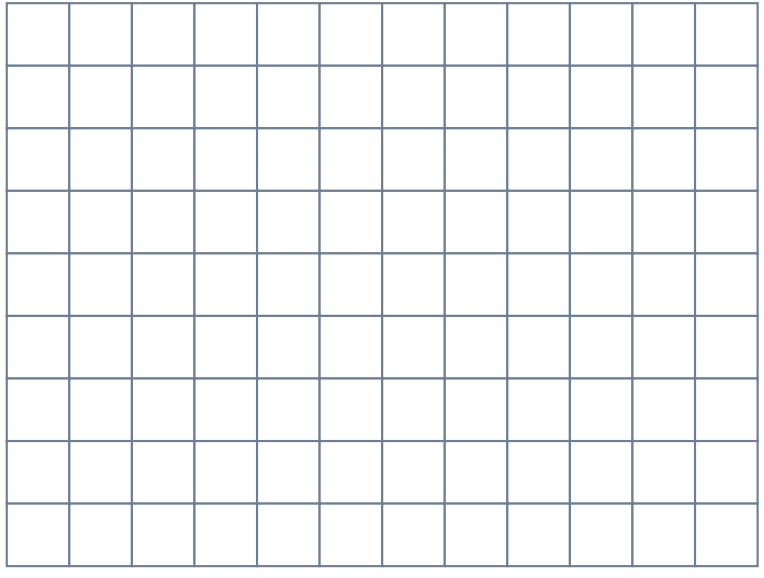


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q3/5

On the grid, draw an isosceles triangle that is not equilateral.

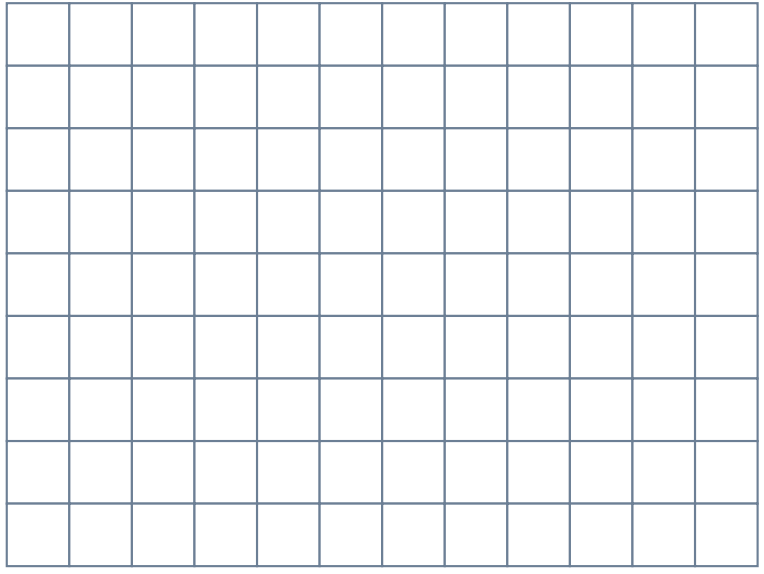


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q4/5

On the grid, draw a triangle with two sides that are each 5 units long and meet at a right angle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q5/5

SET 38 • 7.G.A.2

BACKWARD 1

Measure and draw angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

Use a protractor to draw a 65° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q1/5

One ray of an angle lines up with 0 on a protractor. The other ray passes through 120.

What is the measure of the angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q2/5

Which angle is greater than a right angle?

A. 80°

B. 90°

C. 110°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q3/5

True or false?

A protractor measures angles in degrees.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q4/5

True or false?

A 30° angle is an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • 7.G.A.2

BACKWARD 2

Classify triangles by their angles • Grade 4 • Geometry • 4.G.A.2

A triangle has a 90° angle.

Classify the triangle by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q1/5

A triangle has angles of 100°, 50°, and 30°.

Classify the triangle by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q2/5

What kind of triangle has three angles that are each less than 90°?

A. Obtuse triangle

B. Acute triangle

C. Straight triangle

D. Right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q3/5

True or false?

An obtuse triangle has exactly one obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q4/5

True or false?

A right triangle can have two right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 7.G.A.2

FORWARD 1

Use the Pythagorean theorem to find an unknown side length • Grade 8 • Geometry • 8.G.B.7

A right triangle has legs of 6 cm and 8 cm.

How long is the hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q1/5

A right triangle has a hypotenuse of 13 m and one leg of 5 m.

How long is the other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q2/5

A 10-foot ladder leans against a wall. The top of the ladder is 8 feet above the ground.

How far is the bottom of the ladder from the wall?

A. 6 feet

B. 12.8 feet

C. 18 feet

D. 2 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q3/5

True or false?

A right triangle with legs of 9 and 12 has a hypotenuse of 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q4/5

True or false?

A right triangle with legs of 1 and 1 has a hypotenuse of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q5/5

SET 38 • 7.G.A.2

FORWARD 2

Make formal constructions with a compass and straightedge • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12

Use a compass and a straightedge to construct the perpendicular bisector of segment AB. Leave your construction arcs showing.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q1/5

Use a compass and a straightedge to construct an equilateral triangle with side AB. Leave your construction arcs showing.

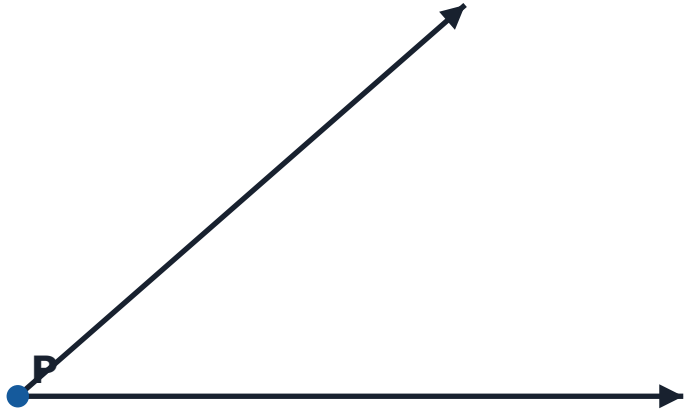


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q2/5

Use a compass and a straightedge to bisect the angle at P. Leave your construction arcs showing.

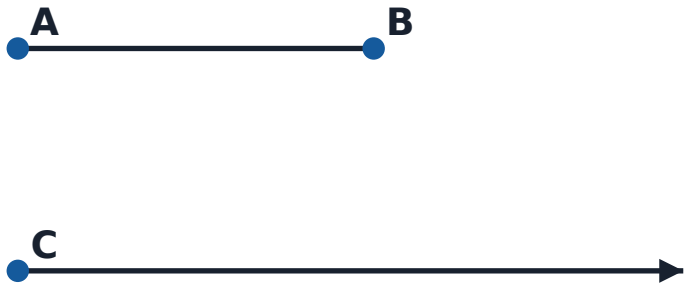


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q3/5

Use a compass and a straightedge to copy segment AB onto the ray, starting at point C. Leave your construction arc showing.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q4/5

True or false?

In a compass-and-straightedge construction, a protractor is used to measure angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q5/5

SET 39 • 7.G.A.2 • GEOMETRY

Decide whether given conditions make one triangle, more than one triangle, or no triangle

MAIN	Grade 7 main question	**7.G.A.2**
BACKWARD 1	Find an unknown angle when angles are joined	**4.MD.C.7**
BACKWARD 2	Find the perimeter of a polygon from its side lengths	**3.MD.D.8**
FORWARD 1	Use the converse of the Pythagorean theorem to test for a right tr…	**8.G.B.6**
FORWARD 2	Explain triangle congruence criteria with rigid motions	**HSG-CO.B.8**

SET 39 • 7.G.A.2

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.2

Different triangles means triangles that are not congruent; position and orientation do not matter.

How many different triangles have sides of 3 cm, 4 cm, and 9 cm?

A. Exactly one triangle

B. More than one triangle

C. No triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q1/5

Different triangles means triangles that are not congruent; position and orientation do not matter.

A triangle has angles of 30°, 60°, and 90°. No side lengths are given.

How many different triangles fit these conditions? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q2/5

Different triangles means triangles that are not congruent; position and orientation do not matter.

How many different triangles have sides of 5 cm, 6 cm, and 7 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q3/5

True or false?

Angles of 80° and 110° cannot both be angles of the same triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q4/5

True or false?

Three side lengths that form a triangle can make two triangles that are not congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q5/5

SET 39 • 7.G.A.2

BACKWARD 1

Find an unknown angle when angles are joined • Grade 4 • Measurement & Data • 4.MD.C.7

Angle ABC is split into two angles that measure 35° and 50°.

What is the measure of angle ABC?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q1/5

A 90° angle is split into two angles. One measures 28°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q2/5

A 120° angle is split into two angles. One measures 45°.

What does the other angle measure?

A. 45°

B. 165°

C. 75°

D. 85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A 45° angle and a 45° angle joined together make a 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q4/5

True or false?

A 120° angle and a 70° angle joined together make a 180° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 7.G.A.2

BACKWARD 2

Find the perimeter of a polygon from its side lengths • Grade 3 • Measurement & Data • 3.MD.D.8

A triangle has sides of 7 cm, 9 cm, and 12 cm.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q1/5

A triangle has a perimeter of 30 inches. Two of its sides are 8 inches and 10 inches.

How long is the third side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q2/5

A triangle has sides of 5 m, 5 m, and 6 m.

What is its perimeter?

A. 11 m

B. 21 m

C. 150 m

D. 16 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q3/5

True or false?

A triangle with sides of 4 feet, 6 feet, and 7 feet has a perimeter of 17 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q4/5

True or false?

A triangle with three 9 cm sides has a perimeter of 18 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • 7.G.A.2

FORWARD 1

Use the converse of the Pythagorean theorem to test for a right triangle • Grade 8 • Geometry • 8.G.B.6

Is a triangle with sides of 7, 24, and 25 a right triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q1/5

Is a triangle with sides of 6, 7, and 9 a right triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q2/5

Which side lengths form a right triangle?

A. 8, 15, 17

B. 4, 5, 6

C. 5, 6, 8

D. 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q3/5

True or false?

A triangle with sides of 9, 40, and 41 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q4/5

True or false?

A triangle with sides of 3, 3, and 4 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 7.G.A.2

FORWARD 2

Explain triangle congruence criteria with rigid motions • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8

Triangles ABC and DEF have AB = DE, BC = EF, and angle B = angle E. A rigid motion moves B onto E and ray BA onto ray ED.

Explain why A lands on D and C lands on F.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q1/5

Triangles PQR and STU have PQ = ST, QR = TU, and PR = SU. A rigid motion moves P onto S and Q onto T, with R and U on the same side of ST.

Explain why R must land on U.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q2/5

Triangles ABC and DEF have angle A = angle D, AB = DE, and angle B = angle E. A rigid motion puts A on D and B on E, with C and F on the same side.

Why must C land on F?

A. All triangles with a side of length AB are congruent.

B. A rigid motion stretches the sides until they match.

C. The third angles add to 180°.

D. The rays from D and E at the equal angles cross at only one point, so C and F are the same point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q3/5

True or false?

Rigid motions (translations, rotations, and reflections) keep side lengths and angle measures the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q4/5

True or false?

Two triangles with all three pairs of angles equal can always be mapped onto each other by a rigid motion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q5/5

SET 40 • 7.G.A.3 • GEOMETRY

Describe the cross sections made by slicing right prisms and pyramids

MAIN	Grade 7 main question	**7.G.A.3**
BACKWARD 1	Identify the faces of solids from their nets	**6.G.A.4**
BACKWARD 2	Name shapes by their number of sides and faces	**2.G.A.1**
FORWARD 1	Find the volumes of cylinders, cones, and spheres	**8.G.C.9**
FORWARD 2	Identify cross sections of three-dimensional objects	**HSG-GMD.B.4**

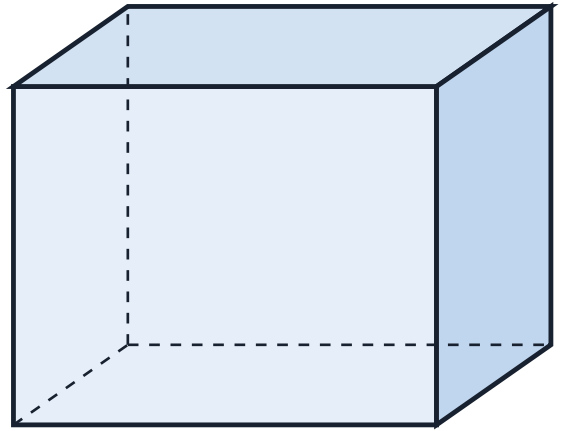
SET 40 • 7.G.A.3

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.3

A right rectangular prism is sliced by a plane parallel to its base.

What is the shape of the cross section?



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q1/5

A right square pyramid is sliced by a plane parallel to its base, strictly between the base and the top vertex.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q2/5

A cube is sliced by a plane that is parallel to one of its faces.

What is the shape of the cross section?

A. A triangle

B. A pentagon

C. A square

D. A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q3/5

True or false?

A right square pyramid is sliced by a plane that passes through its top vertex and the center of its base, perpendicular to the base. The cross section is a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q4/5

True or false?

A right rectangular prism sliced by a plane parallel to its base can have a triangle as its cross section.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q5/5

SET 40 • 7.G.A.3

BACKWARD 1

Identify the faces of solids from their nets • Grade 6 • Geometry • 6.G.A.4

How many faces does a rectangular prism have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q1/5

A net has 1 square and 4 triangles.

What solid does it fold into?

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q2/5

Which solid has a net of 2 triangles and 3 rectangles?

A. Square pyramid

B. Rectangular prism

C. Triangular prism

D. Cube

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q3/5

True or false?

The net of a cube has 6 squares.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q4/5

True or false?

A triangular pyramid has 5 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • 7.G.A.3

BACKWARD 2

Name shapes by their number of sides and faces • Grade 2 • Geometry • 2.G.A.1

How many sides does a pentagon have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q1/5

A shape has 6 sides and 6 angles.

What is the name of the shape?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q2/5

Which shape has 4 sides?

A. Quadrilateral

B. Hexagon

C. Pentagon

D. Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q3/5

True or false?

A cube has 6 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q4/5

True or false?

A triangle has 4 angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • 7.G.A.3

FORWARD 1

Find the volumes of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 2 cm and a height of 9 cm.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q1/5

A cone has a radius of 6 inches and a height of 5 inches.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q2/5

A sphere has a radius of 3 m.

What is its volume in terms of π?

A. 12π cubic meters

B. 108π cubic meters

C. 27π cubic meters

D. 36π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q3/5

True or false?

A cone and a cylinder have the same base and height. The cone's volume is $\frac{1}{3}$ of the cylinder's volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 2 and a height of 5 has a volume of 10π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q5/5

SET 40 • 7.G.A.3

FORWARD 2

Identify cross sections of three-dimensional objects • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

A right circular cylinder is sliced by a plane parallel to its base that cuts through its interior.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q1/5

A right circular cone is sliced by a plane that contains its axis (the line through the apex and the center of the base).

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q2/5

A sphere is sliced by a plane that passes through its interior.

What is the shape of the cross section?

A. A triangle

B. A square

C. An oval that is not a circle

D. A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q3/5

True or false?

Spinning a rectangle around one of its sides makes a cylinder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q4/5

True or false?

A cube can never have a cross section shaped like a hexagon.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q5/5

SET 41 • 7.G.B.4 • GEOMETRY

Find the circumference of a circle

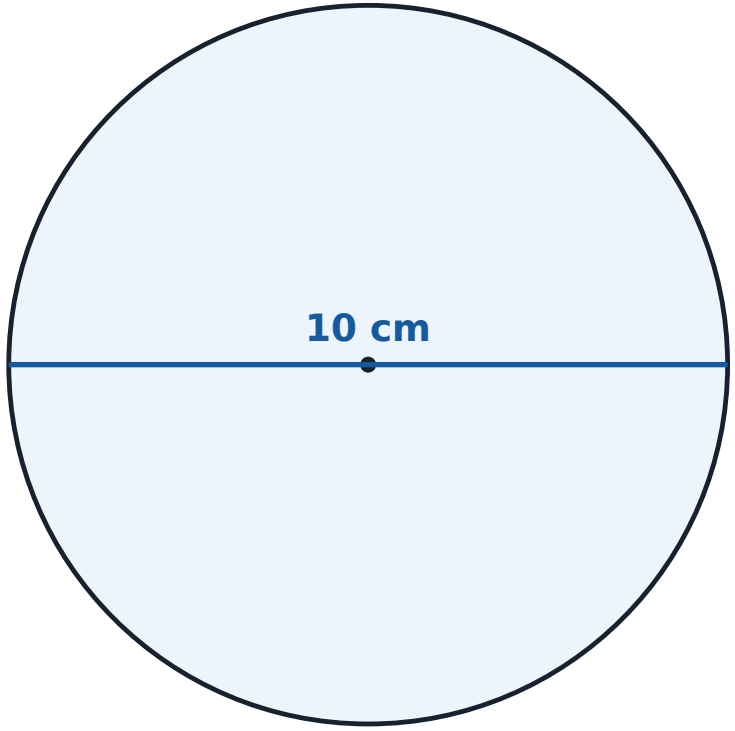
MAIN	Grade 7 main question	**7.G.B.4**
BACKWARD 1	Find the perimeter of a rectangle	**4.MD.A.3**
BACKWARD 2	Multiply decimals	**5.NBT.B.7**
FORWARD 1	Find the volume of a cylinder	**8.G.C.9**
FORWARD 2	Find the length of an arc of a circle	**HSG-C.B.5**

SET 41 • 7.G.B.4

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.4

Find the circumference of the circle. Use 3.14 for π.



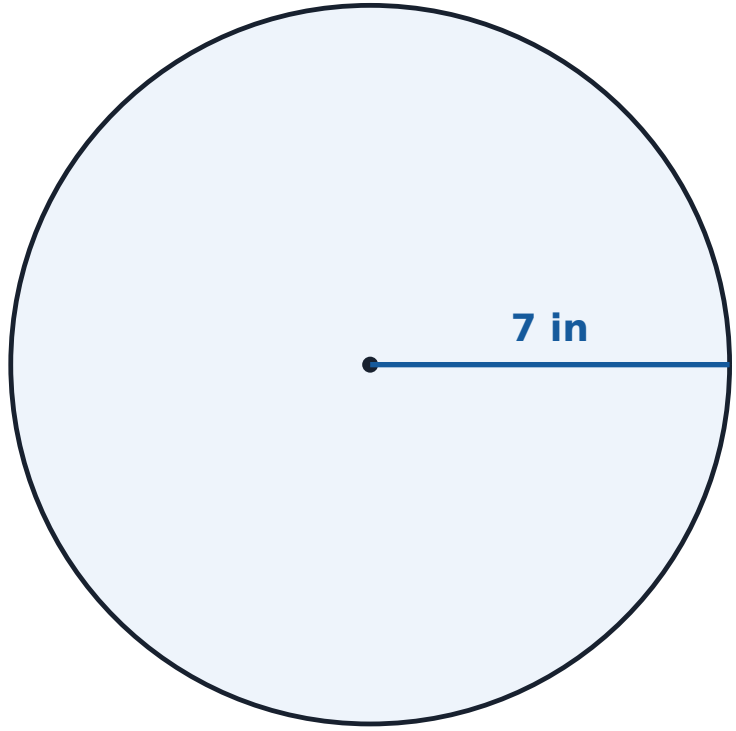
Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q1/5

Find the circumference of the circle. Use $\frac{22}{7}$ for π.



Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q2/5

A circular track has a circumference of 62.8 m.

What is its diameter? Use 3.14 for π.

A. 31.4 m

B. 20 m

C. 10 m

D. 197.2 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q3/5

True or false?

A wheel has a radius of 15 inches. In one full turn it rolls about 94.2 inches. (Use 3.14 for π.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q4/5

True or false?

A circle with a radius of 4 m has a circumference of 4π m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q5/5

SET 41 • 7.G.B.4

BACKWARD 1

Find the perimeter of a rectangle • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 9 m long and 4 m wide.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q1/5

A square has 8 cm sides.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q2/5

A rectangle is 12 feet long and 5 feet wide.

What is its perimeter?

A. 29 feet

B. 17 feet

C. 60 feet

D. 34 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q3/5

True or false?

A rectangle 6 inches long and 3 inches wide has a perimeter of 18 inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q4/5

True or false?

A square with 10 m sides has a perimeter of 100 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 7.G.B.4

BACKWARD 2

Multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

3.14×6

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q1/5

Multiply.

3.14×2.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q2/5

Multiply.

3.14×12

A. 37.68

B. 36.48

C. 376.8

D. 15.14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q3/5

True or false?

$3.14 \times 10 = 31.4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q4/5

True or false?

$3.14 \times 4 = 12.16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • 7.G.B.4

FORWARD 1

Find the volume of a cylinder • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 5 feet and a height of 4 feet.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q1/5

A cylinder has a diameter of 6 cm and a height of 10 cm.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q2/5

A cylinder has a volume of 90π cubic meters and a radius of 3 m.

What is its height?

A. 30 m

B. 3 m

C. 10 m

D. 15 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q3/5

True or false?

Doubling the height of a cylinder doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 1 and a height of 1 has a volume of 2π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q5/5

SET 41 • 7.G.B.4

FORWARD 2

Find the length of an arc of a circle • High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5

A circle has a radius of 6 cm.

Find the length of an arc with a central angle of 90°, in terms of π.

Arc length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q1/5

A circle has a circumference of 40 inches.

How long is an arc with a central angle of 45°?

Arc length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q2/5

A circle has a radius of 10 m.

What is the length of an arc with a central angle of 60°?

A. $\frac{20}{3}\pi$ m

B. $\frac{5}{3}\pi$ m

C. $\frac{10}{3}\pi$ m

D. 60π m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q3/5

Explain why an arc with a central angle of n° in a circle of radius r has a length of $\frac{n}{360} \times 2\pi r$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q4/5

True or false?

In a circle with a radius of 3, an arc with a central angle of 120° has a length of 3π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q5/5

SET 42 • 7.G.B.4 • GEOMETRY

Find the area of a circle

MAIN	Grade 7 main question	**7.G.B.4**
BACKWARD 1	Evaluate powers with whole-number exponents	**6.EE.A.1**
BACKWARD 2	Evaluate a formula	**6.EE.A.2.c**
FORWARD 1	Find the volume of a cone	**8.G.C.9**
FORWARD 2	Find the area of a sector of a circle	**HSG-C.B.5**

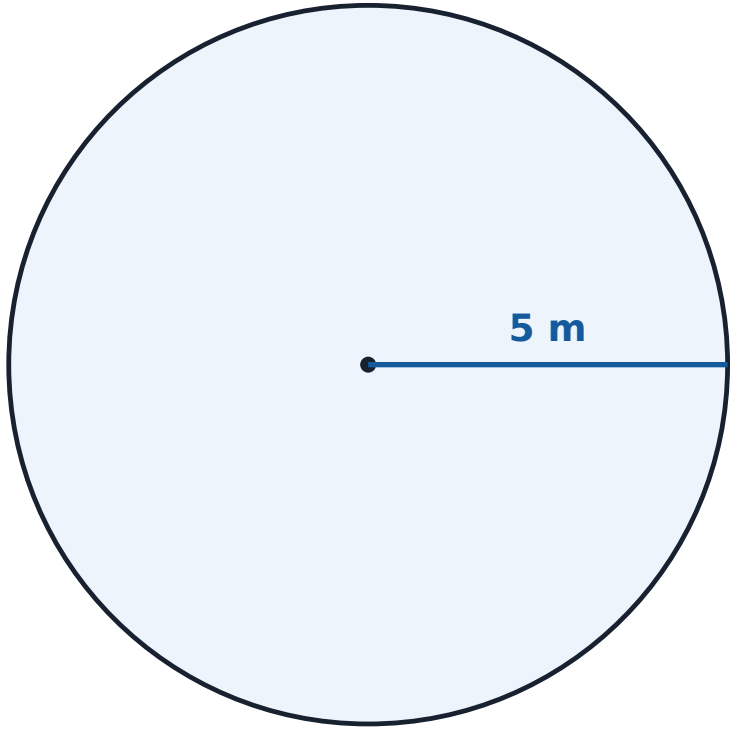
SET 42 • 7.G.B.4

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.4

Find the area of the circle.

Use 3.14 for π.



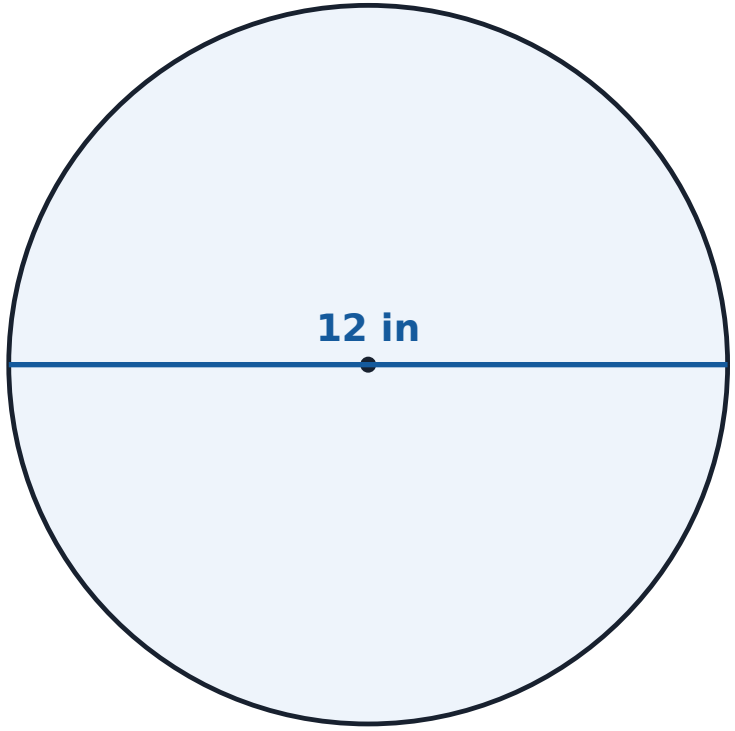
Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q1/5

Find the area of the circle in terms of π.



Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q2/5

A circular rug has a radius of 3 feet.

What is its area? Use 3.14 for π.

A. 113.04 square feet

B. 28.26 square feet

C. 9.42 square feet

D. 18.84 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q3/5

True or false?

A circle with a radius of 10 cm has an area of 100π square centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q4/5

True or false?

A circle with a diameter of 8 m has an area of 64π square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q5/5

SET 42 • 7.G.B.4

BACKWARD 1

Evaluate powers with whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

7^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q1/5

Find the value.

1.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q2/5

Find the value.

4^3

A. 7

B. 64

C. 12

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q3/5

True or false?

$9^2 = 81$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q4/5

True or false?

$6^2 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • 7.G.B.4

BACKWARD 2

Evaluate a formula • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use $A = s^2$.

Find A when s = 11.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q1/5

Use $A = 3.14r^2$.

Find A when r = 2.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q2/5

Use $A = \frac{1}{2}bh$.

Find A when b = 8 and h = 5.

A. 13

B. 20

C. 40

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q3/5

True or false?

Use $V = s^3$. When $s = 2$, $V = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q4/5

True or false?

Use A = lw. When l = 3.5 and w = 2, A = 5.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q5/5

SET 42 • 7.G.B.4

FORWARD 1

Find the volume of a cone • Grade 8 • Geometry • 8.G.C.9

A cone has a radius of 3 cm and a height of 8 cm.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q1/5

A cone has a radius of 6 feet and a height of 7 feet.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q2/5

A cone has a base area of 30 square inches and a height of 9 inches.

What is its volume?

A. 270 cubic inches

B. 90 cubic inches

C. 30 cubic inches

D. 39 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q3/5

True or false?

A cone with a radius of 3 and a height of 3 has a volume of 9π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q4/5

True or false?

The volume of a cone is $\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q5/5

SET 42 • 7.G.B.4

FORWARD 2

Find the area of a sector of a circle • High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5

A circle has a radius of 4 m.

Find the area of a sector with a central angle of 90°, in terms of π.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q1/5

A pizza with a radius of 8 inches is cut into 8 equal slices.

What is the area of one slice, in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q2/5

A circle has a radius of 6 cm.

What is the area of a sector with a central angle of 120°?

A. 12π square centimeters

B. 24π square centimeters

C. 4π square centimeters

D. 36π square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q3/5

Explain why a sector with a central angle of n° in a circle of radius r has an area of $\frac{n}{360} \times \pi r^2$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q4/5

True or false?

In a circle with a radius of 6, a sector with a central angle of 60° has an area of 12π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

High School (Grade 9 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q5/5

SET 43 • 7.G.B.4 • GEOMETRY

Explain the relationship between the circumference and the area of a circle

MAIN	Grade 7 main question	**7.G.B.4**
BACKWARD 1	Find the area of a parallelogram	**6.G.A.1**
BACKWARD 2	Identify equivalent expressions	**6.EE.A.4**
FORWARD 1	Use square roots to solve equations of the form $x^2 = p$	**8.EE.A.2**
FORWARD 2	Give an informal argument for circle and volume formulas	**HSG-GMD.A.1**

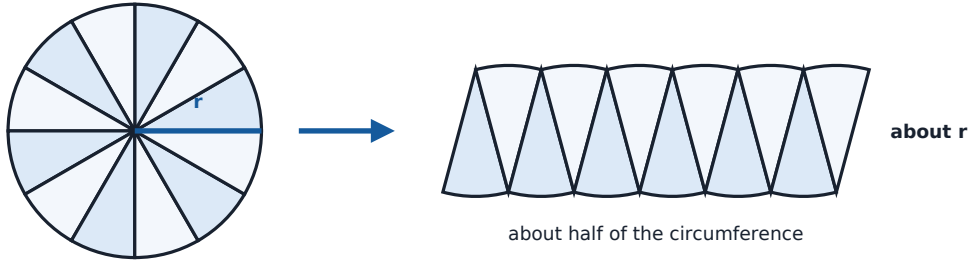
SET 43 • 7.G.B.4

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.4

The circle with radius r is cut into equal wedges and rearranged into a shape close to a parallelogram. Its base is half the circumference.

Write the base in terms of r.



Base: _____

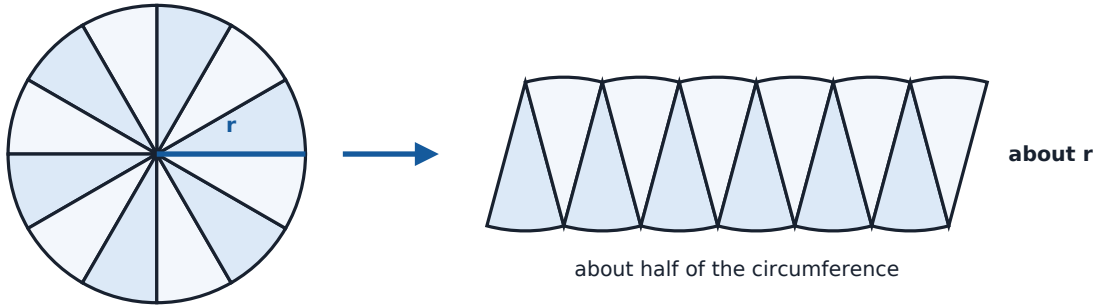
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q1/5

The circle with radius r is cut into equal wedges and rearranged as shown.

Explain how this shows that the area of a circle is πr^2.



Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q2/5

A circle rearranged into a near-parallelogram has a base of πr and a height of r.

What is the area?

A. πr^2

B. $2\pi r^2$

C. πr

D. $2\pi r$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q3/5

True or false?

A circle has circumference C and radius r. Its area equals $\frac{1}{2} \times C \times r$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q4/5

True or false?

A circle with a circumference of 10π has an area of 10π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q5/5

SET 43 • 7.G.B.4

BACKWARD 1

Find the area of a parallelogram • Grade 6 • Geometry • 6.G.A.1

A parallelogram has a base of 9 cm and a height of 4 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

A parallelogram has a base of 12 m and a height of 6.5 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

A parallelogram is cut along a segment from one top corner straight down to the base, at a right angle to the base. The triangle that is cut off is slid to the other end, so its slanted side lines up with the other slanted side.

What shape is made?

A. A pentagon

B. A trapezoid that is not a rectangle

C. A circle

D. A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

A parallelogram and a rectangle with the same base and the same height have the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

A parallelogram has a base of 10 and slanted sides of 6. Its area is 60.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • 7.G.B.4

BACKWARD 2

Identify equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.4

Is $\frac{1}{2} \times 2 \times r$ equivalent to r? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q1/5

Which expression is equivalent to $3 \times r \times r$?

A. $6r$

B. $3 + r^2$

C. $(3r)^2$

D. $3r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q2/5

Write an expression equivalent to $x \cdot x \cdot x$ using an exponent.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q3/5

True or false?

2(4y) and 8y are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q4/5

True or false?

a^2 and $2a$ are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • 7.G.B.4

FORWARD 1

Use square roots to solve equations of the form $x^2 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve for x. Give all solutions.

$x^2 = 64$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q1/5

A circle has an area of 49π square centimeters.

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q2/5

Solve for x.

$x^3 = 27$

A. $x = 3$

B. $x = 9$

C. $x = 6$

D. $x = 81$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q3/5

True or false?

$\sqrt{144} = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q4/5

True or false?

$\sqrt{50}$ is a whole number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q5/5

SET 43 • 7.G.B.4

FORWARD 2

Give an informal argument for circle and volume formulas • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Use the wedge argument to explain why the area of a circle is πr^2.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q1/5

Two stacks of 20 identical coins have the same height. One stack is straight and one is slanted.

Do they have the same volume? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q2/5

The circumference of a circle is $C = \pi d$.

What is the ratio of C to d?

A. 2π

B. π

C. 2

D. r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q3**

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q3/5

True or false?

As a circle is cut into more and more wedges, the rearranged shape gets closer to a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q4/5

True or false?

Cavalieri's principle says that any two solids with the same height have the same volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q5/5

SET 44 • 7.G.B.5 • GEOMETRY

Write and solve equations for supplementary and complementary angles

MAIN	Grade 7 main question	**7.G.B.5**
BACKWARD 1	Find an unknown part of a right angle or a straight angle	**4.MD.C.7**
BACKWARD 2	Solve one-step equations	**6.EE.B.7**
FORWARD 1	Use the angle sum and exterior angles of triangles	**8.G.A.5**
FORWARD 2	Prove angle relationships formed by parallel lines and a transvers…	**HSG-CO.C.9**

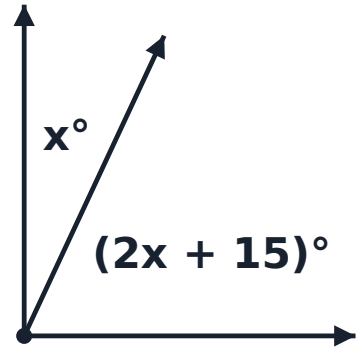
SET 44 • 7.G.B.5

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.5

In the figure, angle ABC is a right angle. A ray splits it into angles of x° and (2x + 15)°.

Write and solve an equation for x.



Equation and answer: _____

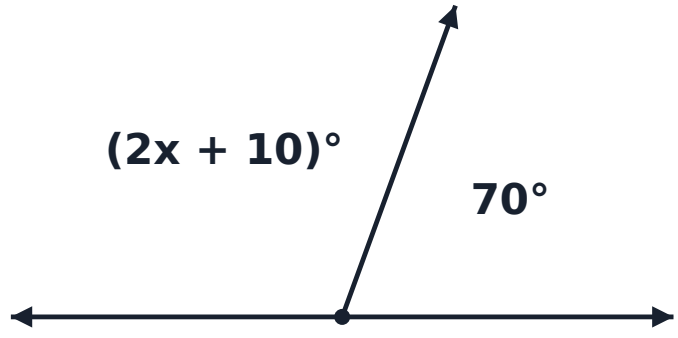
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q1/5

In the figure, angles of $(2x + 10)°$ and $70°$ form a straight line.

Write and solve an equation for x.



Equation and answer: _____

Two complementary angles measure x° and (x + 20)°.

What is the value of x?

A. 45

B. 35

C. 80

D. 70

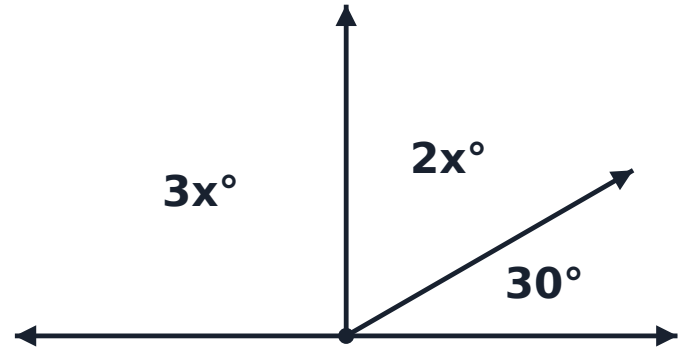
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q3/5

True or false?

In the figure, three angles form a straight line. Then $x = 30$.



A. True

B. False

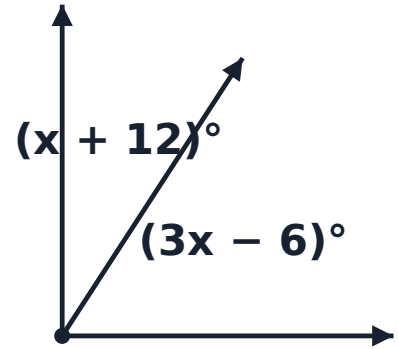
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q4/5

True or false?

In the figure, the two angles form a right angle. Then x = 18.



A. True

B. False

SET 44 • 7.G.B.5

BACKWARD 1

Find an unknown part of a right angle or a straight angle • Grade 4 • Measurement & Data • 4.MD.C.7

A straight angle (180°) is split into two angles. One measures 115°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q1/5

A right angle is split into two angles. One measures 34°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q2/5

A right angle is split into three angles. Two of them measure 20° and 30°.

What does the third angle measure?

A. 50°

B. 60°

C. 40°

D. 10°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q3/5

True or false?

Angles of 72° and 18° together make a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q4/5

True or false?

Angles of 95° and 95° together make a straight angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 7.G.B.5

BACKWARD 2

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$x + 48 = 90$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q1/5

Solve for x.

$3x = 180$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q2/5

Solve for y.

$125 + y = 180$

A. $y = 55$

B. $y = 45$

C. $y = 65$

D. $y = 305$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q3/5

True or false?

a = 27 is the solution of a + 63 = 90.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q4/5

True or false?

$b = 80$ is the solution of $2b = 180$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • 7.G.B.5

FORWARD 1

Use the angle sum and exterior angles of triangles • Grade 8 • Geometry • 8.G.A.5

Two angles of a triangle measure 48° and 75°.

What does the third angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q1/5

An exterior angle of a triangle measures 130°. One of the two remote interior angles measures 60°.

What does the other remote interior angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q2/5

The angles of a triangle measure x°, 2x°, and 3x°.

What is the value of x?

A. 36

B. 30

C. 45

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q3/5

True or false?

The interior angles of a triangle add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q4/5

True or false?

An exterior angle of a triangle equals the sum of all three interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

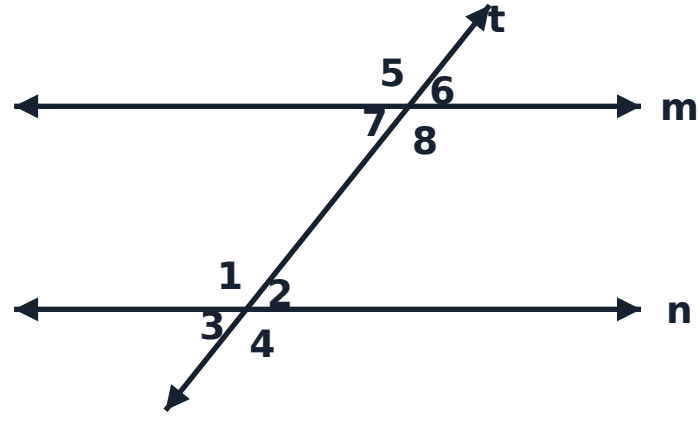
Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • 7.G.B.5

FORWARD 2

Prove angle relationships formed by parallel lines and a transversal • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

In the figure, lines m and n are parallel. Prove that angle 2 = angle 7. Give a reason for each step.



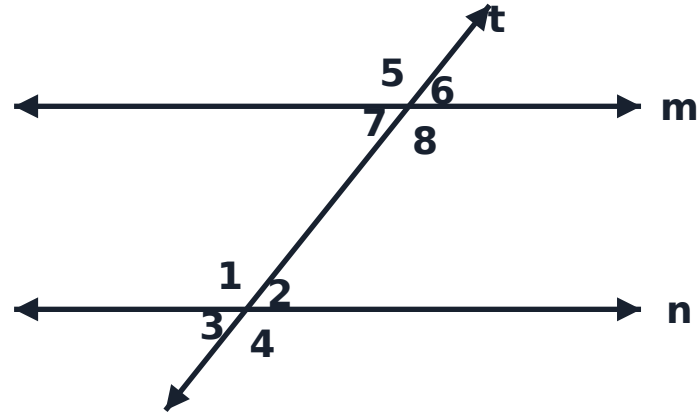
Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q1/5

In the figure, lines m and n are parallel. Prove that angle 2 + angle 8 = 180°. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q2/5

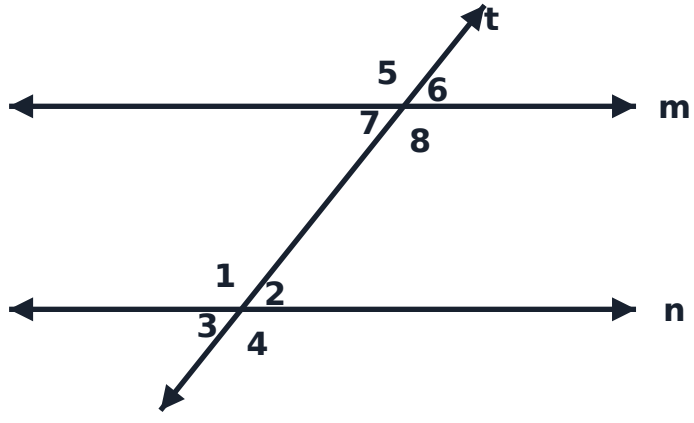
In the figure, lines m and n are parallel. A proof says:

Step 1: angle 1 = angle 5 (corresponding angles)

Step 2: angle 5 = angle 8 (?)

Step 3: angle 1 = angle 8

What is the reason for Step 2?



A. Linear pairs are supplementary.

B. Corresponding angles are congruent.

C. Vertical angles are congruent.

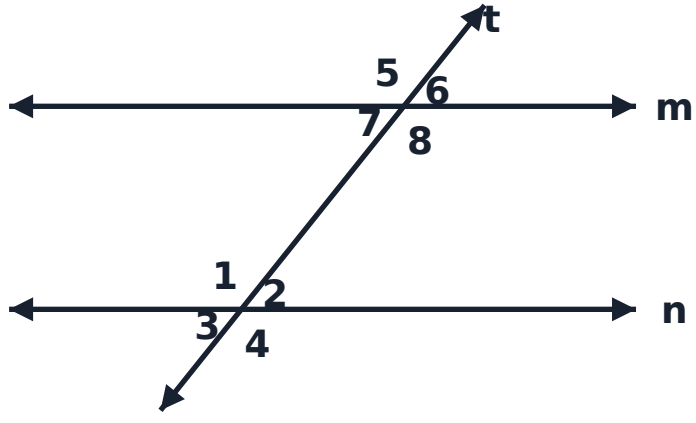
D. Alternate exterior angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q3**

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q3/5

In the figure, lines m and n are parallel and angle 2 = 52°.

Find angle 8 and give a reason for each step.



Answer: _____

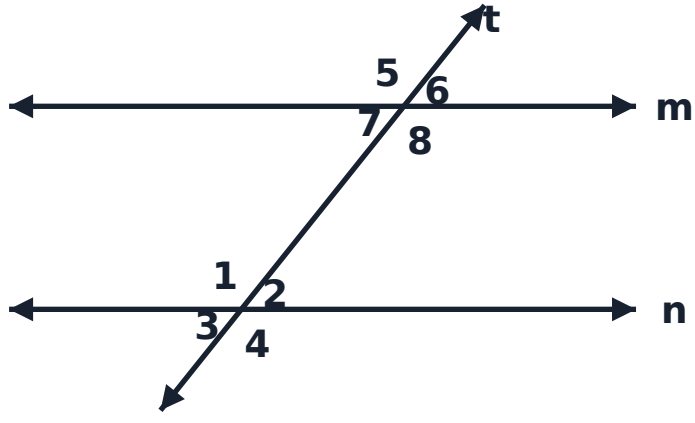
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q4/5

True or false?

In the figure, lines m and n are parallel. "Angle 2 = angle 7 because they look the same size" is a valid proof.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q5/5

SET 45 • 7.G.B.5 • GEOMETRY

Use vertical and adjacent angles to find unknown angles

MAIN	Grade 7 main question	**7.G.B.5**
BACKWARD 1	Identify lines, rays, segments, and angles	**4.G.A.1**
BACKWARD 2	Understand angle measure as a fraction of a full turn	**4.MD.C.5.a**
FORWARD 1	Find angles formed by parallel lines cut by a transversal	**8.G.A.5**
FORWARD 2	Prove that vertical angles are congruent	**HSG-CO.C.9**

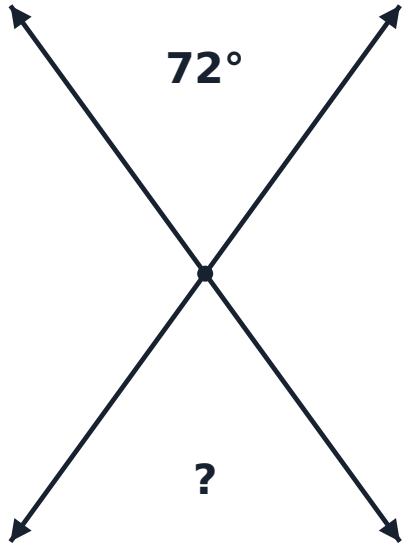
SET 45 • 7.G.B.5

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.5

Two lines intersect. One angle measures 72°.

What does the angle marked with a question mark measure?



Measure: _____

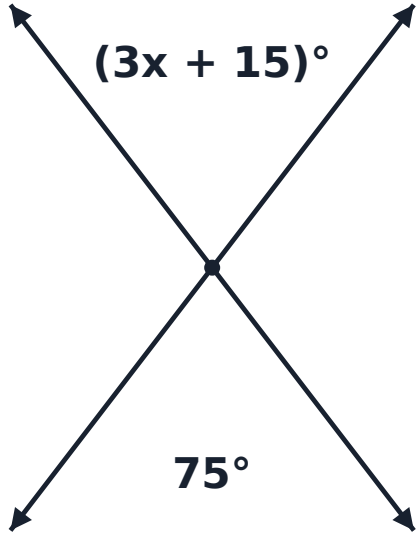
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q1/5

Two lines intersect. The angles shown are vertical angles.

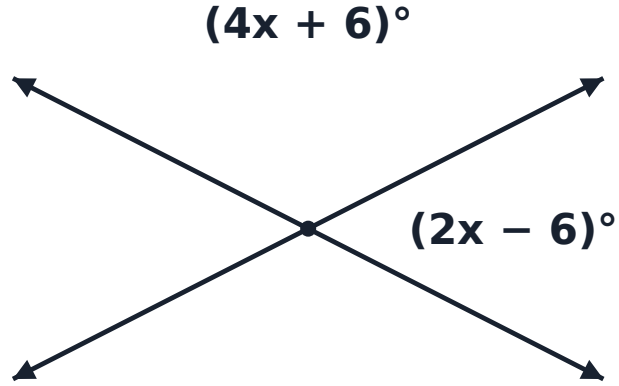
Write and solve an equation for x.



Equation and answer: _____

Two lines intersect. The adjacent angles shown form a straight line.

What is the value of x?



A. 15

B. 25

C. 30

D. 45

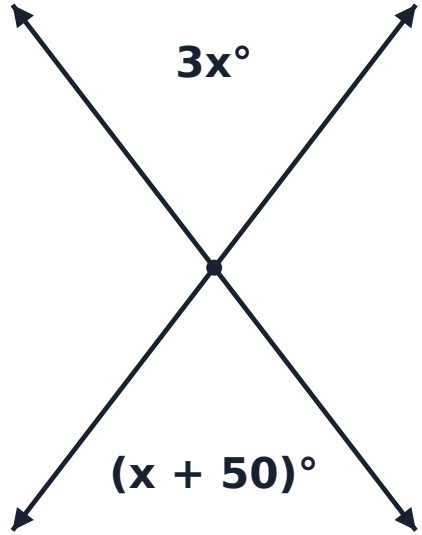
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q3/5

True or false?

Two lines intersect. The angles shown are vertical angles. Each of them measures 75°.



A. True

B. False

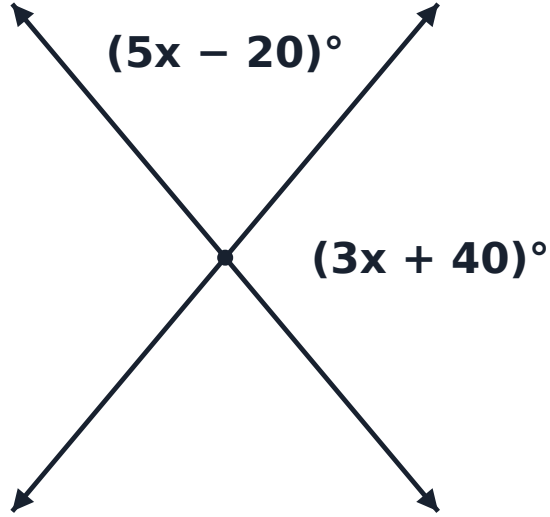
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q4/5

True or false?

Two lines intersect. The adjacent angles shown form a straight line. Then $x = 25$.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q5/5

SET 45 • 7.G.B.5

BACKWARD 1

Identify lines, rays, segments, and angles • Grade 4 • Geometry • 4.G.A.1

How many endpoints does a ray have?

Endpoints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q1/5

What figure is formed by two rays that share an endpoint?

Figure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q2/5

Which figure goes on forever in both directions?

A. Ray

B. Segment

C. Line

D. Angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q3/5

True or false?

Two lines that cross meet at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q4/5

True or false?

A line segment has no endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • 7.G.B.5

BACKWARD 2

Understand angle measure as a fraction of a full turn • Grade 4 • Measurement & Data • 4.MD.C.5.a

How many degrees are in one full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q1/5

An angle turns through $\frac{1}{4}$ of a circle.

What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q2/5

An angle turns through $\frac{1}{2}$ of a circle.

What is its measure?

A. 90°

B. 45°

C. 360°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q3/5

True or false?

An angle that turns through $\frac{1}{360}$ of a circle measures 1°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q4/5

True or false?

An angle that turns through $\frac{1}{3}$ of a circle measures 100°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 7.G.B.5

FORWARD 1

Find angles formed by parallel lines cut by a transversal • Grade 8 • Geometry • 8.G.A.5

Two parallel lines are cut by a transversal. One angle measures 110°.

What does the corresponding angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q1/5

Two parallel lines are cut by a transversal. Alternate interior angles measure $(4x - 10)°$ and $70°$.

Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q2/5

Two parallel lines are cut by a transversal. One same-side interior angle measures 75°.

What does the other measure?

A. 15°

B. 105°

C. 75°

D. 285°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q3/5

True or false?

When parallel lines are cut by a transversal, alternate exterior angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q4/5

True or false?

When parallel lines are cut by a transversal, corresponding angles add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q5/5

SET 45 • 7.G.B.5

FORWARD 2

Prove that vertical angles are congruent • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

Angles 1 and 2 form a linear pair. Angles 2 and 3 form a linear pair.

Explain why angle 1 and angle 3 have the same measure.

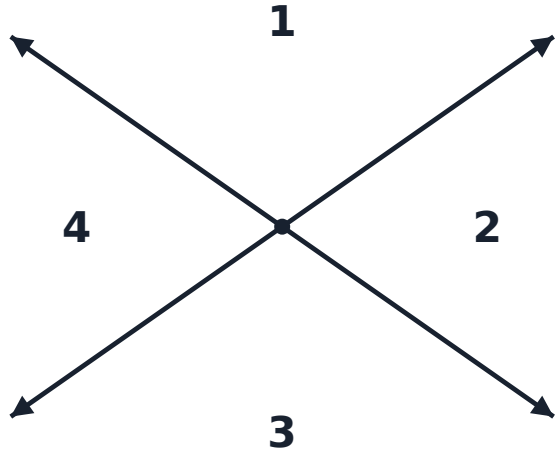
Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q1/5

Two lines intersect, forming angles 1, 2, 3, and 4 in order around the point. Prove that angle 2 = angle 4. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q2/5

A proof that vertical angles are congruent says:

Step 1: angle 1 + angle 2 = 180°

Step 2: angle 2 + angle 3 = 180°

Step 3: angle 1 + angle 2 = angle 2 + angle 3

Step 4: angle 1 = angle 3

What is the reason for Step 4?

A. Corresponding angles are congruent

B. Vertical angles are congruent

C. Definition of a right angle

D. Subtraction property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q3/5

Two lines intersect, and one of the four angles is 90°.

Prove that all four angles are 90°.

Proof: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q4/5

True or false?

"Angles 1 and 3 are vertical angles, and they look equal, so angle 1 = angle 3" is a valid proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q5/5

SET 46 • 7.G.B.6 • GEOMETRY

Find the area of composite figures made of triangles and quadrilaterals

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Find the area of a triangle	**6.G.A.1**
BACKWARD 2	Find areas of rectilinear figures by breaking them into rectangles	**3.MD.C.7.d**
FORWARD 1	Find the volume of a solid made of cylinders, cones, and spheres	**8.G.C.9**
FORWARD 2	Use geometric shapes to model real objects	**HSG-MG.A.1**

SET 46 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

The figure is a 10 m by 6 m rectangle and a triangle with a base of 6 m (the rectangle's side) and a height of 4 m. The parts do not overlap.

What is the total area?



Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q1/5

An L-shaped floor is made of two rectangles that do not overlap. One is 8 feet by 5 feet and the other is 4 feet by 3 feet.

What is the area of the floor?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q2/5

A figure is a 10 cm by 4 cm rectangle with a right triangle attached to one 4 cm side. The legs of the triangle are 4 cm and 3 cm, and the parts do not overlap.

What is the total area?

A. 52 square centimeters

B. 34 square centimeters

C. 46 square centimeters

D. 40 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q3/5

True or false?

A 4 cm by 4 cm square is cut from one corner of a 12 cm by 8 cm rectangle. The area that is left is 80 square centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q4/5

True or false?

A 9 m by 4 m rectangle and a right triangle with legs of 4 m and 3 m are joined without overlapping. Their total area is 48 square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q5/5

SET 46 • 7.G.B.6

BACKWARD 1

Find the area of a triangle • Grade 6 • Geometry • 6.G.A.1

A triangle has a base of 14 cm and a height of 5 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q1/5

A right triangle has legs of 9 m and 6 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q2/5

A triangle has a base of 7 feet and a height of 10 feet.

What is its area?

A. 17 square feet

B. 35 square feet

C. 140 square feet

D. 70 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q3/5

True or false?

A triangle with a base of 8 and a height of 3 has an area of 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q4/5

True or false?

A triangle with a base of 6 and a height of 6 has an area of 36.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • 7.G.B.6

BACKWARD 2

Find areas of rectilinear figures by breaking them into rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.d

A shape is made of a 5-by-3 rectangle and a 2-by-2 square that do not overlap.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q1/5

A shape is made of a 6-by-2 rectangle and a 3-by-4 rectangle that do not overlap.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q2/5

A shape is made of a 4-by-4 square and a 4-by-1 rectangle that do not overlap.

What is its area?

A. 16 square units

B. 20 square units

C. 13 square units

D. 25 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q3/5

True or false?

A 3-by-5 rectangle and a 2-by-5 rectangle that do not overlap have a total area of 25 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q4/5

True or false?

A 7-by-2 rectangle and a 1-by-3 rectangle that do not overlap have a total area of 13 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 7.G.B.6

FORWARD 1

Find the volume of a solid made of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

A cone sits on top of a cylinder. Both have a radius of 3. The cylinder is 10 tall and the cone is 4 tall.

Find the total volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q1/5

A hemisphere (half of a sphere) has a radius of 6 cm.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q2/5

A capsule is a cylinder with radius 2 and height 5 with half of a sphere of radius 2 on each end.

What is its volume?

A. $\frac{52}{3}\pi$

B. 20π

C. $\frac{92}{3}\pi$

D. $\frac{32}{3}\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q3/5

True or false?

A sphere with a radius of 3 has a volume of 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q4/5

True or false?

A cone with a radius of 3 and a height of 4 has a volume of 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 7.G.B.6

FORWARD 2

Use geometric shapes to model real objects • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1

A tree trunk is modeled as a cylinder with a diameter of 2 feet and a height of 30 feet.

Estimate its volume in terms of π. Then give one reason the model might overestimate or underestimate the real volume.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q1/5

A round tabletop is modeled as a circle with a circumference of 9.42 feet.

What is its radius? Use 3.14 for π.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q2/5

A pile of sand is shaped roughly like a cone that is 6 m across at the bottom and 2 m tall.

Which is the best estimate of its volume?

A. About 18π cubic meters

B. About 6π cubic meters

C. About 72π cubic meters

D. About 24π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q3/5

True or false?

A rolled sleeping bag 50 cm long and 20 cm across is modeled as a cylinder. Its volume is about 5,000π cubic centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q4/5

True or false?

To estimate how much cheese covers a pizza, the pizza is best modeled as a sphere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q5/5

SET 47 • 7.G.B.6 • GEOMETRY

Find the volume of right prisms

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Find the volume of a rectangular prism with fractional edges	**6.G.A.2**
BACKWARD 2	Find volume with $V = l \times w \times h$	**5.MD.C.5.b**
FORWARD 1	Compare the volumes of cylinders and prisms	**8.G.C.9**
FORWARD 2	Use volume formulas to solve problems	**HSG-GMD.A.3**

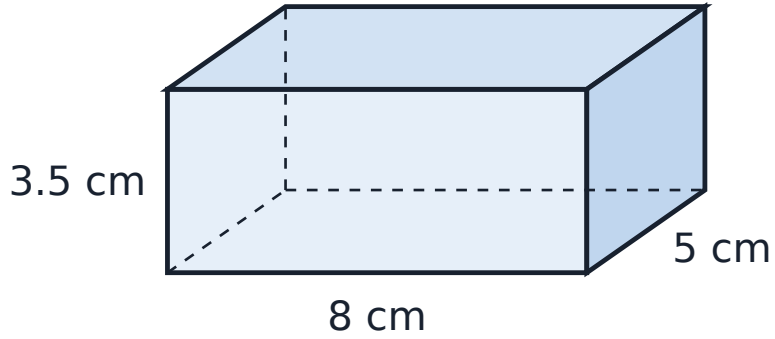
SET 47 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

Find the volume of the right rectangular prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q1/5

A right triangular prism has triangular bases with a base of 6 m and a height of 4 m. The prism is 10 m long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q2/5

The base of a right prism has an area of 24 square inches. The prism is 7.5 inches tall.

What is its volume?

A. 90 cubic inches

B. 180 cubic inches

C. 360 cubic inches

D. 31.5 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q3/5

True or false?

A cube with $\frac{1}{2}$-foot edges has a volume of $\frac{1}{8}$ cubic foot.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q4/5

True or false?

Doubling the height of a right prism doubles the area of its base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q5/5

SET 47 • 7.G.B.6

BACKWARD 1

Find the volume of a rectangular prism with fractional edges • Grade 6 • Geometry • 6.G.A.2

A right rectangular prism is $2\frac{1}{2}$ feet by 2 feet by 4 feet.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q1/5

A cube has $\frac{3}{4}$-inch edges.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q2/5

A right rectangular prism is $\frac{1}{2}$ m by 3 m by 6 m.

What is its volume?

A. 9.5 cubic meters

B. 4.5 cubic meters

C. 9 cubic meters

D. 18 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q3/5

True or false?

A right rectangular prism $1\frac{1}{2}$ cm by 2 cm by 2 cm has a volume of 6 cubic centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q4/5

True or false?

A right rectangular prism $\frac{1}{2}$ foot by $\frac{1}{2}$ foot by 4 feet has a volume of 2 cubic feet.

A. True

B. False

SET 47 • 7.G.B.6

BACKWARD 2

Find volume with V = l × w × h • Grade 5 • Measurement & Data • 5.MD.C.5.b

A rectangular prism is 7 units long, 4 units wide, and 3 units tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q1/5

A box is 10 cm long, 6 cm wide, and 5 cm tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q2/5

A rectangular prism is 9 feet by 2 feet by 3 feet.

What is its volume?

A. 27 cubic feet

B. 14 cubic feet

C. 108 cubic feet

D. 54 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q3/5

True or false?

A cube with 4-inch edges has a volume of 64 cubic inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q4/5

True or false?

A box 5 m by 5 m by 2 m has a volume of 12 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 7.G.B.6

FORWARD 1

Compare the volumes of cylinders and prisms • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 4 inches and a height of 5 inches.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q1/5

A cylinder's base has an area of 12.56 square centimeters. The cylinder is 10 cm tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q2/5

A cylinder has a radius of 1 m and a height of 3 m.

What is its volume in terms of π?

A. π cubic meters

B. 9π cubic meters

C. 6π cubic meters

D. 3π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q3/5

True or false?

The volume of a cylinder is its base area times its height, just like a prism.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 2 and a height of 2 has a volume of 4π.

A. True

B. False

SET 47 • 7.G.B.6

FORWARD 2

Use volume formulas to solve problems • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A pyramid has a square base with 6 m sides. It is 10 m tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q1/5

A cone-shaped cup has a radius of 3 cm and holds 30π cubic centimeters.

How tall is the cup?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q2/5

A sphere has a volume of 288π cubic centimeters.

What is its radius?

A. 6 cm

B. 12 cm

C. 4 cm

D. 8 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q3**

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q3/5

True or false?

A pyramid has $\frac{1}{3}$ the volume of a prism with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q4/5

True or false?

A sphere with a radius of 3 has a volume of 27π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q5/5

SET 48 • 7.G.B.6 • GEOMETRY

Find the surface area of right prisms

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Use a net to find surface area	**6.G.A.4**
BACKWARD 2	Find the area of a rectangle	**4.MD.A.3**
FORWARD 1	Use the Pythagorean theorem in three-dimensional figures — near…	**8.G.B.7**
FORWARD 2	Apply geometric methods to solve design problems	**HSG-MG.A.3**

SET 48 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

Find the surface area of a right rectangular prism that is 6 cm by 4 cm by 3 cm.

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q1/5

Find the surface area of a cube with 5-inch edges.

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q2/5

A right triangular prism has triangular bases with sides of 3 cm, 4 cm, and 5 cm and a right angle between the 3 cm and 4 cm sides. The prism is 10 cm long.

What is its surface area?

A. 126 square centimeters

B. 60 square centimeters

C. 132 square centimeters

D. 120 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q3/5

True or false?

A box 10 in by 2 in by 2 in needs exactly 88 square inches of paper to cover it with no overlap.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q4/5

True or false?

A right rectangular prism 2 m by 3 m by 4 m has a surface area of 24 square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q5/5

SET 48 • 7.G.B.6

BACKWARD 1

Use a net to find surface area • Grade 6 • Geometry • 6.G.A.4

The net of a cube has 6 squares. Each square is 4 cm by 4 cm.

What is the surface area of the cube?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q1/5

The net of a rectangular prism has two 5-by-3 rectangles, two 5-by-2 rectangles, and two 3-by-2 rectangles.

What is the surface area?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q2/5

The net of a square pyramid has a 6-by-6 square and 4 triangles. Each triangle has a base of 6 and a height of 5.

What is the surface area?

A. 186 square units

B. 66 square units

C. 156 square units

D. 96 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q3/5

True or false?

A net shows all the faces of a solid laid flat.

A. True

B. False

True or false?

The net of a rectangular prism has 4 rectangles.

A. True

B. False

SET 48 • 7.G.B.6

BACKWARD 2

Find the area of a rectangle • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 8 cm by 5 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q1/5

A rectangle is 14 inches by 3 inches.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q2/5

A rectangle is 11 m by 6 m.

What is its area?

A. 34 square meters

B. 66 square meters

C. 17 square meters

D. 60 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q3/5

True or false?

A rectangle 9 feet by 2 feet has an area of 18 square feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q4/5

True or false?

A rectangle 5 cm by 5 cm has an area of 20 square centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 7.G.B.6

FORWARD 1

Use the Pythagorean theorem in three-dimensional figures • Grade 8 • Geometry • 8.G.B.7

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A box is 4 units long, 3 units wide, and 12 units tall.

How long is a diagonal of its 4-by-3 base?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q1/5 • NEAREST RELATED

A box is 4 units long, 3 units wide, and 12 units tall.

How long is the diagonal from a bottom corner to the opposite top corner?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q2/5 • NEAREST RELATED

A right square pyramid (its top vertex is directly above the center of its base) has base edges of 6 m and is 4 m tall.

What is the slant height (from the top vertex to the midpoint of a base edge)?

A. 10 m

B. 7 m

C. 5 m

D. 2 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A rectangle that is 6 by 8 has a diagonal of 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

A cube with edges of 1 has a diagonal from one corner to the opposite corner of length 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 48 • 7.G.B.6

FORWARD 2

Apply geometric methods to solve design problems • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A company wants a box shaped like a cube with a volume of 64 cubic inches.

What is the edge length, and how much cardboard covers the box?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q1/5

One gallon of paint covers 350 square feet. The walls of a room total 1,225 square feet.

How many whole gallons should be bought?

Gallons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q2/5

Each box has a volume of 24 cubic centimeters.

Which box uses the least cardboard?

A. 1 cm × 4 cm × 6 cm

B. 1 cm × 1 cm × 24 cm

C. 2 cm × 3 cm × 4 cm

D. 1 cm × 2 cm × 12 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q3/5

True or false?

Among rectangular boxes with the same volume, a cube has the least surface area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q4/5

True or false?

Two boxes with the same volume always have the same surface area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q5/5

SET 67 • 7.G.B.6 • GEOMETRY

Find the volume of a solid made of right prisms

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Add the volumes of two rectangular prisms	**5.MD.C.5.c**
BACKWARD 2	Find the area of a figure made of rectangles	**3.MD.C.7.d**
FORWARD 1	Find the volume of a solid made of a prism and a cylinder	**8.G.C.9**
FORWARD 2	Use volume formulas to solve problems with composite solids	**HSG-GMD.A.3**

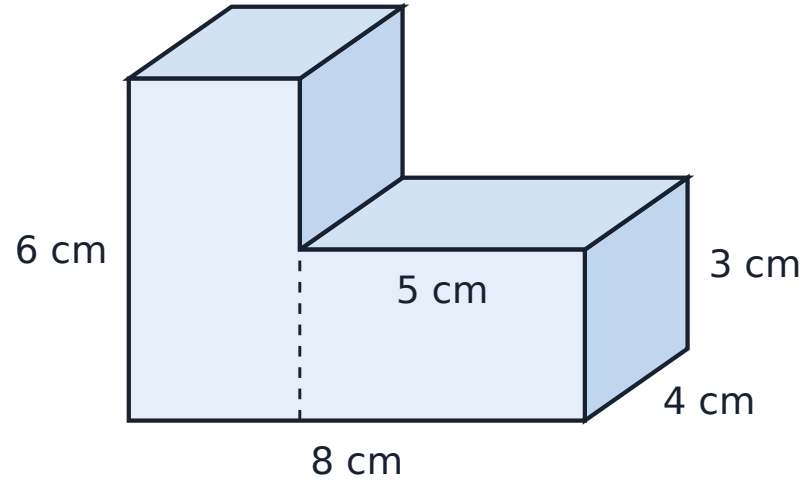
SET 67 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

The solid is made of two right rectangular prisms joined face to face with no gaps or overlaps.

Find its volume.



Volume: _____

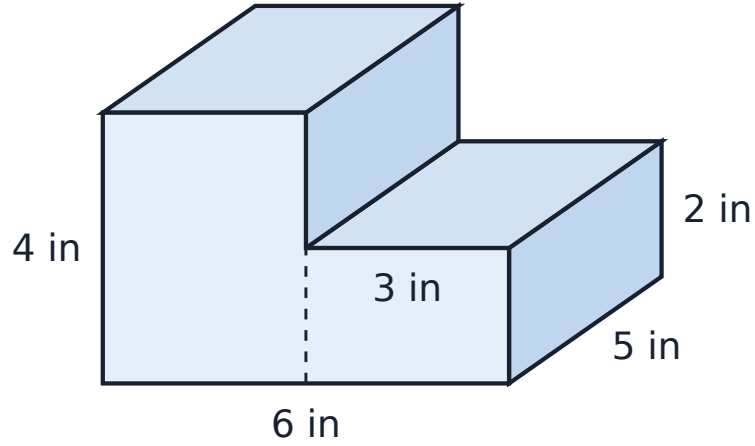
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 67 • MAIN • Q1/5

The step-shaped solid is made of two right rectangular prisms joined face to face with no gaps or overlaps.

Find its volume.



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 67 • MAIN • Q2/5

A cube with 4 cm edges sits on top of a right rectangular prism that is 8 cm by 6 cm by 2 cm. The bottom face of the cube lies flat on the top face of the prism.

What is the total volume?

A. 160 cubic centimeters

B. 64 cubic centimeters

C. 224 cubic centimeters

D. 96 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 67 • MAIN • Q3/5

True or false?

A solid is made of a 10 m by 4 m by 3 m prism and a 2 m by 4 m by 3 m prism joined end to end with no overlap. Its volume is 144 cubic meters.

A. True

B. False

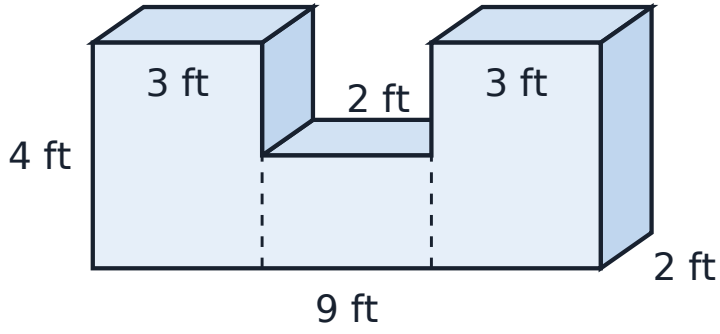
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 67 • MAIN • Q4/5

True or false?

The solid is made of three right rectangular prisms with no gaps or overlaps. Its volume is 72 cubic feet.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 67 • MAIN • Q5/5

SET 67 • 7.G.B.6

BACKWARD 1

Add the volumes of two rectangular prisms • Grade 5 • Measurement & Data • 5.MD.C.5.c

A 3 by 2 by 2 prism and a 4 by 2 by 1 prism are joined with no overlap.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 67 • BACKWARD 1 • Q1/5

A 5 by 3 by 2 prism and a 2 by 3 by 2 prism are joined with no overlap.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 67 • BACKWARD 1 • Q2/5

Two cubes with 3-unit edges are joined face to face.

What is the total volume?

A. 27 cubic units

B. 18 cubic units

C. 54 cubic units

D. 36 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 67 • BACKWARD 1 • Q3/5

True or false?

A prism with a volume of 10 cubic units and a prism with a volume of 6 cubic units are joined with no overlap. The total volume is 16 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 67 • BACKWARD 1 • Q4/5

True or false?

A 2 by 2 by 2 cube and a 2 by 2 by 4 prism are joined with no overlap. The total volume is 12 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 67 • BACKWARD 1 • Q5/5

SET 67 • 7.G.B.6

BACKWARD 2

Find the area of a figure made of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.d

A 4 by 2 rectangle and a 2 by 3 rectangle are joined with no overlap.

What is the total area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 67 • BACKWARD 2 • Q1/5

A 6 by 3 rectangle and a 3 by 3 rectangle are joined with no overlap.

What is the total area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 67 • BACKWARD 2 • Q2/5

A U-shape is made of three rectangles with no overlap: 2 by 4, 4 by 2, and 2 by 4.

What is the total area?

A. 20 square units

B. 16 square units

C. 32 square units

D. 24 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 67 • BACKWARD 2 • Q3/5

True or false?

A 5 by 2 rectangle and a 2 by 2 rectangle joined with no overlap have a total area of 14 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 67 • BACKWARD 2 • Q4/5

True or false?

A 3 by 3 rectangle and a 3 by 1 rectangle joined with no overlap have a total area of 10 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 67 • BACKWARD 2 • Q5/5

SET 67 • 7.G.B.6

FORWARD 1

Find the volume of a solid made of a prism and a cylinder • Grade 8 • Geometry • 8.G.C.9

A cylinder with a radius of 2 cm and a height of 6 cm stands on top of a 10 cm by 10 cm by 4 cm block.

Find the total volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 67 • FORWARD 1 • Q1/5

A cylinder with a radius of 3 inches and a height of 8 inches stands on top of a cube with 8-inch edges.

Find the total volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 67 • FORWARD 1 • Q2/5

A cube with 6 cm edges has a cylinder-shaped hole with a radius of 1 cm drilled straight through it (6 cm long).

What volume is left?

A. 216 + 6π cubic centimeters

B. 216 − 36π cubic centimeters

C. 6π cubic centimeters

D. 216 − 6π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 67 • FORWARD 1 • Q3/5

True or false?

A 2 by 2 by 10 prism with half of a cylinder (radius 1, length 10) on top has a volume of $40 + 5\pi$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 67 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 1 and a height of 5 is stacked on a cylinder with a radius of 1 and a height of 3. The total volume is 15π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 67 • FORWARD 1 • Q5/5

SET 67 • 7.G.B.6

FORWARD 2

Use volume formulas to solve problems with composite solids • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A silo is a cylinder with a radius of 3 m and a height of 10 m, topped with half of a sphere with a radius of 3 m.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 67 • FORWARD 2 • Q1/5

A pencil is a cylinder with a radius of 0.4 cm and a length of 15 cm, with a cone-shaped tip with a radius of 0.4 cm and a height of 1.5 cm.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 67 • FORWARD 2 • Q2/5

A cone with a radius of 3 and a height of 4 is carved out of a cylinder with a radius of 3 and a height of 4.

What volume is left?

A. 24π

B. 48π

C. 36π

D. 12π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F2-Q3**

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 67 • FORWARD 2 • Q3/5

True or false?

A square pyramid with 6 m base edges and a height of 4 m sits on top of a cube with 6 m edges. The total volume is 264 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 67 • FORWARD 2 • Q4/5

True or false?

Two halves of a sphere with a radius of 2 are joined to make a whole sphere. Its volume is $\frac{16}{3}\pi$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 67 • FORWARD 2 • Q5/5

SET 68 • 7.G.B.6 • GEOMETRY

Find the exterior surface area of a solid made of right prisms

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Find the surface area of a rectangular prism	**6.G.A.4**
BACKWARD 2	Find the area of a rectangle by multiplying	**3.MD.C.7.b**
FORWARD 1	Use the Pythagorean theorem to find a slanted edge needed for s…	**8.G.B.7**
FORWARD 2	Apply surface area to design problems	**HSG-MG.A.3**

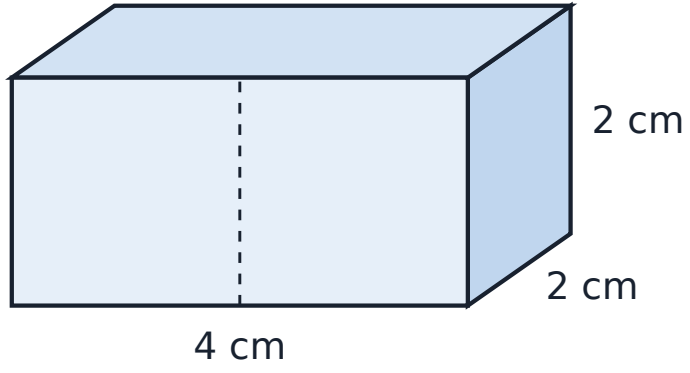
SET 68 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

Two cubes with 2 cm edges are joined face to face.

Find the exterior surface area of the solid. (The faces where the cubes touch are hidden and are not counted.)



Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 68 • MAIN • Q1/5

A cube with 2 m edges sits on top of a 6 m by 6 m by 2 m prism. Its bottom face lies completely on the top face of the prism.

Find the exterior surface area. (Do not count the faces where the solids touch.)

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 68 • MAIN • Q2/5

The solid is made of two right rectangular prisms joined face to face.

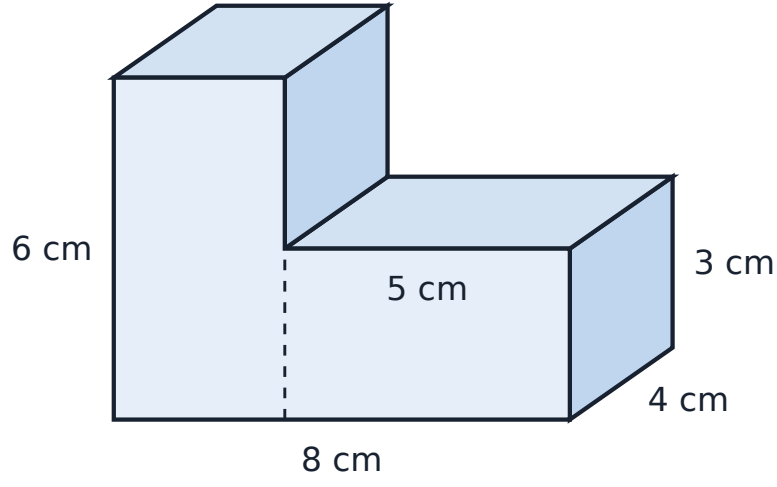
What is its exterior surface area?

A. 132 square centimeters

B. 178 square centimeters

C. 112 square centimeters

D. 210 square centimeters



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 68 • MAIN • Q3/5

True or false?

Two 1 by 1 by 1 cubes joined face to face have an exterior surface area of 10 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 68 • MAIN • Q4/5

True or false?

When two prisms are joined face to face, the exterior surface area is the sum of their two surface areas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 68 • MAIN • Q5/5

SET 68 • 7.G.B.6

BACKWARD 1

Find the surface area of a rectangular prism • Grade 6 • Geometry • 6.G.A.4

Use a net to find the surface area of a 3 by 2 by 1 rectangular prism.

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 68 • BACKWARD 1 • Q1/5

Use a net to find the surface area of a cube with 3-inch edges.

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 68 • BACKWARD 1 • Q2/5

What is the surface area of a 4 by 1 by 1 rectangular prism?

A. 4 square units

B. 16 square units

C. 18 square units

D. 24 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 68 • BACKWARD 1 • Q3/5

True or false?

A 2 by 2 by 1 rectangular prism has a surface area of 16 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 68 • BACKWARD 1 • Q4/5

True or false?

A 5 by 1 by 1 rectangular prism has a surface area of 20 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • 7.G.B.6

BACKWARD 2

Find the area of a rectangle by multiplying • Grade 3 • Measurement & Data • 3.MD.C.7.b

A face of a box is 6 cm by 4 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 68 • BACKWARD 2 • Q1/5

A face of a box is 7 m by 3 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 68 • BACKWARD 2 • Q2/5

A face of a box is 5 inches by 5 inches.

What is its area?

A. 20 square inches

B. 25 square inches

C. 30 square inches

D. 10 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 68 • BACKWARD 2 • Q3/5

True or false?

A 9 by 2 face has an area of 18 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 68 • BACKWARD 2 • Q4/5

True or false?

A 4 by 8 face has an area of 24 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 68 • BACKWARD 2 • Q5/5

SET 68 • 7.G.B.6

FORWARD 1

Use the Pythagorean theorem to find a slanted edge needed for surface area • Grade 8 • Geometry • 8.G.B.7

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A ramp is a right triangular prism. Its triangular ends have legs of 3 m and 4 m.

How wide is the sloped face (the hypotenuse of the end)?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 68 • FORWARD 1 • Q1/5 • NEAREST RELATED

A tent is a triangular prism. Each end is an isosceles triangle with a base of 6 feet and a height of 4 feet.

How long is each slanted side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 68 • FORWARD 1 • Q2/5 • NEAREST RELATED

The end of a prism is a right triangle with legs of 5 and 12.

How long is its hypotenuse?

A. 13

B. 17

C. 7

D. $\sqrt{119}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 68 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A ramp with triangular ends whose legs are 6 and 8 has a sloped face that is 10 units wide.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 68 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

A tent end is an isosceles triangle with a base of 10 and a height of 12. Each slanted side is 12 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 68 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 68 • 7.G.B.6

FORWARD 2

Apply surface area to design problems • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A gift is made of two cubes with 3-inch edges joined face to face.

How much wrapping paper covers its outside exactly?

Paper: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 68 • FORWARD 2 • Q1/5

A shed is a 10 ft by 8 ft by 8 ft rectangular prism (8 ft tall). Its four walls and its flat roof are painted, but not the floor. One quart of paint covers 40 square feet.

How many whole quarts are needed?

Quarts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 68 • FORWARD 2 • Q2/5

Two 1 by 2 by 3 prisms are joined face to face with matching faces.

Which way of joining them gives the least exterior surface area?

A. Joining their 1 by 2 faces

B. Joining their 1 by 3 faces

C. Joining their 2 by 3 faces

D. All three ways give the same surface area

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q3**

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 68 • FORWARD 2 • Q3/5

True or false?

Joining two solids face to face lowers the total exterior surface area by twice the area of the shared face.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 68 • FORWARD 2 • Q4/5

True or false?

A box with a smaller surface area always has a smaller volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 68 • FORWARD 2 • Q5/5

SET 49 • 7.SP.A.1 • STATISTICS & PROBABILITY

Understand random sampling and representative samples

MAIN	Grade 7 main question	**7.SP.A.1**
BACKWARD 1	Recognize a statistical question	**6.SP.A.1**
BACKWARD 2	Report the number of observations	**6.SP.B.5.a**
FORWARD 1	Summarize survey data in a two-way table — nearest related (no…	**8.SP.A.4**
FORWARD 2	Recognize surveys, experiments, and observational studies	**HSS-IC.B.3**

SET 49 • 7.SP.A.1

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.1

A principal wants to know students' favorite lunch.

Which sample is most likely to represent all the students in the school?

A. 20 students in one 7th grade class

B. 20 students on the basketball team

C. The first 20 students in the lunch line

D. 20 students chosen at random from the school list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q1/5

A town surveys people at a gym about how often they exercise.

Is the sample likely to represent the whole town? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q2/5

Describe one way to choose a random sample of 30 students from a school of 600 students.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q3/5

True or false?

In a random sample, every member of the population has the same chance of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q4/5

True or false?

A large sample always represents the population, even if it is not chosen at random.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q5/5

SET 49 • 7.SP.A.1

BACKWARD 1

Recognize a statistical question • Grade 6 • Statistics & Probability • 6.SP.A.1

Is "How tall are the students in my class?" a statistical question? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q1/5

Which is a statistical question?

A. How many hours do 7th graders sleep on school nights?

B. What is 5 × 8?

C. How tall is the school flagpole?

D. What day is it today?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q2/5

Rewrite "How old am I?" as a statistical question.

Question: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q3/5

True or false?

"What are the shoe sizes of the players on the team?" is a statistical question.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q4/5

True or false?

"How many days are in March?" is a statistical question.

A. True

B. False

SET 49 • 7.SP.A.1

BACKWARD 2

Report the number of observations • Grade 6 • Statistics & Probability • 6.SP.B.5.a

How many observations does the dot plot show?



Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q1/5

A survey asks 25 students how many pets they have. Each student gives one answer.

How many observations are there?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q2/5

A table lists the heights of 18 plants, one height for each plant.

How many observations are there?

A. 1

B. 36

C. 18

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q3/5

True or false?

Each dot on a dot plot shows one observation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q4/5

True or false?

A data set of 12 test scores has 24 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q5/5

SET 49 • 7.SP.A.1

FORWARD 1

Summarize survey data in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows the results of a survey.

What fraction of the surveyed students play a sport?

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows the results of a survey.

What fraction of all surveyed students have a job and play a sport?

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows the results of a survey.

Among students who have a job, what fraction play a sport?

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

A. $\frac{3}{10}$ **B.** $\frac{3}{5}$ **C.** $\frac{2}{5}$ **D.** $\frac{3}{20}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, 56% of the students with no job play a sport.

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, more than half of all the students have a job.

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 49 • 7.SP.A.1

FORWARD 2

Recognize surveys, experiments, and observational studies • High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers randomly assign 50 plants to get a new plant food and 50 plants to get none.

Is this a survey, an experiment, or an observational study?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q1/5

A study records the screen time of 200 teens without changing anything they do.

Is this a survey, an experiment, or an observational study?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q2/5

Why is random assignment used in an experiment?

A. To avoid collecting data

B. To make the sample larger

C. To make the groups alike, so differences can be linked to the treatment

D. To let people choose their own group

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q3/5

True or false?

A survey asks a sample of people questions in order to estimate facts about a population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q4/5

True or false?

An observational study can prove that one variable causes another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q5/5

SET 50 • 7.SP.A.2 • STATISTICS & PROBABILITY

Use data from a random sample to draw inferences about a population

MAIN	Grade 7 main question	**7.SP.A.2**
BACKWARD 1	Find what percent a part is of a whole	**6.RP.A.3.c**
BACKWARD 2	Find a missing value in equivalent ratios	**6.RP.A.3.a**
FORWARD 1	Use a linear model to make a prediction — nearest related (no di…	**8.SP.A.3**
FORWARD 2	Use a margin of error to describe a population estimate	**HSS-IC.B.4**

SET 50 • 7.SP.A.2

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.2

In a random sample of 50 students, 18 prefer the art club.

Predict how many of the 800 students in the school prefer the art club.

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q1/5

In a random sample of 40 light bulbs, 2 do not work.

Estimate how many bulbs in a shipment of 2,000 do not work.

Bulbs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q2/5

In a random sample of 60 voters, 33 support a plan.

What is the best estimate of the percent of all voters who support it?

A. 45%

B. 33%

C. 60%

D. 55%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q3/5

True or false?

In a random sample of 25 fish from a lake, 5 are trout. A good estimate is that 20% of the fish in the lake are trout.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q4/5

True or false?

In a random sample of 30 homes, 9 have a dog. A good estimate is that 900 of 1,500 homes have a dog.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q5/5

SET 50 • 7.SP.A.2

BACKWARD 1

Find what percent a part is of a whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

18 out of 50 is what percent?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q1/5

10 is what percent of 40?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q2/5

7 out of 20 is what percent?

A. 35%

B. 7%

C. 20%

D. 70%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q3/5

True or false?

12 out of 48 is 25%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q4/5

True or false?

9 out of 30 is 9%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q5/5

SET 50 • 7.SP.A.2

BACKWARD 2

Find a missing value in equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Find the missing value.

2 : 5 = ? : 100

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q1/5

Find the missing value.

$4 : 9 = 28 : ?$

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q2/5

Which ratio is equivalent to 6 : 15?

A. 3 : 5

B. 6 : 5

C. 2 : 5

D. 1 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q3/5

True or false?

$5 : 8 = 25 : 40$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q4/5

True or false?

$3 : 7 = 9 : 14$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 7.SP.A.2

FORWARD 1

Use a linear model to make a prediction • Grade 8 • Statistics & Probability • 8.SP.A.3

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A line of fit is y = 2.5x + 40, where x is hours studied and y is a test score.

Predict the score for 8 hours of study.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q1/5 • NEAREST RELATED

A line of fit is $y = 2.5x + 40$, where x is hours studied and y is a test score.

What does 2.5 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q2/5 • NEAREST RELATED

A line of fit is $y = -3x + 90$.

What does it predict for $x = 10$?

A. 120

B. 93

C. 87

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

For the line of fit $y = 0.5x + 12$, the predicted y-value at $x = 0$ is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

The line of fit $y = 4x + 1$ predicts $y = 41$ when $x = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 50 • 7.SP.A.2

FORWARD 2

Use a margin of error to describe a population estimate • High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample shows 52% support for a plan, with a margin of error of 4%.

What interval of values is likely for the whole population?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q1/5

A random sample has a mean of 6.8 hours with a margin of error of 0.5 hour.

What interval of values is likely for the population mean?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q2/5

As a random sample gets larger, what usually happens to the margin of error?

A. It gets larger.

B. It becomes 0 once the sample is over 10.

C. It gets smaller.

D. It stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q3/5

True or false?

A margin of error describes how far a sample estimate is likely to be from the population value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q4/5

True or false?

A poll result of 52% with a 4% margin of error proves that more than half of the population supports the plan.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q5/5

SET 51 • 7.SP.A.2 • STATISTICS & PROBABILITY

Use several random samples to judge how much estimates vary

MAIN	Grade 7 main question	**7.SP.A.2**
BACKWARD 1	Find the mean of a data set	**6.SP.B.5.c**
BACKWARD 2	Find the range as a measure of variation	**6.SP.A.3**
FORWARD 1	Describe the association shown in a scatter plot — nearest relate…	**8.SP.A.1**
FORWARD 2	Decide whether a result is consistent with a model by simulation	**HSS-IC.A.2**

SET 51 • 7.SP.A.2

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.2

Four random samples of 20 students give these mean hours of sleep: 7.8, 8.2, 8.0, and 8.4.

What is a good estimate of the mean for all the students? Use the mean of the samples.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q1/5

Four random samples give mean estimates of 7.8, 8.2, 8.0, and 8.4 hours.

What is the range of the estimates?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q2/5

Three random samples of 50 people show that 22%, 26%, and 24% prefer blue.

Which conclusion is best?

A. Exactly 22% of the population prefers blue.

B. Exactly 26% of the population prefers blue.

C. The samples are useless because they differ.

D. About 24% of the population prefers blue.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q3/5

True or false?

Different random samples from the same population usually give somewhat different results.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q4/5

True or false?

Small random samples usually give estimates that vary less than large random samples.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q5/5

SET 51 • 7.SP.A.2

BACKWARD 1

Find the mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

4, 7, 9, 10

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q1/5

Find the mean.

12, 15, 18

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q2/5

What is the mean of 3, 3, 6, and 8?

A. 5

B. 6

C. 3

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q3/5

True or false?

The mean of 2, 4, 6, and 8 is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q4/5

True or false?

The mean of 10, 20, and 60 is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 7.SP.A.2

BACKWARD 2

Find the range as a measure of variation • Grade 6 • Statistics & Probability • 6.SP.A.3

Find the range.

12, 18, 9, 15

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q1/5

Find the range.

3.5, 7, 4.25, 6

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q2/5

Which data set has the greatest range?

A. 1, 3, 2

B. 7, 9, 8

C. 4, 6, 5

D. 2, 10, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q3/5

True or false?

The range of 20, 25, and 30 is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q4/5

True or false?

A data set whose values are all the same has a range of 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • 7.SP.A.2

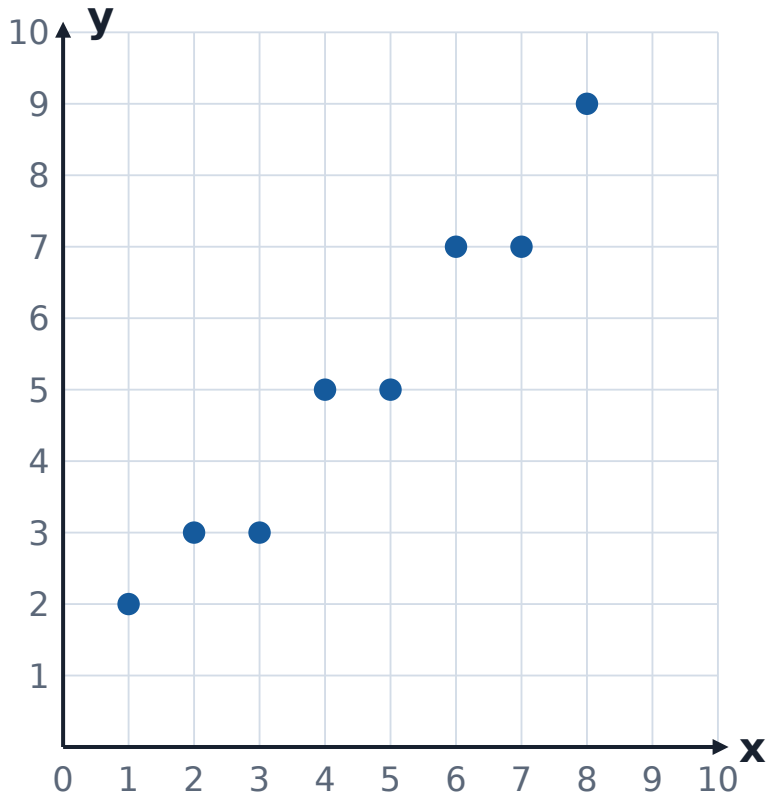
FORWARD 1

Describe the association shown in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

What kind of association does the scatter plot show?

Association: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q1/5 • NEAREST RELATED

A scatter plot shows hours of practice (x) and free throws made (y). The points rise from left to right in a roughly straight pattern.

Describe the association.

Association: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q2/5 • NEAREST RELATED

A scatter plot shows the age of a car (x) and its value (y).

What association is most likely?

A. Negative association

B. Every point on one line

C. No association

D. Positive association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In a scatter plot with no association, the points show no clear trend.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

Points that fall from left to right in a scatter plot show a positive association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 51 • 7.SP.A.2

FORWARD 2

Decide whether a result is consistent with a model by simulation • High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A coin is said to be fair. In 100 simulated sets of 20 fair flips, 15 or more heads happened only 2 times. A real coin gives 15 heads in 20 flips.

Is the fair-coin model believable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q1/5

Simulations of a spinner model usually give 3 to 7 reds in 20 spins. A real spinner gives 5 reds in 20 spins.

What does this show?

A. The spinner is proven unfair.

B. The result is impossible.

C. The result is consistent with the model.

D. The model is proven wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q2**

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q2/5

In 100 simulations of a model, a certain result happened 40 times.

Is that result unusual for the model? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q3/5

True or false?

A result that rarely happens in simulations of a model suggests that the model may be wrong.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q4/5

True or false?

A result that happens even once in 1,000 simulations is a common result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q5/5

SET 69 • 7.SP.A.2 • STATISTICS & PROBABILITY

Generate several same-size random samples with random digits and see how the estimates vary

MAIN	Grade 7 main question	**7.SP.A.2**
BACKWARD 1	Write a part of a small sample as a percent	**6.RP.A.3.c**
BACKWARD 2	Find the mean of a small data set	**6.SP.B.5.c**
FORWARD 1	Summarize categorical data from two random samples in a two-…	**8.SP.A.4**
FORWARD 2	Use the spread of simulated samples to give an interval for a popu…	**HSS-IC.B.4**

SET 69 • 7.SP.A.2

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.2

A school has 100 students numbered 00 to 99. Students 00 to 39 chose art as their favorite class.

Use these random two-digit numbers to choose a sample of 5 students:

07 52 31 88 14

What percent of the sample chose art?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 69 • MAIN • Q1/5

A school has 100 students numbered 00 to 99. Students 00 to 39 chose art as their favorite class.

Use these random two-digit numbers to choose a sample of 5 students:

45 63 02 97 71

What percent of the sample chose art?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 69 • MAIN • Q2/5

Students 00 to 39 of a school of 100 chose art. Four random samples of 5 students were chosen with random digits:

Sample 1: 07 52 31 88 14

Sample 2: 45 63 02 97 71

Sample 3: 38 19 60 25 84

Sample 4: 50 11 76 93 66

What is the range of the four sample percents?

A. 60 percentage points

B. 0 percentage points

C. 20 percentage points

D. 40 percentage points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 69 • MAIN • Q3/5

Students 00 to 39 of a school of 100 chose art, so the true percent is 40%. Four random samples of 5 students gave 60%, 20%, 60%, and 20%.

What is the mean of the four sample percents? How does it compare with the true percent, and how much do single samples vary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 69 • MAIN • Q4/5

True or false?

Random samples of 5 students gave estimates from 20% to 60%. Random samples of 50 students from the same school would usually give estimates that are closer together.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 69 • MAIN • Q5/5

SET 69 • 7.SP.A.2

BACKWARD 1

Write a part of a small sample as a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write 3 out of 5 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 69 • BACKWARD 1 • Q1/5

Write 2 out of 5 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 69 • BACKWARD 1 • Q2/5

What percent is 4 out of 5?

A. 80%

B. 20%

C. 45%

D. 4%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 69 • BACKWARD 1 • Q3/5

True or false?

1 out of 5 is 20%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 69 • BACKWARD 1 • Q4/5

True or false?

3 out of 5 is 35%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 69 • BACKWARD 1 • Q5/5

SET 69 • 7.SP.A.2

BACKWARD 2

Find the mean of a small data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

60, 20, 60, 20

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 69 • BACKWARD 2 • Q1/5

Find the mean.

30, 50, 40

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 69 • BACKWARD 2 • Q2/5

What is the mean of 20, 40, 40, and 60?

A. 50

B. 35

C. 45

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 69 • BACKWARD 2 • Q3/5

True or false?

The mean of 10, 30, and 50 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 69 • BACKWARD 2 • Q4/5

True or false?

The mean of 20, 20, and 80 is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 69 • BACKWARD 2 • Q5/5

SET 69 • 7.SP.A.2

FORWARD 1

Summarize categorical data from two random samples in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows random samples of 30 students from each grade.

What percent of the Grade 7 sample chose art?

	Chose art	Did not	Total
Grade 7 sample	12	18	30
Grade 8 sample	9	21	30
Total	21	39	60

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 69 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows random samples of 30 students from each grade.

What percent of the Grade 8 sample chose art?

	Chose art	Did not	Total
Grade 7 sample	12	18	30
Grade 8 sample	9	21	30
Total	21	39	60

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 69 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows random samples of 30 students from each grade.

Which conclusion do the samples best support?

	Chose art	Did not	Total
Grade 7 sample	12	18	30
Grade 8 sample	9	21	30
Total	21	39	60

A. Every Grade 7 student prefers art.

B. A greater share of Grade 7 students than Grade 8 students may prefer art.

C. No Grade 8 student prefers art.

D. Exactly 40% of all Grade 7 students prefer art.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F1-Q3**

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 69 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of choosing art in the whole sample of 60 students is 0.35.

	Chose art	Did not	Total
Grade 7 sample	12	18	30
Grade 8 sample	9	21	30
Total	21	39	60

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 69 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, more Grade 8 students than Grade 7 students chose art.

	Chose art	Did not	Total
Grade 7 sample	12	18	30
Grade 8 sample	9	21	30
Total	21	39	60

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 69 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 69 • 7.SP.A.2

FORWARD 2

Use the spread of simulated samples to give an interval for a population percent • High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

Simulations show that the middle 95% of random samples of 50 students land within 10 percentage points of the true percent. A real random sample of 50 gives 44%.

What interval is likely to contain the population percent?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 69 • FORWARD 2 • Q1/5

Simulations show that the middle 95% of random samples of 200 people land within 7 percentage points of the true percent. A real random sample of 200 gives 58%.

What interval is likely to contain the population percent?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 69 • FORWARD 2 • Q2/5

In simulations, how does the spread of sample percents change as the samples get larger?

A. It gets larger.

B. It stays the same.

C. It gets smaller.

D. It becomes 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 69 • FORWARD 2 • Q3/5

True or false?

An interval of 34% to 54% means the population percent is likely between those values, but it is not certain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 69 • FORWARD 2 • Q4/5

True or false?

Simulated samples of 50 vary by about 10 percentage points. So one real sample of 50 always lands exactly on the population percent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 69 • FORWARD 2 • Q5/5

SET 52 • 7.SP.B.3 • STATISTICS & PROBABILITY

Describe how much two distributions overlap using the difference in means as a multiple of the MAD

MAIN	Grade 7 main question	**7.SP.B.3**
BACKWARD 1	Find the mean absolute deviation (MAD)	**6.SP.B.5.c**
BACKWARD 2	Describe the shape of a distribution	**6.SP.A.2**
FORWARD 1	Compare groups using relative frequencies in a two-way table —…	**8.SP.A.4**
FORWARD 2	Compare the center and spread of two data sets	**HSS-ID.A.2**

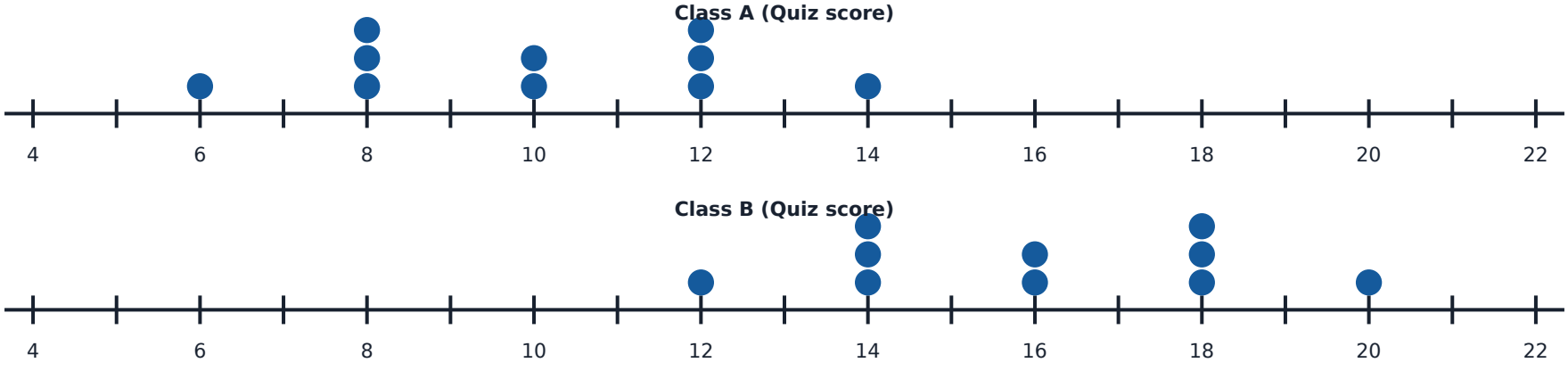
SET 52 • 7.SP.B.3

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.B.3

The dot plots show quiz scores for two classes. The means are 10 and 16, and each data set has a MAD of 2.

Write the difference in the means as a multiple of the MAD.



Answer: _____

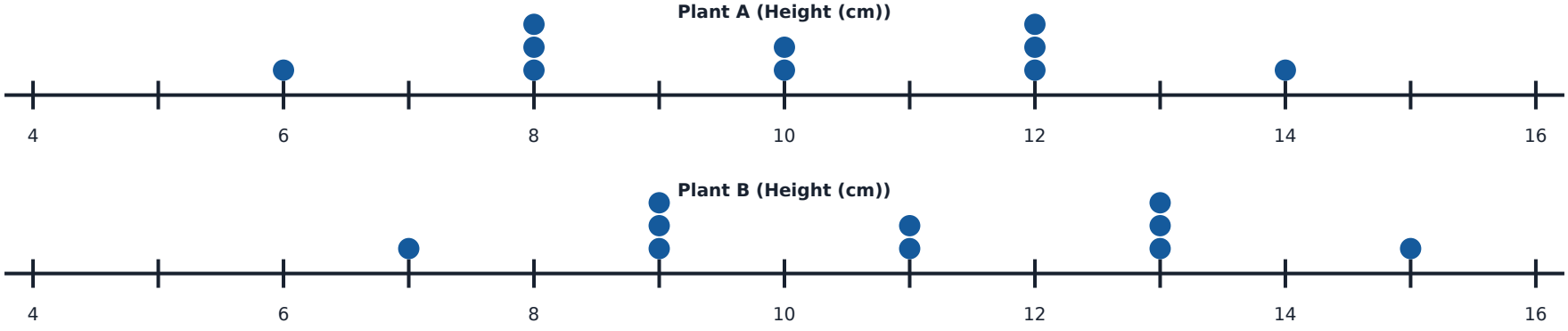
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q1/5

The dot plots show the heights of two kinds of plants, in centimeters. The means are 10 and 11, and each data set has a MAD of 2.

Write the difference in the means as a multiple of the MAD, and describe how much the dot plots overlap.



Answer: _____

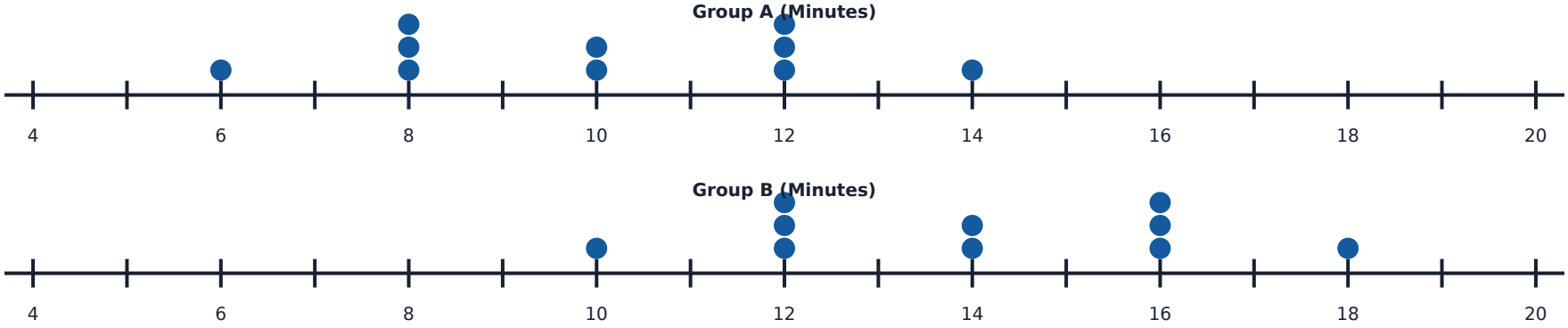
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q2/5

The dot plots show the minutes two groups spent reading. The means are 10 and 14, and each data set has a MAD of 2.

Which statement describes the dot plots?



A. The means differ by 2 MADs, and the plots overlap somewhat.

B. The means are the same, so the plots overlap completely.

C. The means differ by 2 MADs, so the plots cannot share any values.

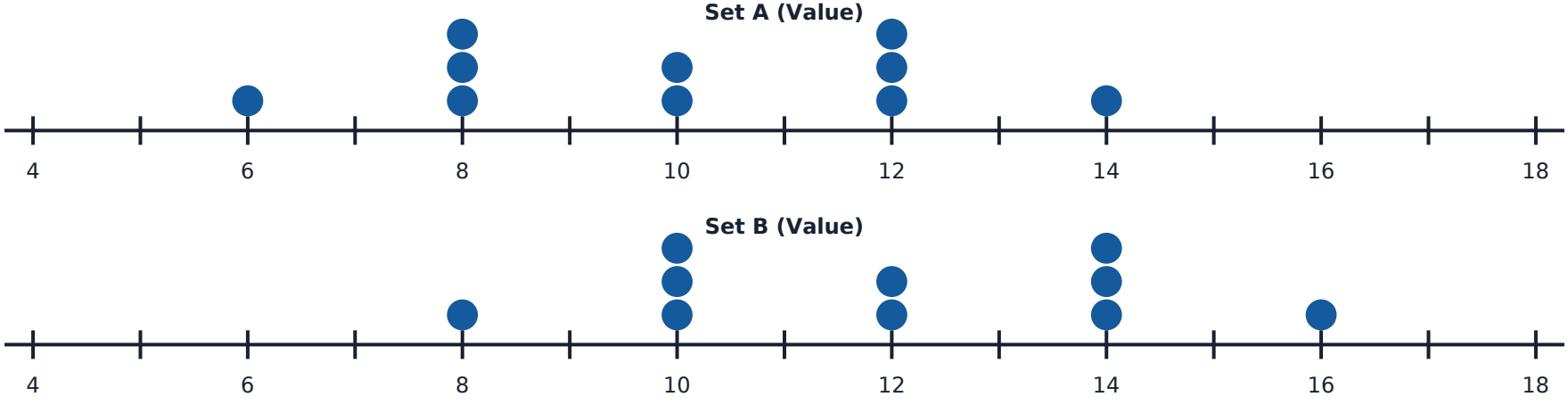
D. The means differ by 4 MADs, and the plots do not overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-M-Q3**

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q3/5

True or false?

The dot plots show two data sets with means of 10 and 12, and each has a MAD of 2. The means differ by 1 MAD, and the dot plots overlap a lot.



A. True

B. False

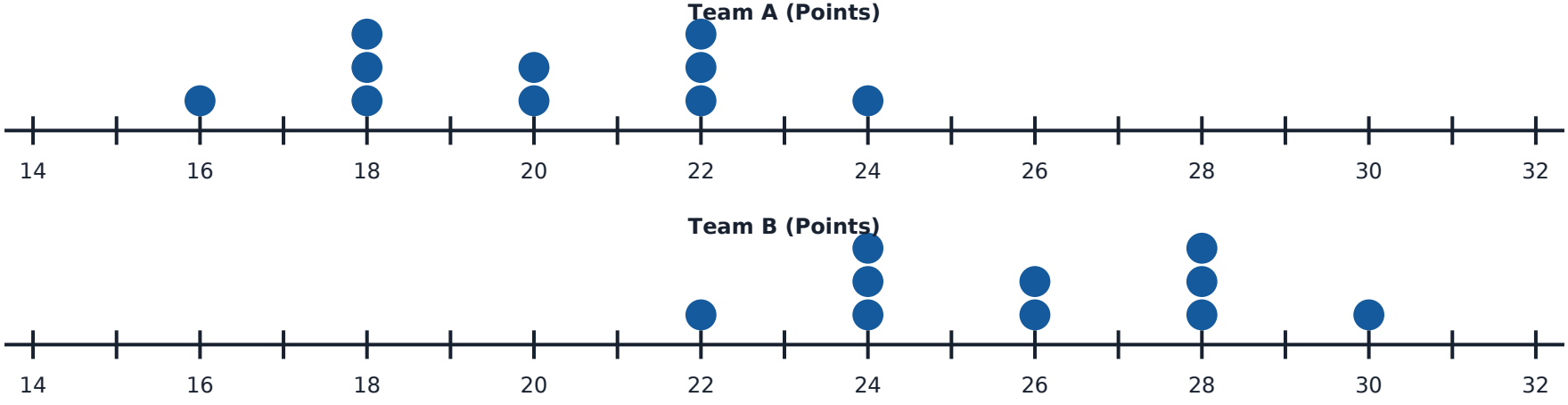
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q4/5

True or false?

The dot plots show two data sets with means of 20 and 26, and each has a MAD of 2. The means differ by 3 MADs, and the two dot plots share no values.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q5/5

SET 52 • 7.SP.B.3

BACKWARD 1

Find the mean absolute deviation (MAD) • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the MAD.

2, 4, 6, 8

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q1/5

Find the MAD.

5, 5, 9, 13

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q2/5

What is the MAD of 10, 10, 10, 10?

A. 0

B. 40

C. 1

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q3/5

True or false?

The MAD of 1, 3, 5, 7 is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q4/5

True or false?

The MAD of 0 and 10 is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 7.SP.B.3

BACKWARD 2

Describe the shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

A dot plot has most of its dots on the left and a long tail to the right.

How is its shape described?

A. Uniform

B. Skewed right

C. Skewed left

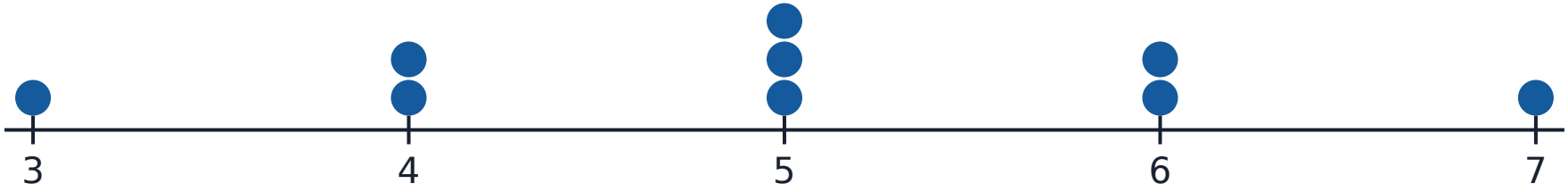
D. Symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q1/5

Is the distribution in the dot plot symmetric?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q2/5

A dot plot has a tall stack of dots in the middle and short, equal stacks on each side.

What is the shape of the distribution?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q3/5

True or false?

A distribution that looks the same on both sides of its center is symmetric.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q4/5

True or false?

A dot plot with all of its dots stacked above one value is skewed left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • 7.SP.B.3

FORWARD 1

Compare groups using relative frequencies in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows how students get to school.

What percent of sixth graders walk?

	Walks	Rides	Total
Sixth grade	24	36	60
Seventh grade	15	45	60
Total	39	81	120

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows how students get to school.

What percent of seventh graders walk?

	Walks	Rides	Total
Sixth grade	24	36	60
Seventh grade	15	45	60
Total	39	81	120

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows how students get to school.

Is there evidence that grade and walking are related?

	Walks	Rides	Total
Sixth grade	24	36	60
Seventh grade	15	45	60
Total	39	81	120

A. No, because each grade has 60 students.

B. No, because more students ride than walk in both grades.

C. Yes, because 40% of sixth graders walk but only 25% of seventh graders do.

D. Yes, because 24 is greater than 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

Relative frequencies make it possible to compare groups of different sizes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

If 40% of sixth graders and 40% of seventh graders walk, there is a strong relationship between grade and walking.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 52 • 7.SP.B.3

FORWARD 2

Compare the center and spread of two data sets • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Data set A has a mean of 20 and a standard deviation of 2. Data set B has a mean of 20 and a standard deviation of 6.

Which data set is more spread out?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q1/5

Data set A has a median of 45 and an IQR of 10. Data set B has a median of 52 and an IQR of 10.

Compare the centers and the spreads.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q2/5

Which measure of spread is best for a skewed data set with outliers?

A. Interquartile range

B. Mean

C. Range

D. Standard deviation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q3/5

True or false?

Standard deviation measures how spread out the data are around the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q4/5

True or false?

Two data sets with the same mean must have the same standard deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q5/5

SET 53 • 7.SP.B.4 • STATISTICS & PROBABILITY

Compare two populations using measures of center and variability from random samples

MAIN	Grade 7 main question	**7.SP.B.4**
BACKWARD 1	Find the interquartile range	**6.SP.B.5.c**
BACKWARD 2	Read a box plot	**6.SP.B.4**
FORWARD 1	Identify clusters and outliers in a scatter plot — nearest related (n…	**8.SP.A.1**
FORWARD 2	Interpret differences in center and spread, including the effect of…	**HSS-ID.A.3**

SET 53 • 7.SP.B.4

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.B.4

Random samples of quiz scores: Class A has a median of 78 and an IQR of 10. Class B has a median of 70 and an IQR of 10.

What do the samples suggest about the two classes? Use both the center and the spread.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q1/5

A random sample of 7th graders has a mean of 6.5 hours of screen time and a MAD of 1.2. A random sample of 8th graders has a mean of 7.7 hours and a MAD of 1.2.

Write the difference in means as a multiple of the MAD. What does this suggest about the two populations?

Answer: _____

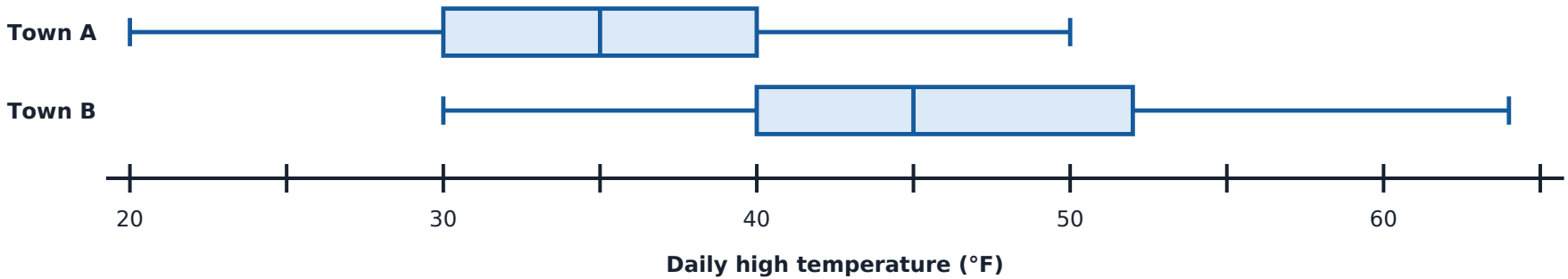
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q2/5

The box plots show random samples of daily high temperatures in two towns.

Which statement is true?



A. The two towns have the same range.

B. The IQRs differ by 10 degrees.

C. The median for Town B is 10 degrees greater than the median for Town A.

D. Town A has the greater median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q3/5

True or false?

Random samples from two populations have means of 20 and 30, and each has a MAD of 2. The samples suggest that values in the second population tend to be greater.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q4/5

True or false?

A random sample from population A has a mean of 52 and one from population B has a mean of 48, each with a MAD of 10. The samples prove that every member of population A has a greater value than every member of population B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q5/5

SET 53 • 7.SP.B.4

BACKWARD 1

Find the interquartile range • Grade 6 • Statistics & Probability • 6.SP.B.5.c

A data set has Q1 = 12 and Q3 = 20.

What is the IQR?

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q1/5

Find the IQR.

2, 4, 6, 8, 10, 12, 14, 16

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q2/5

A box plot has Q1 = 30 and Q3 = 55.

What is the IQR?

A. 30

B. 85

C. 25

D. 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q3/5

True or false?

The IQR is the difference between Q3 and Q1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q4/5

True or false?

A data set with Q1 = 7 and Q3 = 15 has an IQR of 22.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

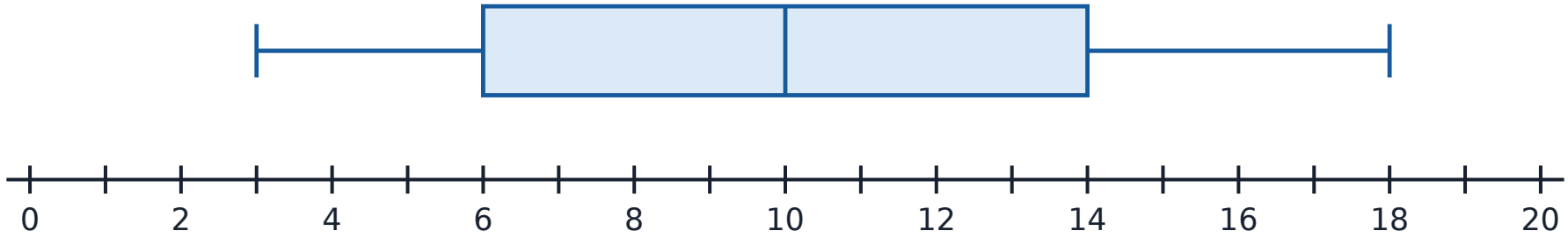
Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q5/5

SET 53 • 7.SP.B.4

BACKWARD 2

Read a box plot • Grade 6 • Statistics & Probability • 6.SP.B.4

What is the median of the data in the box plot?



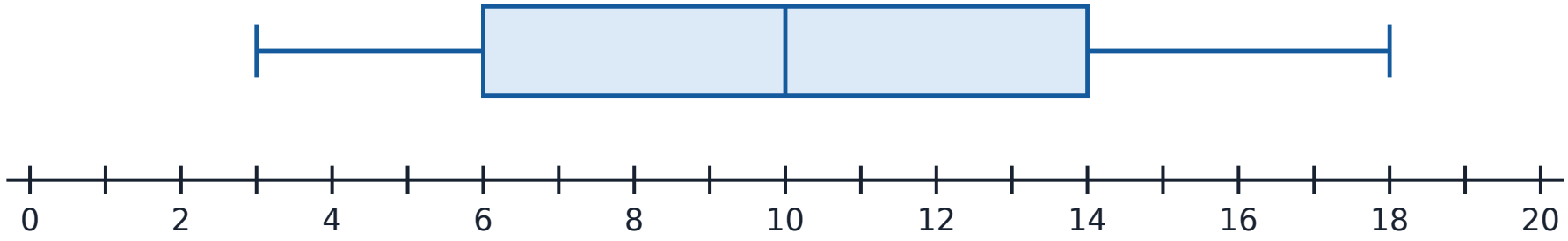
Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q1/5

What is the first quartile (Q1) of the data in the box plot?



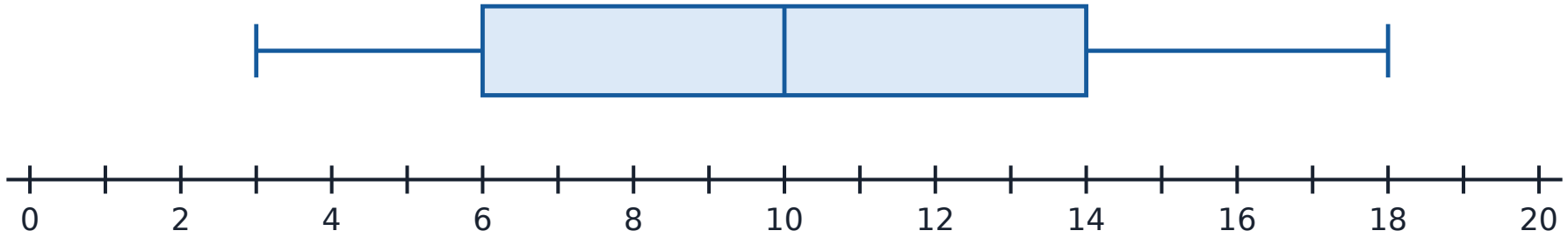
Q1: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q2/5

What is the maximum of the data in the box plot?



A. 14

B. 18

C. 3

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q3/5

True or false?

The box of a box plot shows the middle half of the data.

A. True

B. False

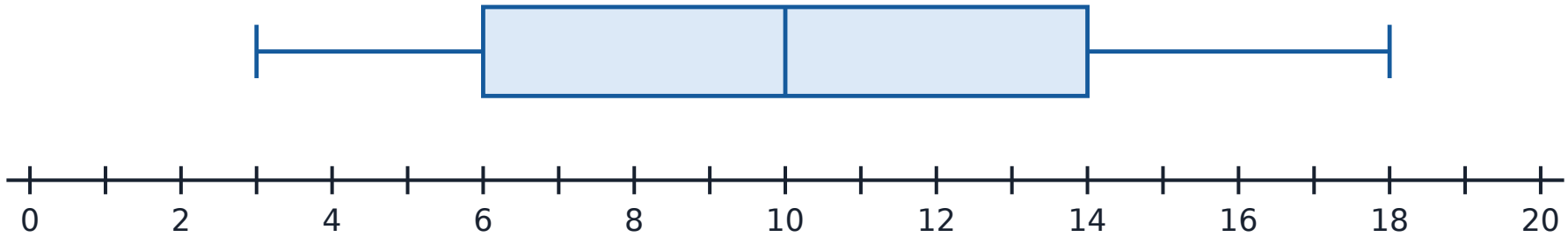
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q4/5

True or false?

The minimum of the data in the box plot is 6.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • 7.SP.B.4

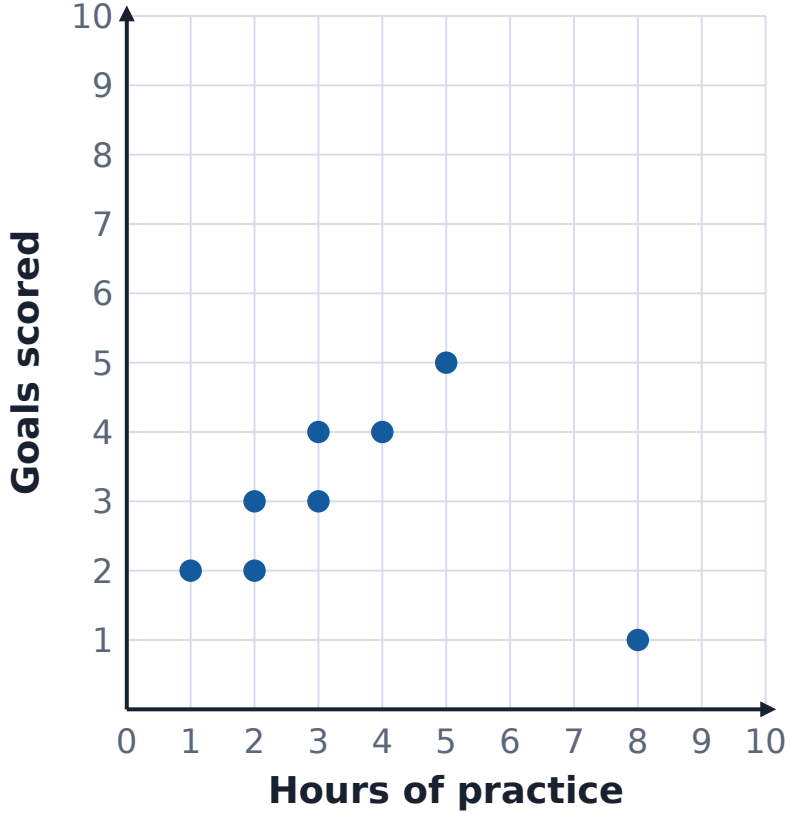
FORWARD 1

Identify clusters and outliers in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Which point in the scatter plot is an outlier?

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q1/5 • NEAREST RELATED

A scatter plot has a group of points close together.

What is the group called?

A. A cluster

B. An outlier

C. An intercept

D. A slope

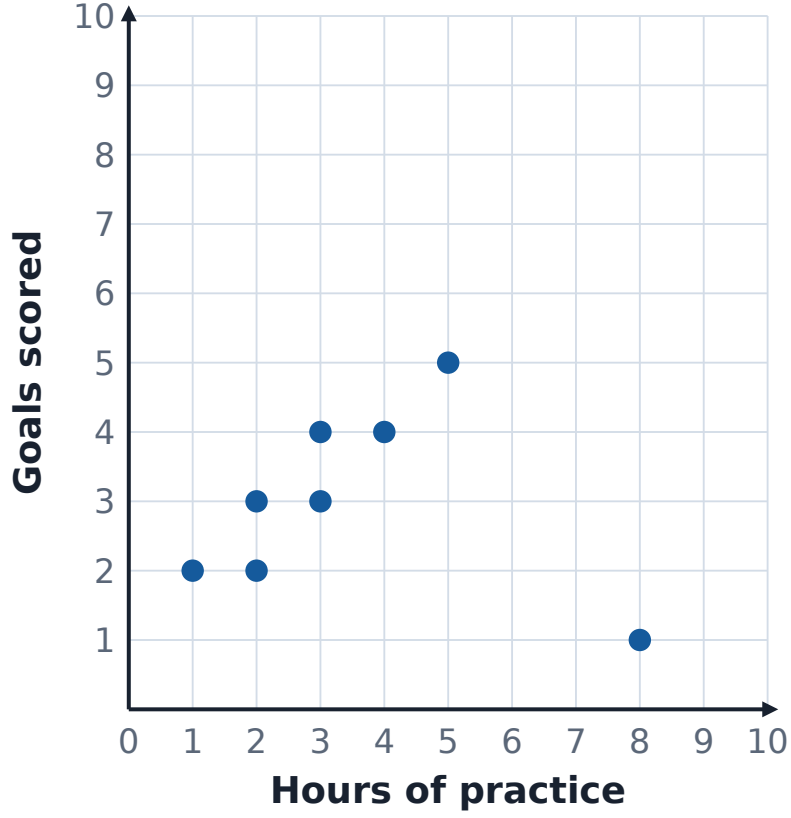
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q2/5 • NEAREST RELATED

Leaving out the outlier, describe the association in the scatter plot.

Association: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

An outlier is a point that is far from the overall pattern of the data.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

Every scatter plot has exactly one outlier.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 53 • 7.SP.B.4

FORWARD 2

Interpret differences in center and spread, including the effect of outliers • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

A data set is 4, 5, 5, 6, 7, 40.

Which better describes its center, the mean or the median? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q1/5

A data set is 4, 5, 5, 6, 7, 40.

Find the mean with the outlier and the mean without it. Round a mean that does not end to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q2/5

Which measure is affected most by an outlier?

A. Mean

B. IQR

C. Mode

D. Median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q3/5

True or false?

The median is resistant to outliers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q4/5

True or false?

Removing a very large outlier makes the mean larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q5/5

SET 54 • 7.SP.C.5 • STATISTICS & PROBABILITY

Understand probability as a number from 0 to 1 that shows how likely an event is

MAIN	Grade 7 main question	**7.SP.C.5**
BACKWARD 1	Compare two fractions	**4.NF.A.2**
BACKWARD 2	Place fractions on a number line from 0 to 1	**3.NF.A.2**
FORWARD 1	Find relative frequencies in a two-way table — nearest related (no…	**8.SP.A.4**
FORWARD 2	Describe unions, intersections, and complements of events as sub…	**HSS-CP.A.1**

SET 54 • 7.SP.C.5

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.5

Which probability describes an event that is likely to happen?

A. 0.1

B. 0.8

C. 0.5

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q1/5

A bag has 5 red marbles and no other marbles.

What is the probability of drawing a red marble?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

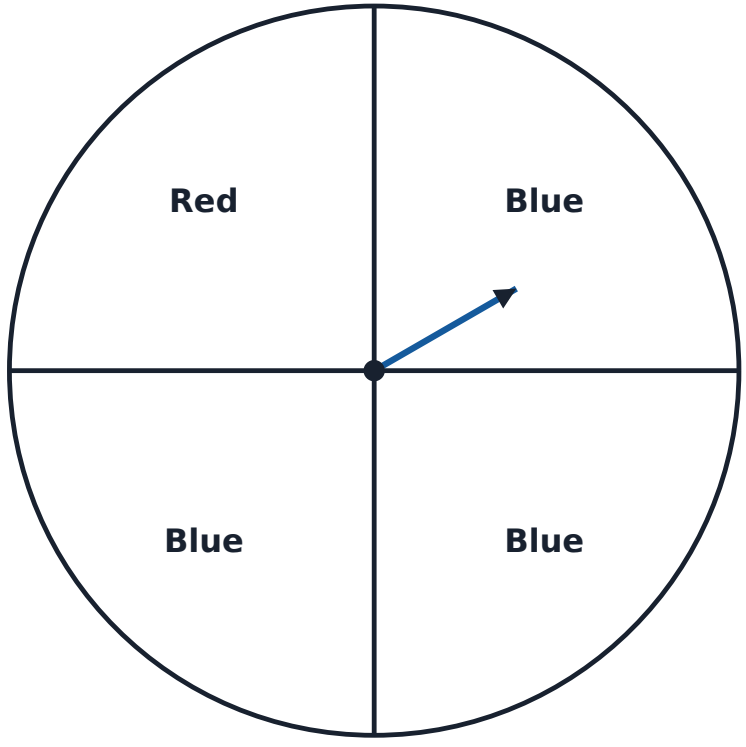
ID S54-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q2/5

The spinner has 4 equal sections.

Is landing on Blue likely, unlikely, or neither likely nor unlikely?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q3/5

True or false?

An event with a probability of $\frac{1}{2}$ is just as likely to happen as not to happen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q4/5

True or false?

A probability of 1.3 means that an event is very likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q5/5

SET 54 • 7.SP.C.5

BACKWARD 1

Compare two fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which is greater, $\frac{3}{8}$ or $\frac{2}{3}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q1/5

Write >, <, or =.

$\frac{5}{10}$ ____ $\frac{1}{2}$

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q2/5

Which fraction is less than $\frac{1}{2}$?

A. $\frac{3}{4}$

B. $\frac{5}{8}$

C. $\frac{3}{8}$

D. $\frac{4}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q3/5

True or false?

$\frac{7}{12} > \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q4/5

True or false?

$\frac{2}{5} > \frac{3}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • 7.SP.C.5

BACKWARD 2

Place fractions on a number line from 0 to 1 • Grade 3 • Number & Operations–Fractions • 3.NF.A.2

Plot $\frac{3}{4}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q1/5

A number line from 0 to 1 is split into 4 equal parts.

What fraction is at the first tick mark after 0?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q2/5

Which fraction is halfway between 0 and 1 on a number line?

A. $\frac{1}{2}$

B. $\frac{1}{4}$

C. $\frac{2}{1}$

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q3/5

True or false?

$\frac{4}{4}$ is at the same point as 1 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{3}$ is farther from 0 than $\frac{2}{3}$ is.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • 7.SP.C.5

FORWARD 1

Find relative frequencies in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows the results of a survey.

Among Grade 9 students, what is the relative frequency of playing music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows the results of a survey.

Among Grade 10 students, what is the relative frequency of playing music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows the results of a survey.

What is the relative frequency of all students who play music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. 0.3 **B.** 0.18 **C.** 0.12 **D.** 0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, students in both grades play music at the same rate.

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, Grade 9 students are more likely than Grade 10 students to play music, because 18 > 12.

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 54 • 7.SP.C.5

FORWARD 2

Describe unions, intersections, and complements of events as subsets of a sample space • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A card is drawn at random from cards numbered 1 to 10. Event A is "an even number." Event B is "a number greater than 6."

List the outcomes in "A and B" (the intersection).

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q1/5

A card is drawn at random from cards numbered 1 to 10. Event A is "an even number." Event B is "a number greater than 6."

List the outcomes in "A or B" (the union).

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q2/5

A card is drawn at random from cards numbered 1 to 10. Event A is "an even number."

Which set of outcomes is the complement of A ("not A")?

A. 1, 3, 5, 7, 9

B. 2, 4, 6, 8, 10

C. 7, 8, 9, 10

D. 1, 2, 3, 4, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q3/5

True or false?

A card is drawn at random from cards numbered 1 to 10. Event A is "an even number" and event B is "a number greater than 6." Then P(A or B) = $\frac{7}{10}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q4/5

True or false?

A card is drawn at random from cards numbered 1 to 10. Event A is "an even number" and event B is "a number greater than 6." The intersection of A and B has more outcomes than their union.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q5/5

SET 55 • 7.SP.C.6 • STATISTICS & PROBABILITY

Estimate a probability from relative frequency and make predictions

MAIN	Grade 7 main question	**7.SP.C.6**
BACKWARD 1	Use a rate to make a prediction	**6.RP.A.3.b**
BACKWARD 2	Compare decimals to thousandths	**5.NBT.A.3.b**
FORWARD 1	Use relative frequencies in a two-way table to make predictions…	**8.SP.A.4**
FORWARD 2	Find conditional probabilities from a two-way table	**HSS-CP.A.4**

SET 55 • 7.SP.C.6

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.6

A spinner landed on green 36 times in 120 spins.

Estimate the probability of landing on green.

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q1/5

A basketball player made 42 of her last 60 free throws.

Predict how many she will make in her next 150 free throws.

Free throws: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q2/5

A coin was flipped 200 times and landed heads 94 times.

What is the relative frequency of heads?

A. 0.94

B. 0.47

C. 0.53

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q3/5

True or false?

A fair number cube (numbered 1 to 6) is rolled 300 times. About 50 of the rolls should be a 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q4/5

True or false?

A spinner landed on red 9 out of 10 times. The probability of red must be exactly 0.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q5/5

SET 55 • 7.SP.C.6

BACKWARD 1

Use a rate to make a prediction • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

A team scores 3 goals every 2 games.

At this rate, how many goals does it score in 10 games?

Goals: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q1/5

4 out of every 10 apples in a crate are green.

At this rate, how many of 250 apples are green?

Apples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q2/5

A machine fills 6 bottles every 4 seconds.

At this rate, how many bottles does it fill in 30 seconds?

A. 45

B. 20

C. 24

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q3/5

True or false?

At a rate of 2 out of every 5, there are 40 out of every 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q4/5

True or false?

At a rate of 3 out of every 8, there are 30 out of every 64.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q5/5

SET 55 • 7.SP.C.6

BACKWARD 2

Compare decimals to thousandths • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b

Which is greater, 0.47 or 0.5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q1/5

Write >, <, or =.

0.305 ___ 0.35

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q2/5

Which decimal is the greatest?

A. 0.059

B. 0.62

C. 0.506

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q3/5

True or false?

0.250 = 0.25

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q4/5

True or false?

$0.09 > 0.1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q5/5

SET 55 • 7.SP.C.6

FORWARD 1

Use relative frequencies in a two-way table to make predictions • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows the results of a survey.

Among teens, what is the relative frequency of liking rain?

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q1/5 • NEAREST RELATED

Use the adults row of the table.

Predict how many of 200 adults like rain.

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

Adults: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows the results of a survey.

Which group has the greater relative frequency of liking rain?

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

A. Adults

B. Teens

C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of liking rain among all 100 people is 0.27.

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of disliking rain among adults is 0.42.

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 55 • 7.SP.C.6

FORWARD 2

Find conditional probabilities from a two-way table • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

The table shows students' favorite pets. A student is chosen at random.

Find P(dog | Grade 9).

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q1/5

The table shows students' favorite pets. A student is chosen at random.

Find P(Grade 10 | cat).

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q2/5

The table shows students' favorite pets. A student is chosen at random.

Find P(cat).

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

A. $\frac{1}{2}$ **B.** $\frac{9}{16}$ **C.** $\frac{7}{16}$ **D.** $\frac{3}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q3/5

True or false?

In the table, P(dog | Grade 10) = $\frac{1}{2}$.

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q4/5

True or false?

In the table, P(Grade 9 | dog) = $\frac{25}{40}$.

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q5**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q5/5

SET 70 • 7.SP.C.6 • STATISTICS & PROBABILITY

Collect data from a chance process and use it to estimate a probability

MAIN	Grade 7 main question	**7.SP.C.6**
BACKWARD 1	Organize data with up to four categories	**2.MD.D.10**
BACKWARD 2	Write fractions with denominators of 10 or 100 as decimals	**4.NF.C.6**
FORWARD 1	Record results of two chance processes in a two-way table — near…	**8.SP.A.4**
FORWARD 2	Use collected data to judge whether two events seem independe…	**HSS-CP.A.2**

SET 70 • 7.SP.C.6

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.6

Use these random digits as rolls of a fair number cube. Use only the digits 1 to 6, and skip 0, 7, 8, and 9.

4 8 2 6 6 1 9 3 5 6 0 2 4 6 1 7 3 6 5 2

Record the first 15 rolls. What is the relative frequency of rolling a 6?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 70 • MAIN • Q1/5

A student records 20 flips of a coin:

H T T H H H T H T T H H T H H T H H T H

What is the relative frequency of heads? Use it to predict the number of heads in 200 flips.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 70 • MAIN • Q2/5

Roll a number cube (or use a number-cube app) 30 times. Record each result in a tally table. Then find the relative frequency of rolling an even number.

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 70 • MAIN • Q3/5

A student records 50 spins of a spinner: 18 red, 22 blue, and 10 green.

What is the relative frequency of blue?

A. 0.36

B. 0.44

C. 0.22

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 70 • MAIN • Q4/5

True or false?

A student rolls a fair number cube 30 times. Her data must show exactly 5 threes, or she made a mistake.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 70 • MAIN • Q5/5

SET 70 • 7.SP.C.6

BACKWARD 1

Organize data with up to four categories • Grade 2 • Measurement & Data • 2.MD.D.10

A spinner gives these results: R, B, R, G, R, B, R, G, B, R.

How many times did it land on R?

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

Grade 2 • Measurement & Data • 2.MD.D.10 • Set 70 • BACKWARD 1 • Q1/5

A spinner gives these results: R, B, R, G, R, B, R, G, B, R.

How many results are there in all?

Results: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

Grade 2 • Measurement & Data • 2.MD.D.10 • Set 70 • BACKWARD 1 • Q2/5

A tally chart shows red 4, blue 3, and green 3.

How many results are there in all?

A. 7

B. 4

C. 10

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q3

Grade 2 • Measurement & Data • 2.MD.D.10 • Set 70 • BACKWARD 1 • Q3/5

True or false?

A tally chart can be used to record results as they happen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

Grade 2 • Measurement & Data • 2.MD.D.10 • Set 70 • BACKWARD 1 • Q4/5

True or false?

Red has 5 tally marks and blue has 3. Blue has 2 more than red.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

Grade 2 • Measurement & Data • 2.MD.D.10 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • 7.SP.C.6

BACKWARD 2

Write fractions with denominators of 10 or 100 as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{6}{10}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 70 • BACKWARD 2 • Q1/5

Write $\frac{44}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 70 • BACKWARD 2 • Q2/5

Which decimal is equal to $\frac{35}{100}$?

A. 3.5

B. 0.035

C. 35

D. 0.35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 70 • BACKWARD 2 • Q3/5

True or false?

$\frac{8}{10} = 0.8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 70 • BACKWARD 2 • Q4/5

True or false?

$\frac{5}{100} = 0.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • 7.SP.C.6

FORWARD 1

Record results of two chance processes in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A coin was flipped and a number cube rolled 40 times. The table shows the results.

What is the relative frequency of heads?

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 70 • FORWARD 1 • Q1/5 • NEAREST RELATED

A coin was flipped and a number cube rolled 40 times. The table shows the results.

Among the heads results, what is the relative frequency of an even number?

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 70 • FORWARD 1 • Q2/5 • NEAREST RELATED

A coin was flipped and a number cube rolled 40 times. The table shows the results.

Which result happened most often?

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

A. Tails and odd

B. Tails and even

C. Heads and even

D. Heads and odd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 70 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of tails and odd is 0.3.

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 70 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, the relative frequencies of the four results add to more than 1.

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 70 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 70 • 7.SP.C.6

FORWARD 2

Use collected data to judge whether two events seem independent • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

A coin was flipped and a number cube rolled 40 times. The table shows the results.

Estimate P(heads), P(even), and P(heads and even).

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

Estimates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q1**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 70 • FORWARD 2 • Q1/5

A coin was flipped and a number cube rolled 40 times. The table shows the results.

Is P(heads and even) close to P(heads) × P(even)? What does this suggest?

	Even	Odd	Total
Heads	11	9	20
Tails	8	12	20
Total	19	21	40

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q2**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 70 • FORWARD 2 • Q2/5

Events A and B are independent.

Which statement must be true?

A. $P(A) = P(B)$

B. $P(A \text{ and } B) = P(A) + P(B)$

C. $P(A \text{ or } B) = 1$

D. $P(A \text{ and } B) = P(A) \times P(B)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 70 • FORWARD 2 • Q3/5

True or false?

A fair coin is flipped and a fair number cube numbered 1 to 6 is rolled. P(heads and 6) = $\frac{1}{12}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 70 • FORWARD 2 • Q4/5

True or false?

Events A and B are independent, with P(A) = 0.5 and P(B) = 0.4. Then P(A and B) = 0.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 70 • FORWARD 2 • Q5/5

SET 56 • 7.SP.C.7.a • STATISTICS & PROBABILITY

Develop a uniform probability model and use it to find probabilities

MAIN	Grade 7 main question	**7.SP.C.7.a**
BACKWARD 1	Write equivalent fractions	**4.NF.A.1**
BACKWARD 2	Write a fraction for part of a whole	**3.NF.A.1**
FORWARD 1	Use a function that assigns a probability to each outcome — near…	**8.F.A.1**
FORWARD 2	Use the addition rule for probability	**HSS-CP.B.7**

SET 56 • 7.SP.C.7.a

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.7.a

A fair number cube has faces numbered 1 to 6.

What is the probability of rolling a number greater than 4?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

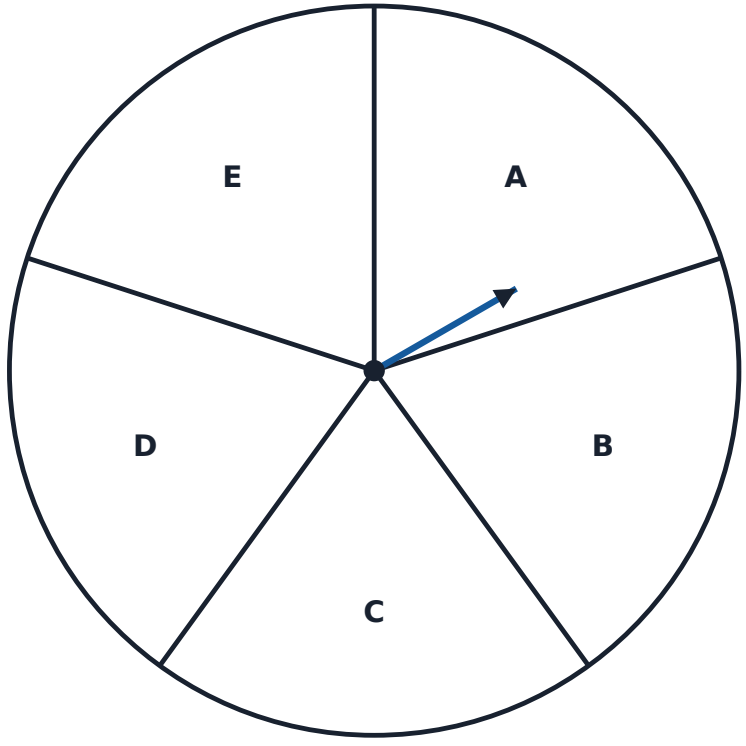
ID S56-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q1/5

The spinner has 5 equal sections.

What is the probability of landing on a vowel?

Probability: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q2/5

A class has 12 girls and 18 boys. One student is chosen at random.

What is the probability that a girl is chosen?

A. $\frac{2}{5}$

B. $\frac{2}{3}$

C. $\frac{1}{12}$

D. $\frac{3}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q3/5

True or false?

In a uniform probability model, every outcome has the same probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q4/5

True or false?

A letter is chosen at random from the word MATH.

The probability of choosing a vowel is $\frac{1}{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q5/5

SET 56 • 7.SP.C.7.a

BACKWARD 1

Write equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Write a fraction equivalent to $\frac{3}{12}$ with a denominator of 4.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q1/5

Write a fraction equivalent to $\frac{8}{12}$ with a denominator of 3.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q2/5

Which fraction is equivalent to $\frac{4}{6}$?

A. $\frac{2}{3}$

B. $\frac{6}{4}$

C. $\frac{4}{12}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q3/5

True or false?

$\frac{5}{10} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q4/5

True or false?

$\frac{2}{8} = \frac{1}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • 7.SP.C.7.a

BACKWARD 2

Write a fraction for part of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

A spinner has 8 equal parts. 3 of the parts are red.

What fraction of the spinner is red?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q1/5

A bag has 6 equal-size tiles. 1 tile is blue.

What fraction of the tiles is blue?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q2/5

A pizza is cut into 4 equal slices. 3 slices are eaten.

What fraction of the pizza is eaten?

A. $\frac{3}{6}$

B. $\frac{4}{3}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q3/5

True or false?

One of 6 equal parts is $\frac{1}{6}$ of the whole.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q4/5

True or false?

Two of 6 equal parts is $\frac{2}{4}$ of the whole.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • 7.SP.C.7.a

FORWARD 1

Use a function that assigns a probability to each outcome • Grade 8 • Functions • 8.F.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A spinner has 4 equal sections, A, B, C, and D, and a fair pointer. A probability model assigns a probability to each outcome.

Write the model as pairs (outcome, probability). Is it a function? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q1/5 • NEAREST RELATED

A rule assigns (red, 0.5), (blue, 0.3), and (red, 0.2).

Is this rule a function from outcomes to probabilities? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q2/5 • NEAREST RELATED

A fair number cube numbered 1 to 6 is rolled. The function P gives the probability of each outcome.

What is P(4)?

A. $\frac{4}{6}$

B. $\frac{1}{6}$

C. $\frac{1}{4}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A probability model gives exactly one probability to each outcome, just as a function gives exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In a function, one input can have two different outputs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 56 • 7.SP.C.7.a

FORWARD 2

Use the addition rule for probability • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

A card is chosen at random from cards numbered 1 to 10.

Find P(even or greater than 7).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q1/5

P(A) = 0.4, P(B) = 0.5, and P(A and B) = 0.2.

Find P(A or B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q2/5

A fair number cube numbered 1 to 6 is rolled.

What is P(less than 3 or odd)?

A. $\frac{5}{6}$

B. $\frac{2}{3}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q3/5

2025

True or false?

If A and B cannot both happen, then P(A or B) = P(A) + P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q4/5

True or false?

P(A or B) is always equal to P(A) + P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q5/5

SET 57 • 7.SP.C.7.b • STATISTICS & PROBABILITY

Develop a probability model from observed frequencies

MAIN	Grade 7 main question	**7.SP.C.7.b**
BACKWARD 1	Read a scaled bar graph	**3.MD.B.3**
BACKWARD 2	Write a part of a whole as a percent	**6.RP.A.3.c**
FORWARD 1	Construct a two-way table from survey counts — nearest related…	**8.SP.A.4**
FORWARD 2	Summarize categorical data with relative frequencies	**HSS-ID.B.5**

SET 57 • 7.SP.C.7.b

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.7.b

A bottle cap is tossed 80 times. It lands top-up 52 times.

Estimate the probability that it lands top-up.

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q1/5

A spinner with three unequal sections is spun 200 times: red 90 times, blue 70 times, and green 40 times.

Write an estimated probability for each color.

Probabilities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q2/5

A spinner landed on red 90 times, blue 70 times, and green 40 times in 200 spins.

Which is the best prediction for 1,000 spins?

A. About 900 reds

B. About 333 reds

C. About 90 reds

D. About 450 reds

True or false?

A probability estimated from observed frequencies can change when more trials are done.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q4/5

True or false?

A model built from 10 tosses is just as reliable as a model built from 1,000 tosses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q5/5

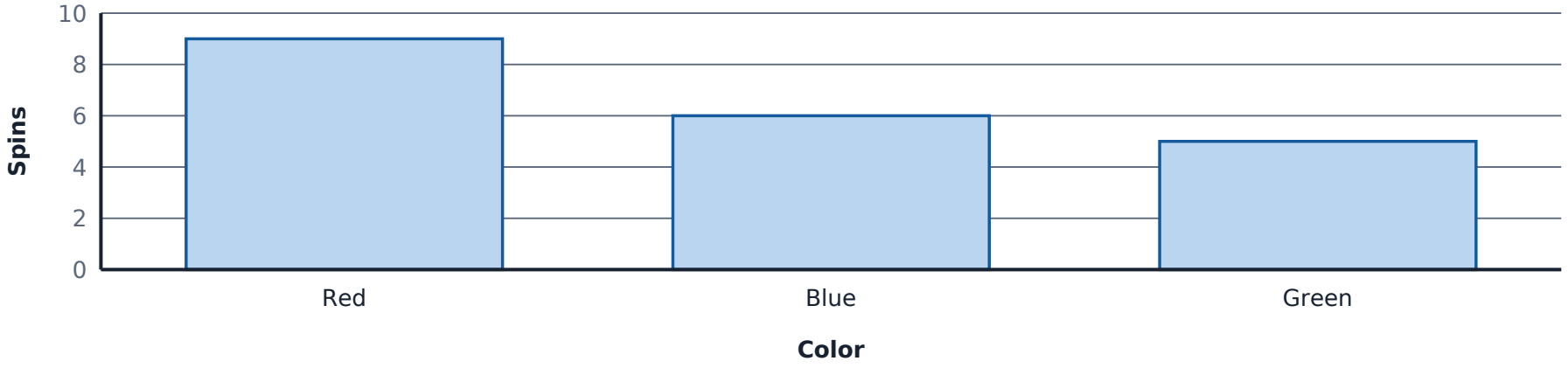
SET 57 • 7.SP.C.7.b

BACKWARD 1

Read a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

The bar graph shows the results of some spins.

How many spins landed on red?



Spins: _____

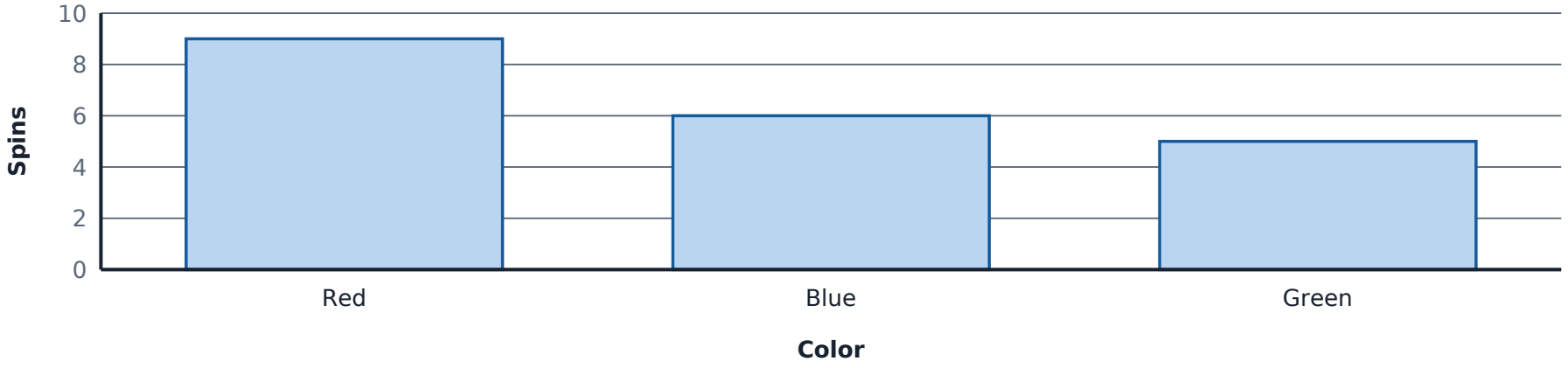
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q1/5

The bar graph shows the results of some spins.

How many more spins landed on red than on green?



Spins: _____

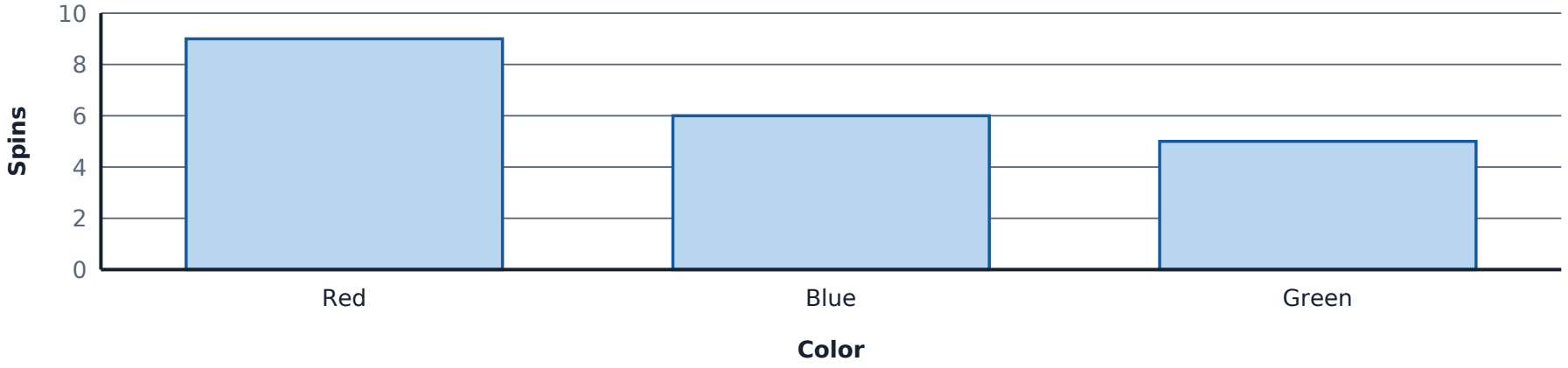
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q2/5

The bar graph shows the results of some spins.

How many spins are shown in all?



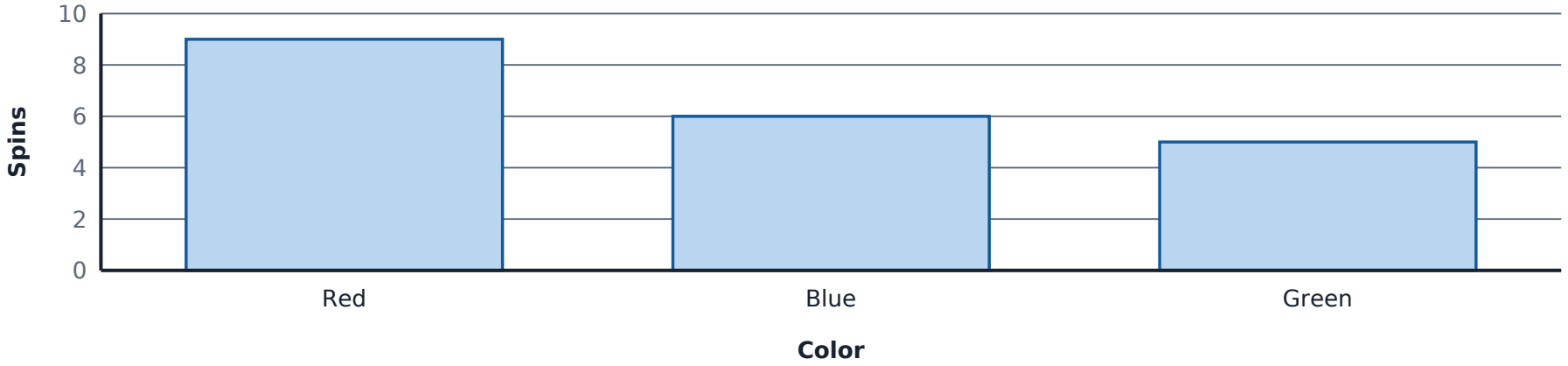
A. 9 **B.** 11 **C.** 15 **D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-B1-Q3**

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q3/5

True or false?

The bar graph shows that blue was spun 6 times.



A. True

B. False

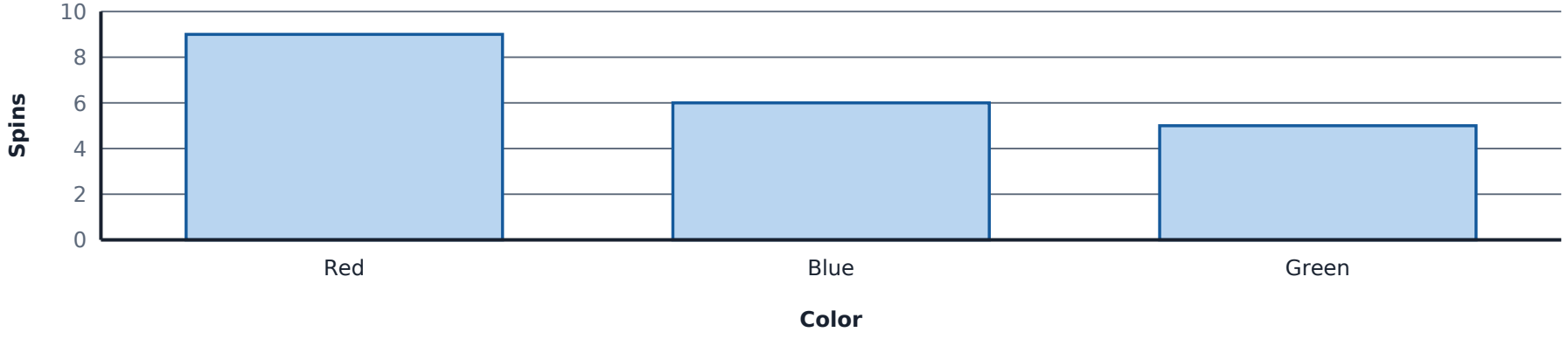
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q4/5

True or false?

The bar graph shows that green was spun more times than blue.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • 7.SP.C.7.b

BACKWARD 2

Write a part of a whole as a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write 45 out of 60 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q1/5

Write 13 out of 50 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q2/5

What percent is 72 out of 120?

A. 72%

B. 48%

C. 60%

D. 40%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q3/5

True or false?

9 out of 36 is 25%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q4/5

True or false?

14 out of 40 is 14%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q5/5

SET 57 • 7.SP.C.7.b

FORWARD 1

Construct a two-way table from survey counts • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

50 students were surveyed. 30 have a phone. 18 of the students with a phone also have a tablet. 8 of the students without a phone have a tablet.

Complete the two-way table.

	Tablet	No tablet	Total
Phone			
No phone			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q1/5 • NEAREST RELATED

40 people were surveyed. 15 have a dog. 6 of the dog owners also have a cat. 10 of the people without a dog have a cat.

Complete the two-way table.

	Cat	No cat	Total
Dog			
No dog			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q2/5 • NEAREST RELATED

60 students were surveyed. 35 play a sport. 20 of the sport players also play an instrument. 30 of the students play an instrument.

Complete the two-way table.

	Instrument	No instrument	Total
Sport			
No sport			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q3/5 • NEAREST RELATED

80 adults were surveyed. 48 drink coffee. 12 of the coffee drinkers also drink tea. 20 adults drink neither coffee nor tea.

Complete the two-way table.

	Tea	No tea	Total
Coffee			
No coffee			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q4/5 • NEAREST RELATED

100 students were surveyed: 45 in Grade 7 and 55 in Grade 8. 27 of the Grade 7 students and 22 of the Grade 8 students ride the bus.

Complete the two-way table.

	Bus	No bus	Total
Grade 7			
Grade 8			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 57 • 7.SP.C.7.b

FORWARD 2

Summarize categorical data with relative frequencies • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

The table shows the results of a survey.

What is the joint relative frequency of Grade 9 students who play music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q1/5

The table shows the results of a survey.

What is the marginal relative frequency of students who do not play music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q2/5

The table shows the results of a survey.

What is the conditional relative frequency of playing music among Grade 10 students?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. 0.7 **B.** 0.4 **C.** 0.3 **D.** 0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q3/5

True or false?

A marginal relative frequency uses a row total or a column total divided by the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q4/5

True or false?

A conditional relative frequency always divides by the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q5/5

SET 58 • 7.SP.C.7 • STATISTICS & PROBABILITY

Compare probabilities from a model with observed frequencies and explain any differences

MAIN	Grade 7 main question	**7.SP.C.7**
BACKWARD 1	Find a fraction of a whole number	**5.NF.B.4.a**
BACKWARD 2	Divide whole numbers to find equal shares	**4.NBT.B.6**
FORWARD 1	Compare a linear model's predictions with observed data — near…	**8.SP.A.3**
FORWARD 2	Judge whether a result is consistent with a model by using a simu…	**HSS-IC.A.2**

SET 58 • 7.SP.C.7

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.7

A fair number cube is rolled 60 times, and a 5 comes up 14 times.

How many 5s does the model predict? How does the observed count compare?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q1/5

A coin is claimed to be fair. It is flipped 100 times and lands heads 61 times.

Compare this with the model. Give a possible reason for the difference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q2/5

A spinner has 4 equal sections, one of them yellow. In 80 spins, it lands on yellow 35 times.

Which conclusion is best?

A. The model predicts 40 yellows.

B. Yellow came up much more than the expected 20 times, so the spinner may not be fair.

C. Yellow should come up 35 times in every 80 spins.

D. The results match the model exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q3/5

True or false?

A good probability model does not have to match observed frequencies exactly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q4/5

True or false?

A coin that lands heads 7 times in 10 flips must be unfair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q5/5

SET 58 • 7.SP.C.7

BACKWARD 1

Find a fraction of a whole number • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Find the value.

$\frac{1}{6} \times 60$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q1/5

What is $\frac{1}{4}$ of 80?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q2/5

What is $\frac{2}{5}$ of 50?

A. 25

B. 125

C. 20

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{3}$ of 90 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q4/5

True or false?

$\frac{3}{4}$ of 40 is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • 7.SP.C.7

BACKWARD 2

Divide whole numbers to find equal shares • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Divide.

240 ÷ 4

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q1/5

Divide.

$300 \div 6$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q2/5

Divide.

1,000 ÷ 8

A. 105

B. 150

C. 125

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q3/5

True or false?

$500 \div 5 = 100$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q4/5

True or false?

$720 \div 3 = 24$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • 7.SP.C.7

FORWARD 1

Compare a linear model's predictions with observed data • Grade 8 • Statistics & Probability • 8.SP.A.3

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A model is $y = 2x + 1$. At $x = 4$, the observed value is $y = 11$.

How far is the observed value from the prediction?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q1/5 • NEAREST RELATED

A model is $y = 5x$. At $x = 3$, the observed value is $y = 13$.

Is the model's prediction too high or too low, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which observed point is closest to the model $y = x + 3$?

A. (2, 7)

B. (2, 9)

C. (2, 3)

D. (2, 5.2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A linear model can be useful even when the data points do not all lie on the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

If one data point is far from the line of fit, the model must be thrown out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q5/5 • NEAREST RELATED

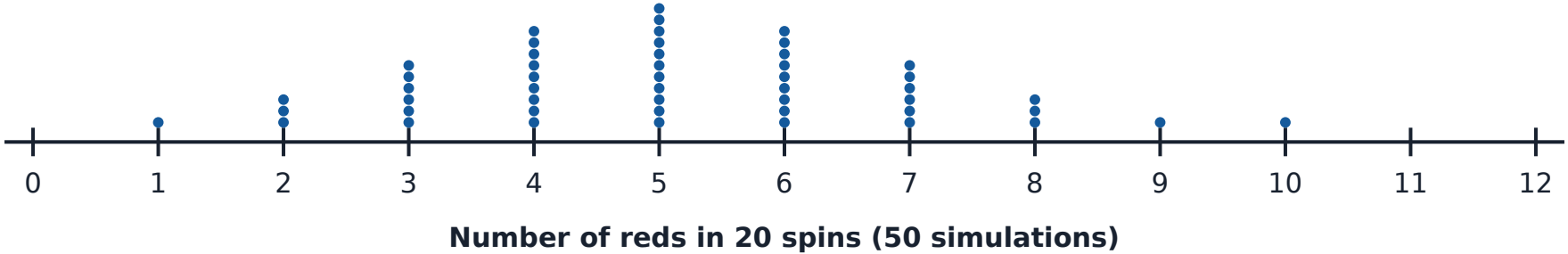
SET 58 • 7.SP.C.7

FORWARD 2

Judge whether a result is consistent with a model by using a simulation • High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A spinner is claimed to land on red 25% of the time. The dot plot shows the number of reds in each of 50 simulated sets of 20 spins under that model. A real spinner lands on red 11 times in 20 spins.

Is the result consistent with the model? Use the simulation.



Answer: _____

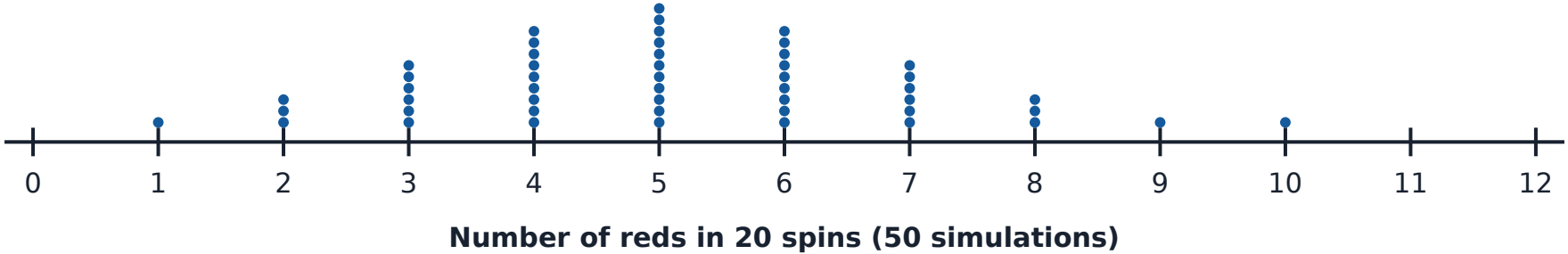
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q1/5

A spinner is claimed to land on red 25% of the time. The dot plot shows the number of reds in each of 50 simulated sets of 20 spins under that model. A real spinner lands on red 6 times in 20 spins.

Is the result consistent with the model? Use the simulation.



Answer: _____

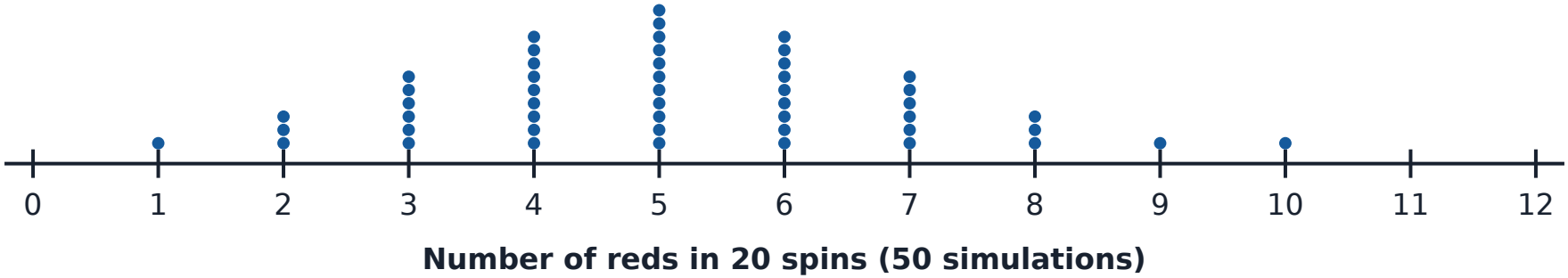
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q2/5

The dot plot shows the number of reds in each of 50 simulated sets of 20 spins of a spinner that lands on red 25% of the time.

Which real result would be the most surprising under this model?



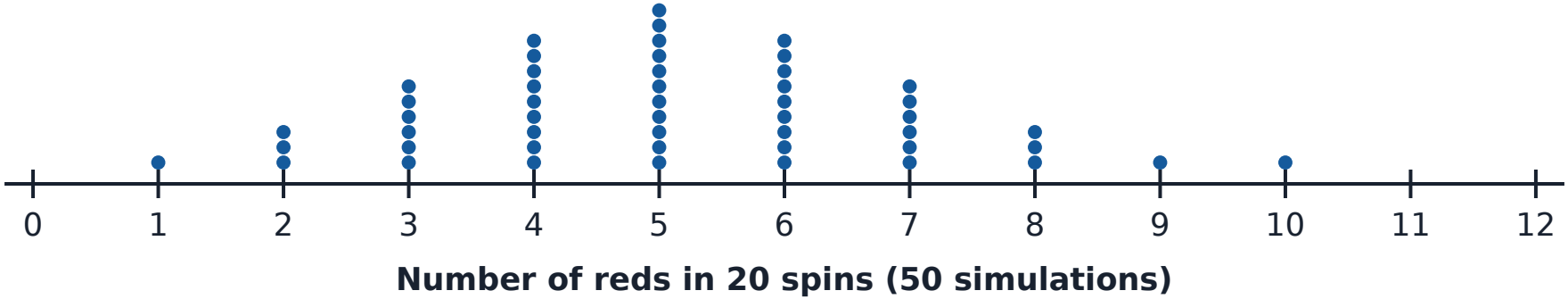
A. 6 reds **B.** 4 reds **C.** 5 reds **D.** 10 reds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q3/5

True or false?

In the simulation shown, 9 or more reds happened in 2 of the 50 sets, about 4% of the time.



A. True

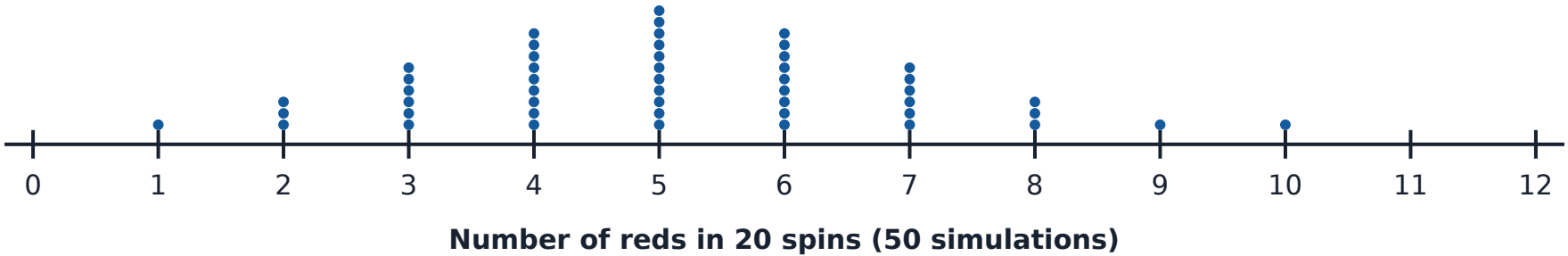
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q4**

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q4/5

True or false?

A real spinner gives 8 reds in 20 spins. Because 8 is more than the expected 5, the simulation shown proves the 25% model is wrong.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q5/5

SET 59 • 7.SP.C.8.a • STATISTICS & PROBABILITY

Find the probability of a compound event as the fraction of outcomes in the sample space

MAIN	Grade 7 main question	**7.SP.C.8.a**
BACKWARD 1	Multiply fractions	**5.NF.B.4.a**
BACKWARD 2	Count pairs with an array: one row for each first choice	**3.OA.A.3**
FORWARD 1	Find joint relative frequencies in a two-way table — nearest relate…	**8.SP.A.4**
FORWARD 2	Use the multiplication rule for independent events	**HSS-CP.A.2**

SET 59 • 7.SP.C.8.a

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.8.a

A fair coin is flipped and a fair number cube numbered 1 to 6 is rolled.

What is the probability of getting heads and a 6?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q1/5

Two fair coins are flipped.

What is the probability that both land tails?

Probability: _____

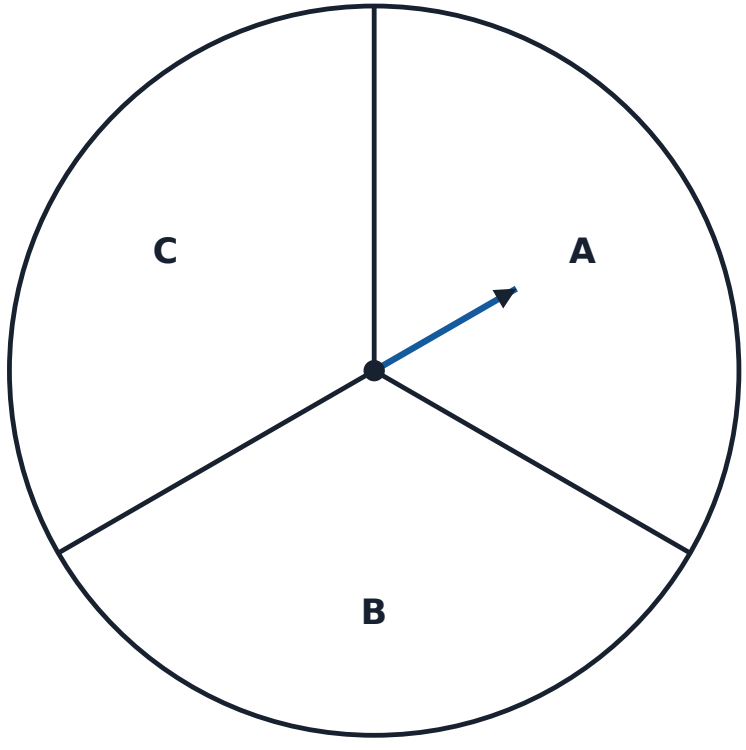
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q2/5

The spinner has 3 equal sections and a fair pointer. It is spun twice, and the spins do not affect each other.

What is the probability of landing on A both times?



A. $\frac{2}{3}$ **B.** $\frac{1}{6}$ **C.** $\frac{1}{9}$ **D.** $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q3/5

True or false?

Two fair number cubes numbered 1 to 6 are rolled. The probability that the sum is 7 is $\frac{6}{36} = \frac{1}{6}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q4/5

True or false?

Two fair coins are flipped. The probability of getting one head and one tail is $\frac{1}{4}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q5/5

SET 59 • 7.SP.C.8.a

BACKWARD 1

Multiply fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$\frac{1}{2} \times \frac{1}{6}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q1/5

Multiply.

$\frac{2}{3} \times \frac{3}{4}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q2/5

Multiply.

$\frac{1}{4} \times \frac{1}{4}$

A. $\frac{2}{4}$

B. $\frac{1}{8}$

C. $\frac{1}{16}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q4/5

True or false?

$\frac{1}{5} \times \frac{1}{5} = \frac{2}{5}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q5

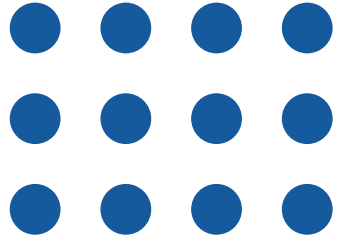
Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q5/5

SET 59 • 7.SP.C.8.a

BACKWARD 2

Count pairs with an array: one row for each first choice • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

A café has 3 kinds of sandwiches and 4 kinds of drinks. The array has one row for each sandwich and one column for each drink; each dot is one sandwich-and-drink pair.



How many pairs are there?

Pairs: _____

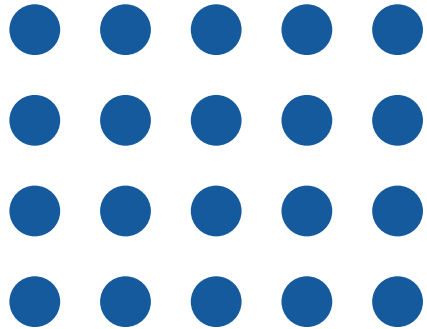
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q1/5

Ty has 4 shirts and 5 pairs of pants. The array has one row for each shirt and one column for each pair of pants; each dot is one outfit.

How many outfits are there?



Outfits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q2/5

A shop has 6 ice cream flavors and 2 kinds of cones. Each flavor can go with each cone, so there are 6 rows with 2 choices in each row.

How many flavor-and-cone choices are there?

A. 12

B. 8

C. 3

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q3/5

True or false?

3 rows of 7 chairs is 21 chairs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q4/5

True or false?

An array has one row for each of 5 hats and one column for each of 3 scarves. It shows 8 hat-and-scarf pairs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • 7.SP.C.8.a

FORWARD 1

Find joint relative frequencies in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows 100 trials of flipping a coin and rolling a number cube.

What is the relative frequency of an even number and heads?

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows 100 trials of flipping a coin and rolling a number cube.

What is the relative frequency of tails?

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows 100 trials of flipping a coin and rolling a number cube.

Which result happened most often?

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

A. Even and tails

B. Odd and tails

C. Even and heads

D. Odd and heads

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of an odd number and tails is 0.23.

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

The relative frequencies of the four results in the table add to 2.

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 59 • 7.SP.C.8.a

FORWARD 2

Use the multiplication rule for independent events • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

Events A and B are independent. P(A) = 0.3 and P(B) = 0.5.

Find P(A and B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q1/5

A fair coin is flipped 3 times, and the flips are independent.

What is the probability that all 3 flips are heads?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q2/5

A fair number cube numbered 1 to 6 is rolled twice, and the rolls are independent.

What is the probability of rolling two 6s?

A. $\frac{1}{6}$

B. $\frac{1}{36}$

C. $\frac{1}{12}$

D. $\frac{2}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q3/5

True or false?

Events A and B are independent when P(A and B) = P(A) × P(B).

A. True

B. False

True or false?

Drawing two cards from a deck without putting the first card back are independent events.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q5/5

SET 60 • 7.SP.C.8.b • STATISTICS & PROBABILITY

Represent the sample space of a compound event with a list, a table, or a tree diagram

MAIN	Grade 7 main question	**7.SP.C.8.b**
BACKWARD 1	Use repeated addition to find the total in an array	**2.OA.C.4**
BACKWARD 2	Form ordered pairs from two numerical patterns	**5.OA.B.3**
FORWARD 1	Organize two categorical variables in a two-way table — nearest…	**8.SP.A.4**
FORWARD 2	Find a conditional probability as P(A and B) ÷ P(B)	**HSS-CP.A.3**

SET 60 • 7.SP.C.8.b

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.8.b

A coin is flipped and a spinner with sections 1, 2, and 3 is spun.

List the sample space.

Sample space: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q1/5

A café offers 3 sandwiches (ham, egg, tuna) and 2 drinks (milk, juice).

Make an organized list of every sandwich-and-drink pair.

Sample space: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q2/5

Draw a tree diagram for choosing a shirt (red or blue) and then pants (black, tan, or gray). Write the outcome at the end of each branch.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q3/5

Complete the table to show the sample space for flipping a coin and then rolling a number cube numbered 1 to 6.

	1	2	3	4	5	6
H						
T						

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q4/5

A first coin and a second coin are flipped.

Make an organized list of the sample space. Then list the outcomes in the event "exactly one head."

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q5/5

SET 60 • 7.SP.C.8.b

BACKWARD 1

Use repeated addition to find the total in an array • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

An array has 3 rows with 4 objects in each row.

Write an addition equation for the total.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q1/5

An array has 2 rows with 5 objects in each row.

Write an addition equation for the total.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q2/5

Which equation gives the total of an array with 4 rows of 3?

A. 4 + 3 = 7

B. 3 + 3 + 3 + 3 = 12

C. 3 + 3 + 3 = 9

D. 3 + 4 + 3 = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q3/5

True or false?

An array with 3 rows of 5 has 5 + 5 + 5 = 15 objects.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q4/5

True or false?

An array with 2 rows of 4 has 4 + 2 = 6 objects.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • 7.SP.C.8.b

BACKWARD 2

Form ordered pairs from two numerical patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 4.

Write the first 3 ordered pairs (A, B).

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q1/5

Pattern A starts at 1 and adds 1. Pattern B starts at 3 and adds 3.

Write the first 3 ordered pairs (A, B).

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q2/5

Pattern A is 0, 3, 6, 9. Pattern B is 0, 6, 12, 18.

Which ordered pair (A, B) comes from these patterns?

A. (3, 3)

B. (12, 6)

C. (6, 9)

D. (6, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q3/5

True or false?

In the patterns A: 0, 5, 10 and B: 0, 10, 20, each term in B is twice the matching term in A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q4/5

True or false?

The patterns A: 0, 2, 4 and B: 0, 3, 6 make the ordered pair (4, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • 7.SP.C.8.b

FORWARD 1

Organize two categorical variables in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

In a class, 14 students have a sibling and 10 do not. 8 of the students with a sibling have a pet. 3 of the students without a sibling have a pet.

Complete the two-way table.

	Pet	No pet	Total
Sibling			
No sibling			
Total			

A class has 30 students. 18 like soccer. 12 of the soccer fans also like basketball. 4 of the students who do not like soccer like basketball.

Complete the two-way table.

	Basketball	No basketball	Total
Soccer			
No soccer			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q2/5 • NEAREST RELATED

50 shoppers were counted. 20 used a coupon. 15 of the coupon users paid by card. 25 of the shoppers paid by card.

Complete the two-way table.

	Card	Cash	Total
Coupon			
No coupon			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q3/5 • NEAREST RELATED

A team played 36 games. It played 20 at home and won 14 of them. It won 22 games in all.

Complete the two-way table.

	Won	Lost	Total
Home			
Away			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q4/5 • NEAREST RELATED

90 students were surveyed. 40 walk to school, and 25 of them eat breakfast at home. 30 of the students who do not walk eat breakfast at home.

Complete the two-way table.

	Breakfast	No breakfast	Total
Walk			
Do not walk			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 60 • 7.SP.C.8.b

FORWARD 2

Find a conditional probability as P(A and B) ÷ P(B) • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 0.12 and P(B) = 0.4.

Find P(A | B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q1/5

Two fair coins are flipped. At least one landed heads.

What is the probability that both landed heads?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q2/5

A fair number cube numbered 1 to 6 is rolled, and the result is even.

What is the probability that it is a 6?

A. $\frac{1}{6}$

B. $\frac{1}{2}$

C. $\frac{1}{3}$

D. $\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q3/5

True or false?

If A and B are independent and P(B) > 0, then P(A | B) = P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q4/5

True or false?

P(A | B) is always equal to P(B | A).

A. True

B. False

SET 61 • 7.SP.C.8.c • STATISTICS & PROBABILITY

Design and use a simulation to estimate the probability of a compound event

MAIN	Grade 7 main question	**7.SP.C.8.c**
BACKWARD 1	Display data in a dot plot	**6.SP.B.4**
BACKWARD 2	Compare fractions with the same numerator or the same denomin…	**3.NF.A.3.d**
FORWARD 1	Construct and interpret a scatter plot — nearest related (no direct…	**8.SP.A.1**
FORWARD 2	Use a simulation to decide whether a difference between treatme…	**HSS-IC.B.5**

SET 61 • 7.SP.C.8.c

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.8.c

A player makes 60% of her free throws, and each shot is independent of the others.

Describe how to use random digits 0 to 9 to simulate one trial of 2 free throws.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q1/5

A player makes 60% of her free throws; each shot is independent. In the simulation, the digits 0 to 5 mean a make and 6 to 9 mean a miss. Each pair of digits is one trial of 2 shots:

32 70 15 86 41 09 55 63 27 84

In how many trials are both shots made? Estimate the probability that both shots are made.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q2/5

Each of 3 children is equally likely to be a boy or a girl, and the children are independent.

Which tool can simulate the 3 children?

A. Spin a spinner with 3 unequal parts once

B. Flip a fair coin 3 times

C. Draw one card from a deck

D. Roll one number cube once

Each of 3 children is independently equally likely to be a boy or a girl. In the simulation, an even digit means a girl and an odd digit means a boy. Each group of 3 digits is one family:

204 317 856 993 120 448 715 362 091 574

Estimate the probability that a family has at least 2 girls.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q4/5

True or false?

A simulation of 3 children uses even digits for girls and odd digits for boys. Its trials are 246, 135, and 802. Exactly 1 of these 3 trials shows at least 2 girls.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q5/5

SET 61 • 7.SP.C.8.c

BACKWARD 1

Display data in a dot plot • Grade 6 • Statistics & Probability • 6.SP.B.4

Make a dot plot of the numbers of heads in 8 trials:

1, 2, 2, 3, 1, 2, 0, 2



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q1/5

The dot plot shows the number of heads in 10 trials.

In how many trials were there 2 or more heads?



Trials: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q2/5

Which display shows each data value as a dot above a number line?

A. Dot plot

B. Histogram

C. Bar graph

D. Box plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q3/5

True or false?

A dot plot shows every value in a data set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q4/5

True or false?

A box plot shows every value in a data set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • 7.SP.C.8.c

BACKWARD 2

Compare fractions with the same numerator or the same denominator • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

Which is greater, $\frac{3}{8}$ or $\frac{5}{8}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q1/5

Write >, <, or =.

$\frac{2}{3}$ ___ $\frac{2}{6}$

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q2/5

Which fraction is less than $\frac{4}{6}$?

A. $\frac{5}{6}$

B. $\frac{6}{6}$

C. $\frac{3}{6}$

D. $\frac{4}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{4} < \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q4/5

True or false?

$\frac{3}{8} > \frac{4}{8}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q5/5

SET 61 • 7.SP.C.8.c

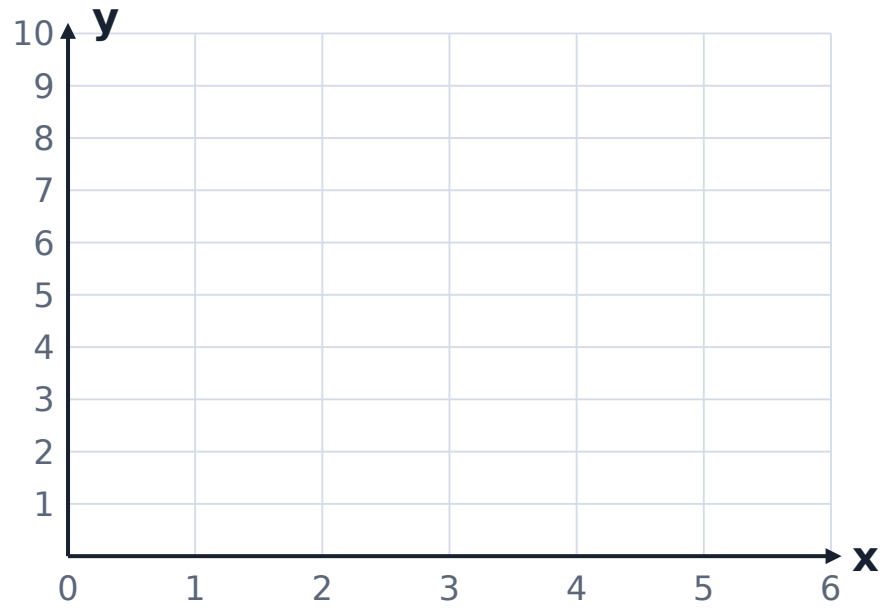
FORWARD 1

Construct and interpret a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Make a scatter plot of the data in the table.

x	1	2	3	4	5
y	2	4	5	7	9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows x-values 1, 2, 3, 4, 5 and y-values 2, 4, 5, 7, 9.

Describe the association between x and y.

Association: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q2/5 • NEAREST RELATED

What does each point in a scatter plot stand for?

A. An outlier

B. One pair of values from the data

C. The mean of the data

D. The line of fit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A scatter plot shows the relationship between two numerical variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

All the points in a scatter plot must lie on a straight line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 61 • 7.SP.C.8.c

FORWARD 2

Use a simulation to decide whether a difference between treatments is significant • High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In an experiment, 40 similar plants were randomly assigned: 20 got a new plant food (Group A) and 20 got none (Group B). Group A grew a mean of 12 cm and Group B a mean of 9 cm. The data were re-randomized 100 times to simulate no treatment effect; a difference of 3 cm or more happened 2 times.

Is there evidence that the plant food made a difference? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q1/5

In the same kind of randomized experiment (plants randomly assigned to plant food or none), the data were re-randomized 100 times to simulate no treatment effect. A difference of 3 cm or more happened 40 times.

What does this suggest about the plant food?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q2/5

Why are the data re-randomized many times in the simulation?

A. To remove outliers

B. To see what differences happen by chance alone

C. To make the mean larger

D. To make the treatment work better

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F2-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q3/5

True or false?

Random assignment helps make two treatment groups alike before an experiment begins.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q4/5

True or false?

A difference between two groups in one experiment is always caused by the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q5/5